

This is an excerpt from Frank Elavsky's dissertation on *Tool-making as an Intervention on the Accessibility of Interactive Data Experiences*, which can be accessed in full at this archival link (once published):

<http://reports-archive.adm.cs.cmu.edu/anon/hcii/CMU-HCII-26-103.pdf>

This document contains the following sections:

- Abstract
- Table of contents
- Chapter 1: What do we mean by “accessibility” and “tools?”
- Chapter 2: Background & Related Work
- Chapter 3: Overview of Contributions
- Chapter 6: *Skeleton*: Visual Authoring of Non-visual Data Experiences
- Chapter 7: *Cross-perception*: Rethinking Input Design Towards Blind Analytical Interaction
- Chapter 9: Findings
- Chapter 10: Discussion & Future Work
- References

This document does not contain the following chapters:

- Chapter 4: *Chartability*: Heuristics as a Tool and Resource
- Chapter 5: *Data Navigator*: Low-level Tooling for Creating Navigable Data Structures
- Chapter 8: *Software*: Enabling Personalization of Interactive Data Representations for Users with Disabilities
- Chapter 12: The *Software* Option Taxonomy
- Chapter 11: Biographical Sketch

Contents

Abstract	vii
I Introduction	1
1 What do we mean by “accessibility” and “tools?”	3
1.1 Is “accessible visualization” really an oxymoron?	3
1.2 On <i>tools</i> , <i>tool-making</i> , and <i>human-tool</i> interaction	5
2 Background & Related Work	7
2.1 Tools and the People Who Use Them	7
2.1.1 Studying Practitioners and Their Work	7
2.1.2 How Tools Use Us	7
2.2 The Conversation Between Accessibility and Data Visualization	9
2.2.1 A Short History Across Modalities	9
2.2.2 The Split Between Research and Practice	10
2.2.3 Empirical Depth, but a Lack of Applied Work in Production	11
2.3 Critical Voices and Work Along the Periphery	12
2.4 Why Navigation, Interaction, and Personalization	13
2.4.1 Navigation	14
2.4.2 Interaction	14
2.4.3 Personalization	14
2.5 What to Take Away	15
3 Overview of Contributions	17
III Navigation: Making Data Structures Traversable	23
6 <i>Skeleton</i>: Visual Authoring of Non-visual Data Experiences	25
6.1 Preface	25
6.1.1 Context	25
6.2 Overview	27
6.3 Related Work	29
6.3.1 Visualizing Non-visual Structure	29

6.3.2	Non-visual Data Experiences	29
6.3.3	Authoring Non-visual Data Experiences	30
6.4	Co-design Foundations	31
6.4.1	Geologic Map	32
6.4.2	Design System Library	33
6.4.3	Open Source Visualization Library	33
6.4.4	Infrastructure from Practice	34
6.5	<i>Skeleton</i> : System Design	40
6.5.1	Staging: Input and Preparation for Authoring	40
6.5.2	Edit: Interacting with Topology, Layout, and Semantics	41
6.5.3	Test: Debugging Interaction Interactively	43
6.5.4	System Implementation: Technical Details	44
6.6	User Study	46
6.6.1	Participants	48
6.6.2	Procedure	48
6.6.3	Analysis	49
6.7	Results	49
6.7.1	Seeing Navigation Made Structural Problems Legible as Design Problems	49
6.7.2	Practitioners Developed a Designerly Interest in What Constitutes Good Navigation	50
6.7.3	Iteration Was Substantive, Self-directed, and Concentrated on Semantics	51
6.7.4	Seeing Navigation Prompted Practitioners to Reconsider the Architecture of Their Own Charts	52
6.7.5	Experiencing Keyboard Navigation Surfaced a Broader Range of Users and Input Technologies	52
6.8	Discussion	54
6.8.1	Visibility as a Precondition for Iteration	54
6.8.2	From Compliance to Design	54
6.8.3	Bespoke Visualizations as an Unaddressed Accessibility Research Problem	55
6.8.4	What Visualization Owes Accessibility	55
6.9	Limitations and Future Work	56
6.10	Conclusion	57
6.11	Postface	58

IV Interaction: Exploring New Possibilities for Blind Data Science 59

7	<i>Cross-perception</i>: Rethinking Input Design Towards Blind Analytical Interaction	61
7.1	Preface	61
7.1.1	Context	61
7.2	Overview	63
7.3	Related Work	65
7.3.1	Cross-Filtering and Interactive Visual Analysis	65
7.3.2	Non-Visual Data Representations	65

7.3.3	Access-Oriented Data Interaction	66
7.3.4	Tactile and Haptic <i>Input</i>	66
7.4	Formalizing Cross-Perception	67
7.4.1	A Design Vignette: Fuzzy Search and Dual-Task Comparison in Natu- ralistic Blind Reading	67
7.4.2	Design Goals	68
7.5	Prototype: The Cross-Feelter	69
7.5.1	Interaction Flow	69
7.5.2	Hardware Design	70
7.5.3	Analytical Environment	72
7.6	Data Exploration Study	74
7.6.1	Study Design	74
7.6.2	Participants	75
7.6.3	Datasets	75
7.6.4	Device Familiarization: Video Interaction Task	75
7.6.5	Introduction to Tactile Histograms and Cross-Filtering	76
7.6.6	Data Tasks	76
7.6.7	Measures and Analysis	77
7.7	Results	77
7.7.1	Objective Results	78
7.7.2	Subjective Results	79
7.8	Discussion	79
7.8.1	Research Question Responses	80
7.8.2	Accuracy and Precision Considerations	80
7.8.3	The Case for Analytical Interaction as a Primary Research Site	81
7.8.4	Toward Hardware and Physicalization in Accessible Data Interaction	81
7.9	Future Use Cases	82
7.9.1	Cross-Perception Without a Powered Display: The Pre-Rendered Inter- action Cube	82
7.9.2	Sequential Data Navigation with Continuous Positional Feedback: Felter- Only Chart Interaction	83
7.9.3	Sonification Navigation via Physical Rail Position	84
7.9.4	Multi-Dimensional Spatial Interaction: Rails as Input and Output	84
7.10	Limitations	84
7.11	Conclusion	85
7.12	Postface	86

VI Conclusion 87

9 Findings 89

9.1	Access Barriers Are Produced for Practitioners, Both With and Without Disabilities	89
9.2	Tools Can Reduce the Work Required to Make Things Accessible	90
9.3	Making the Invisible Structurally Visible Changes Practice	90

9.4	Friction From a Tool Can Be Productive	91
9.5	Tool-use and Tool-making Are Generative	92
9.6	The Hardest Barriers Required Intervention at the Substrate	93
9.7	What Tool-making Did Not Do	94
9.7.1	State of the Art does not Necessitate Adoption	94
9.7.2	Tooling does not <i>Solve</i> Access Friction	95
9.7.3	Authorship by People with Disabilities Remains Unresolved	95
10	Discussion & Future Work	97
10.1	What is a “tool?” A reflection on the social and material identity of tools	97
10.2	“Applied” accessibility work and <i>low research</i>	98
10.3	Who is responsible for repair?	99
	References	101

List of Figures

6.1	<i>Skeleton</i> being used to build a navigation structure over a line chart. A: a dual tree visualization of nodes and edges built for a line chart with 3 lines, spanning over 12 months. B: a direct-manipulation canvas that resides over a static png image of a line chart. C: a skeleton-like structure of nodes, edges, and group outlines that correspond to the schema shown in (A), overlaid on the visual space of the chart in (B). D: controls for rapidly scaffolding the nodes, edges, and their shapes shown in (C), using Vega’s visualization engine for automatically generating spatial properties.	27
6.2	Our visual design work in Figma over a static geologic infographic map of Wisconsin. We use visual forms and illustrations over and beside the map to communicate flows, structure, navigation styles, and interaction patterns.	32
6.3	Input data transformed into a navigable structure using the <i>Dimensions API</i> and visualized with our <i>Inspector</i> gadget (left). The input chart (middle). The navigable structure is transformed and drawn over the chart using the <i>Scaffolding Engine</i> (right).	41
6.4	Group label pattern builder, including an array of aggregate summary options, template formatter field, and preview.	43
6.5	Computed layout for marks via Vega’s visualization engine plus group outlines (left) and with padding estimation using our non-ML computer vision approach (right) for various types of common visualizations.	47
6.6	Re-creation of P8’s moment of realization, placing nodes manually: not every element in their voronoi pie chart <i>should</i> be navigable.	53

7.1	Interaction and perception in one space while being able to perceive output in a separate space is the cornerstone of cross-filtering. Our prototype <i>cross-feelter</i> (left) can manipulate a visual cross-filter on one visualization (middle) in order to produce output in a separate visualization, as a tactile graphic (right).	63
7.2	A. Our prototype <i>cross-feelter</i> . B. USB output (to computer). C. 3D printed rail cover. D. 3D printed knobs that consolidate rails into one. E. 2 motorized linear potentiometers. F. Arduino board. G. 3D printed casing.	69
7.3	Wiring schematic of the cross-feelter. A. Teensy 4.0 microcontroller. B. DRV8833 dual H-bridge motor driver. C. Motorized fader, rail 0. D. Motorized fader, rail 1. E. Host computer, connected over USB. Wire colors distinguish 5V motor power, the 3.3V supply, ground, wiper position lines, touch-sense lines, and motor drive lines.	72
7.4	Our analytical environment. A. a 1 to 1 translation between visual and tactile that represents a 60x40 pixel-to-pin array. B. A tactile version of the filter location (if on the chart being focused). C. Cross-filtering controls, including text inputs for screen-reader only manipulation. D. Refreshable tactile display output. E. Example renderings of the other 2 charts.	73
7.5	Results. Completion rate and accuracy are counts. Task time, data queries (computational), and data queries (spoken) are objective, observational measures. Anxiety, stress, cognitive load, and enjoyment are Likert-scale results from 1 (“very little”) to 5 (“quite a lot”).	78
7.6	Speculative future use cases for our <i>cross-perception</i> approach, contributed by our participants.	83

Abstract

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Skeleton, Cross-perception, and Softeware, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework, contributed first and mapped the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softeware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosystems.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part I

Introduction

Chapter 1

What do we mean by “accessibility” and “tools?”

This thesis is a body of research situated within the existing research area focused on making interactive data representations more accessible for people with disabilities. Much of the work in this existing area is situated within the context of making interactive data *visualizations* accessible, particularly (but not exclusively) for people who are blind. Our work, contributed here in this thesis, is focused on using *tools* as a specific intervention and sub-area of study for making interactive data *representations* accessible for people with disabilities, broadly speaking. (“Representations” here is an intentionally broader term than “visualizations,” which are exclusively visual representations of data.)

Before we begin, two things must be understood up front, or else the rest of this thesis could be interpreted with disruptive assumptions: we must interrogate the phrase “making visualizations accessible” and unpack why *tools* are a meaningful area of study.

1.1 Is “accessible visualization” really an oxymoron?

The first assumption that must be disrupted is perhaps the motivating cornerstone of this research, which is that the phrase “making visualizations accessible,” while a noble goal, is not the semantically correct phrasing nor precisely what describes our work. This can be misleading. We do use the phrase “accessible visualization” but will admit that this seems to confuse certain people with very particular opinions about things. We will clear this up.

Villains of our field’s past have written incendiary and ableist perspectives on why “no forms of data visualization, not just dashboards jam-packed with graphics, can be made fully accessible to someone who is blind,” and that “[a blind man] will never be able to analyze data as I do visually, because many aspects of vision cannot be duplicated by his other senses” [52]. However, this position misunderstands what the goal of accessibility is, and arguably even what the goal of visualization itself is.

Making visualizations accessible *isn’t* about the visualization, it’s about making the outcomes of the visualization accessible.

Visualizations are ubiquitous and paramount for decision-making. However, the *artifact* that is a visualization is not even the goal of the act of visualizing: developing understanding, insight, confidence, and communication among and between human beings are the goals of visualization. Visualization is about making data easier to use for all kinds of things. Yes, our visual system enables us more than any other form of sensory cognition that we have [26, 62, 176]. But we aren’t trying to make sight itself accessible. We are trying to make it possible for people to make meaningful decisions, gain valuable information, build conjectures, and effectively communicate with others.

Many, many people who I, Frank, have spoken to over the course of my career, even before embarking on this thesis journey, misunderstand this simple fact: making a “visualization” accessible *isn’t* about the visualization itself but rather making what the visualization is meant to

accomplish accessible. It's about equal outcomes, not equal interactions with an artifact.

People with disabilities are no small portion of the world's population. In the United States, 26% of people self-report living with at least one disability that affects their daily lives [138] and all of us will eventually age into disability (if we are lucky to live a long life).

People with disabilities deserve to participate fully in life. They deserve financial independence. They also deserve loving care and interdependence [11, 92, 131]. People with disabilities have a right to make informed decisions, to know about the status of a global pandemic, and to have an understanding of local and national politics [50]. While we use visualizations to navigate all of these domains, the goal is not to make the charts and graphs themselves somehow equally useful to all people. That would be a false measurement of success.

Our goal then, is measured by the success of lives led by people with disabilities [184]. Many other measurements are just metrics along the journey towards that goal. We then ask: Can people with disabilities also use data to live full lives? Can they make *fast* decisions based on data? Meaningful, careful, *slow* decisions? Communicate complex ideas? Crunch and clean data, develop models, find errors, and build hypotheses? Can they have memorable, immersive, beautiful, aesthetic experiences with data too [92]? Ultimately, our aim is not accessible "visualizations" but equitable and accessible *outcomes*, everything an interactive data experience *accomplishes* for the people who use it.

Again, if the goal of accessible visualization were about visualizations themselves, then the correct course of action would be one framed by the medical model of disability [124]: that there is a normative state of behavior and capability (in this case, it would be "normal" to be able to read a visualization and make a decision) and any deviation from that norm must be corrected. This framing first assumes that the visualization should not be altered or improved. And then this framing puts the burden on the bodies of people with disabilities: that they must be "fixed" and given sight or brought to some equivalent state as someone who is "healthy," normal, and sighted. Plenty of scholars have already discussed why this framing is a problem, not only because it places undue burden on people with disabilities and produces pathologies and hierarchies of disability, but also because it is fundamentally not economically or ethically feasible.

So we then turn to other models of disability, such as the social model. The social model is heavily discussed by disability scholars and is not the end-game or last and total way of thinking about disability [124, 145, 147, 174, 209]. But the core motivation is that society, not medicine, is also a path towards solving problems that people with disabilities face. A few important concepts and concretely actionable things come from the social model that can help motivate the work of this thesis.

First, we look to the historical birth of the social model of disability: in the 504 sit-ins that took place in the United States in 1977. Cities had curbs and curbs are a barrier for people who use wheelchairs. So protests happened because decisions were being made without people with disabilities at the table. In this instance, people acknowledged that political power was an exclusive club and fought to ensure their cry "nothing about us, without us!" materialized.

And this leads us to the first and most-foundational philosophical framing for this thesis: that our *artifacts*, these things we've created from curbs to data visualizations, can become *barriers* for people with disabilities. And it is then the artifact, not the body of the person with a disability, where disability is produced in this model. Rather than a comparison to a normative state as a

way to frame disability (the medical model), we instead must observe and evaluate material outcomes based on human-made problems.

So, the social model is framed around society “solving” inequities: we get involved and make political and legal change tangible. But a second model also emerges from within the social model: one where we can now frame *who is first responsible* for repair: the curb designers and implementers.

And knowing who is first responsible for access leads us into the moral and ethical imperative that motivates this thesis: the builders and makers of visualizations are ultimately the ones who provide exclusive value for only a subset of people: those *without* disabilities. **We must first change how builders and makers do their work.**

So the phrase “accessible visualization” is really about recognizing that visualizations produce barriers for people. That means that it is our ethical imperative, as builders and makers, to fix them. And that act of fixing barriers leads us away from mere visual representations of data into a wide variety of other senses and interaction modalities. There are many paths forward towards fuller and more-equitable lives led by people with disabilities.

1.2 On *tools*, *tool-making*, and *human-tool* interaction

Then the act of making becomes immensely important: we, the builders and makers of our world, need to get things right; there is a risk involved when making things that we will exclude people with disabilities. We need to make sure that we build a better world than the one we have now. We must care for new things we create and tend to the repair and maintenance of what we’ve already made. And this ethical imperative leads us to the topic of *tools* and *tool-making*.

So the second thing that must be understood before we embark on this thesis is that *tools* are not the same as *solutions* or *applications*. Sometimes tools can be used to *solve* things and are certainly, in ideal circumstances, *applied* in various contexts. But understanding the role of the “tool” in human-tool interaction is paramount for engaging in the work of making anything accessible for people with disabilities.

We use tools to shape our world, break old things, and make new things. But a tool, like the hammer (as an example), does not inherently *solve* something like homelessness. But a hammer can be used to build homes if there are social policies in place and proper resources allocated. This means that for the success of tooling, there is often a larger material, social, legal, and policy reality that supports and necessitates those tools. This thesis will not be focusing on changing the upstream dependencies, but optimistically operating as if they were true (or will be true in time).

However, in some cases, tools can *destroy*. The hammer has a claw and can easily pry apart boards and tear down homes. So tools carry potential to do all kinds of things, both good and bad, and how a tool is used is often open-ended, variable, and heavily dependent on socio-technical realities. Tools participate in personal and political agendas [205] and are sometimes, for this reason, regulated or made proprietary and controlled by powerful entities [67, 198].

So tools are not without any sort of ethics. We cannot just blame tool-users for outcomes when much of a tool depends on these larger systems and structures. Technologies (tools included) encode the assumptions and biases of their *creators* as much as, if not more than, their

users. Tools that build things for others to use can be loaded with assumptions about what people are *able* to do [206] and also rules and guardrails about what anyone downstream from that tool’s design *should* do [67, 197]. These assumptions, biases, and rules *limit*, *enforce*, *magnify*, *exclude*, and *enable* what a tool-user is capable of.

Tools for visualizing data are a perfect case study in this problem: virtually every major data visualization library, application, or software ever made was made entirely with the assumption that data should be transformed into visual representations. This is a reasonable assumption, since virtually all of the tool-makers are sighted and visualization is incredibly helpful to our cognition of and communication with data [56].

So data visualization, as a field, has focused its tool-making efforts on reducing the difficulty involved in visualizing data. Some visualization tools are concise [154], others are lower level but much more expressive [19]. Tool-making in visualization has focused on making it easier to scaffold a wide variety of interactions both with the visualizations as well as with their underlying data models [82].

However, as time has moved on, people began to speak out about color-vision deficiency in data visualization. Some people, primarily those with X/Y chromosomes (largely men) who are of European ancestry, have a deficiency in their ability to perceive certain colors. Then a plethora of research arose that began to look into the barriers that folks who are colorblind face in data visualization. As a result, our practices and tools improved. We began to educate practitioners, develop new color palettes, researched new methods for testing our designs, and built new systems for handling automatic color encoding. Our tools evolved.

But now data visualizations have arguably become ubiquitous in daily life. By comparison, we have far more tools now for making visualizations quickly and easily than we do for representing data in non-visual ways. We also have far more research, relatively speaking, into how sighted end users interact with visualizations.

So this thesis engages gaps that arise in this space: Practitioners face immense challenges when crafting accessible data experiences. We first need to educate practitioners on what accessibility barriers actually are in interactive visualizations. Then, we must help them engage the hardest barriers in this work and create building blocks that help them to construct navigable data experiences, build design frameworks that can inform entirely new kinds of data interaction, and develop software systems for end-user personalization and agency. Our research seeks to advance the state of the art in tools that assist in accessible data interaction while also using tool-making as an intervention that helps us to better understand and characterize *why* and *how* data practitioners face barriers themselves in this work.

Chapter 2

Background & Related Work

This chapter sets up the story that the rest of this thesis engages. It makes three moves. First, we ground the idea that tools are not neutral instruments: tools shape the work and the workers who use them, which is why tool-making is a meaningful site for accessibility intervention. Second, we survey the long conversation between accessibility and data visualization and its history across modalities, including engaging critical voices that intersect with visualization and accessibility work which inform, support, and hold in tension the work of this thesis. Lastly, we argue that the hardest remaining challenges cluster in three domains, navigation, interaction, and personalization, and we signpost which chapters engage each of these.

2.1 Tools and the People Who Use Them

2.1.1 Studying Practitioners and Their Work

Human-computer interaction has a long tradition of studying the people who make things. Ethnographic and case-study methods immerse researchers in maker, DIY, and assistive technology communities to understand how practitioners actually work, especially when the work is new and unfamiliar to them [91, 93, 94]. Participatory design and co-creation bring practitioners and end users directly into the research process, surfacing how designers shift their thinking when they encounter novel problems [72, 152, 175]. And a growing body of design-based research aims not merely to fill gaps in existing practice but to extend what practitioners are capable of: scaffolding experimentation, supporting learning, and amplifying creative and technical expression [31, 70, 100, 105, 187].

This thesis sits inside that tradition but takes a particular methodological stance: we study practitioners *through their tools*. Each project in this thesis builds a tool, places it in front of working practitioners, and observes what the tool reveals about why accessibility work is hard for them. Sometimes the tool is a resource practitioners apply in their real environments, sometimes a system they build with, and sometimes a probe whose main purpose is to make an invisible part of their work observable. Tool-making here is both an intervention and an instrument of inquiry. That stance only makes sense, however, if tools really do shape work in a deep way. So before surveying the accessibility and visualization literature, we want to engage that claim seriously.

2.1.2 How Tools Use Us

We tend to talk about using tools. We talk much less about how tools use us. A chair is a tool for sitting, but a lifetime of chairs trains a body into the chair’s posture. A pickaxe is a tool for extracting ore, but the mine organizes the miner: their hours, their movements, their risks, and what counts as a day’s work. Tools come with a direction already built in, and the people who take them up inherit that direction. As a participant in Hohman et al’s work on model compression in practice remarked, “Better tools make it easy to do correct things and harder

to do incorrect things“ [88]. Tools are made precisely with the intention to shape us and our behavior.

Sara Ahmed gives this intuition theoretical teeth. In her phenomenology of orientation, objects are not passive things waiting to be used: they direct bodies, making some actions feel near and natural while other actions fall out of reach, and the social world is full of what she calls straightening devices, arrangements that keep bodies aligned with the paths already laid down for them [2]. In her later work she traces how *use* itself is formative: things become shaped by their use, “fit” for some users and not others, and a history of use leaves a trace that directs how a thing will be used next [3]. Bowker and Star make a parallel argument at the scale of infrastructure: the classification systems and standards embedded in our information systems are largely invisible scaffolding, and every standard valorizes some point of view while silencing another [20]. Assumptions about ability are among the assumptions most commonly baked in: tools encode what their makers believe people are able to do [206].

Visualization research has begun to examine its own tools in these terms. McNutt’s survey of 57 JSON-style domain-specific languages for visualization shows that each grammar embodies contested decisions about who its user is and what that user may express, with recurring tensions between formal and colloquial specification and between competing user types [128]. His dissertation develops the larger argument: the design of visualization languages, and of the tools that surround them, directly shapes and can either enhance or constrain end-user agency, motivating systems that offer assistance while respecting the user’s intent [129]. In other words, the field’s own instruments are straightening devices. Every charting grammar that assumes a sighted reader, every renderer that emits unstructured pixels, quietly directs practitioners away from accessible outcomes, not by forbidding them but by making them feel out of reach.

Tool users are not powerless in this relationship. A rich literature on appropriation documents how people adapt tools beyond their designers’ intent, and how that emergent use can be a source of design knowledge in its own right [39, 40, 151, 182], with some work building general theory from the study of emergent and generative tool use [9, 13]. HCI’s systems community has likewise developed methods for building and studying tools deliberately: piggybacking on existing systems to study an improvement in context [68, 85], and contributing toolkits whose value is argued through demonstration, usage, technical performance, or heuristics. Ledo et al., analyzing 68 toolkit papers, found these four evaluation strategies dominate, with demonstration by far the most common [112]. That finding will matter later in this chapter: a field can produce many demonstrated toolkits while producing very little maintained infrastructure.

The takeaway from this section is the premise the whole thesis rests on: tools direct work. If the tools of data visualization currently straighten practitioners toward inaccessible outcomes, then re-orienting those tools is not a cosmetic fix. It is an intervention at the place where practice is formed. The rest of this chapter examines what the field has learned about accessible data experiences, and why that learning has so far failed to re-orient the tools.

2.2 The Conversation Between Accessibility and Data Visualization

Accessibility and data representation have been in conversation far longer than the modern visualization community has existed, and that conversation has two important threads. The first is a history of techniques: a steady accumulation of ways to carry data to people through senses and devices other than a sighted eye on a screen. The second is a critical lineage, largely led by disabled scholars and disability studies, that questions how that technical work is framed, who leads it, and who it serves. This thesis tries to take both seriously, so we survey them in turn.

2.2.1 A Short History Across Modalities

Tactile representation is the oldest thread. Tactile maps and graphics for blind readers date to at least the 1830s [71], and tactile sensory substitution for charts has been studied in the computational sciences since the early 1980s [158]. Practice in this modality is unusually mature: tactile and braille standards are robust and well-maintained [8], although their assumptions (embossed paper, fixed layouts) transfer only partially to digital, interactive contexts. Research has worked on that seam, for example by augmenting tactile graphics with audio annotations [7], and industry accessibility teams have published candid experience reports on what it takes to produce tactile graphics inside a real design workflow [36]. Recent work continues to refine the perceptual foundations of non-visual structure [130].

Sonification has a similarly long arc, and because it is a substantial and active research community in its own right, it deserves explicit treatment here. Mansur et al.’s sound graphs demonstrated in 1985 that line data mapped to pitch over time could communicate trends to blind listeners [123]. Subsequent decades built out the empirical base: cross-modal comparisons of auditory and visual graphs [55], non-visual presentation of structured information [23], techniques for scanning and comparing multiple data series by ear [127], and interactive sonification of geo-referenced data [211]. The community consolidated its methods and theory in the Sonification Handbook [84]; Walker and Nees’s theory chapter formalizes sonification as a mapping problem from data dimensions to auditory dimensions, emphasizes context cues that play the role axis ticks play in vision, and stresses that interaction, not just playback, shapes what listeners can extract [193]. Sonification research has also been a site of methodological progress, including co-design with blind and low vision collaborators [35].

What is most relevant for this thesis is the recent trajectory: sonification is increasingly integrated into multimodal, interactive systems rather than offered as a standalone substitution. VoxLens adds summaries, sonification, and voice queries to web visualizations through a JavaScript plug-in, and in a study with 21 screen reader users improved information-extraction accuracy by 122% and interaction time by 36% over conventional interaction [164]. Chart Reader, co-designed with screen reader users, combines structured navigation with sonification and author-configurable data insights [186]. Auditory maps have reached usable maturity for wayfinding and choropleth reading [15]. Notably, sonification is also one of the few accessible-visualization techniques to cross into production: Apple shipped Audio Graphs in 2021 as a VoiceOver feature with a public developer API [6]. And Umwelt re-frames the relationship en-

tirely, treating sonification, textual structure, and visualization as coequal representations derived from a shared abstract model in an accessible authoring environment [215]. Sonification’s trajectory, from output substitution toward interactive structure and authoring, previews the argument this chapter is building.

Textual description is the most widely deployed modality, and research has moved it well past “write some alt text.” Lundgard and Satyanarayan’s model of semantic content distinguishes the levels of meaning a description can carry, and shows that blind and sighted readers value different levels [116]. Jung et al. offer evidence-based guidance on description length, plain language, and the ordering of information [103]. Kim et al. collected the questions screen reader users actually ask of visualizations, pointing toward more dialogic, query-driven access [109], and Chundury et al. deepen our understanding of how sensory substitution choices should be grounded in users’ practices [30].

Navigation and screen reader interaction is the youngest thread and the one closest to this thesis. Foundational work made statistical and mathematical content traversable by screen reader [63, 173]. Studies of screen reader users’ experiences with two-dimensional digital content, including visualizations, documented both the depth of exclusion and the strategies users develop anyway [156, 163]. Zong et al. then gave the area its clearest design vocabulary: structure (how a visualization’s entities are organized for traversal), navigation (the operations that move through that structure), and description (what is spoken at each position), showing through co-design that screen reader users want rich, structured exploration rather than a single summary [213]. Olli operationalized part of this insight as an open-source adapter that renders structured traversal for existing charting libraries [17]. A small number of production systems, notably Highcharts, SAS Graphics Accelerator, and Visa Chart Components, have shipped meaningful accessibility functionality [86, 153, 188]; they remain exceptions, and Chapter 4 examines that practitioner tooling landscape in detail.

Across all four modalities the same pattern recurs: each began with converting visual output into another channel, and each has progressively rediscovered that access is not a conversion problem but an interaction problem, turning toward structure, navigation, interactivity, and user control. The research itself has shifted accordingly: where early decades produced perception studies and one-off substitution systems, the last several years have produced co-designed interactive systems, design dimensions, and authoring environments. That convergence is part of why we argue, later in this chapter, that the remaining hard problems are not modality problems at all.

2.2.2 The Split Between Research and Practice

Before turning to the critical literature, it is worth naming that research and practice participate in accessibility at very different speeds, through very different methods, and with very different artifacts.

Accessibility *practice* organizes itself around standards. The Web Content Accessibility Guidelines carry legal and policy weight across much of the world [189], and a profession of accessibility specialists exists largely to translate such guidelines into concrete implementations. One example of this work would be ensuring that interactive elements carry functional semantics that assistive technologies can interpret, which is a foundational, simple, and necessary requirement for accessibility [191].

Standards are powerful precisely because they are consensus artifacts, or *politically mediated documents*, and for the same reason they are inherently reactive: developing, vetting, and formalizing them takes years, so they chronically trail the technologies they govern and are subject to intervention by individuals and organizations. Interactive data visualization sits at a painful point in that lag. Charts are semantically dense, interaction-heavy, and rendered through technologies the guidelines were not written for, so a practitioner who consults the standards in good faith finds little that speaks to their actual artifact. Robust evaluation in practice therefore leans on expert judgment and, in the best cases, direct involvement of people with disabilities, both of which are scarce resources [142].

Technical accessibility and assistive technology *research* meanwhile (outside of the previously mentioned work on modalities and our empirical record), moves quickly and publishes minimally-viable prototypes. These prototypes are an artifact practice largely cannot consume because it requires technical translation and methodological generalization from the small scope of an artifact into a system-wide implementation [54]. A practitioner cannot install a paper; a paper must be read and the artifact carefully dissected before being reconstructed or replicated. The result is a conversation in which researchers accumulate evidence and demonstrations on one side, practitioners wait on standards that lag by a decade on the other, and very little crosses between them. This split is not unique to visualization, but visualization makes it unusually visible, and closing it from the tooling side is the project of this thesis.

2.2.3 Empirical Depth, but a Lack of Applied Work in Production

What exists, first, is empirical breadth. Surveys and meta-analyses have mapped the research space and its biases: accessibility research broadly concentrates on blindness and on computer output [118], and visualization accessibility research mirrors that concentration, with vision-related work dominating while motor, cognitive, neurological, and vestibular access remain nearly unstudied [110, 125, 203]. Even within the well-studied territory the breadth has a shape: blind and low vision people are routinely researched as one population despite using different assistive technologies and different interaction practices [181], and the prototypes the field celebrates are overwhelmingly built for screen reader interaction in particular.

Second, the field has produced exemplary prototypes. The systems surveyed above, such as VoxLens, Chart Reader, Olli, and Umwelt, are rigorous, validated, and often co-designed with disabled collaborators.

What barely exists is the other half of the pipeline: practitioner tools in production environments, and practitioner action at scale. Studies of deployed visualizations and of the people who build them find that accessible outcomes remain rare and that practitioners struggle even when motivated, gathering fragmented guidance themselves with little way to judge its quality [102, 165]. Mainstream visualization tooling has invested its enormous momentum elsewhere, in expressiveness, scale, and interactive speed [19, 82, 154], with accessibility absent from core abstractions and left to individual developers, libraries, or projects. Research prototypes, meanwhile, rarely become infrastructure: toolkit research is evaluated primarily by demonstration [112], and most demonstrated systems are archived with their papers.

This asymmetry is the load-bearing observation of this chapter: **the field has a breadth of empirical work, yet it lacks practitioner tools in production environments and practitioner**

action. A practitioner today inherits strong evidence about what good access looks like but material change remains far behind. The natural next question is to investigate why this is the case, and whether and how tool-making can play a role in practitioner action.

2.3 Critical Voices and Work Along the Periphery

The second register of the conversation is critical, and although it is sometimes treated as peripheral to systems research, it sets the terms under which systems research should be read. We treat it here as load-bearing. This section will be brief, but essential.

The disability rights movement crystallized its central demand in the principle “nothing about us without us:” people with disabilities must hold power in the decisions and designs that concern them [29]. HCI and accessibility research have been working through what that demand means for our field’s methods, authorship, and self-conception [87, 174, 177, 178], including calls for a crip HCI and crip visualization that engages disability on disabled people’s own terms [92, 201] and concrete practices such as crediting disabled participant-contributors by name [212].

Several concepts from this literature directly shape this thesis. Mingus names *access intimacy*, the felt sense of someone genuinely getting your access needs, to insist that access is relational rather than merely technical and logistical [131]; Bennett et al. build on that disability justice lineage to propose *interdependence* as a frame for assistive technology research, countering the field’s fixation on independence [11]. Hamraie and Fritsch’s crip technoscience manifesto insists that disabled people are experts and designers of everyday life, and re-frames access not as a finished state but as friction, something contested and negotiated [77]. Hamraie’s history of universal design develops *access-knowledge*, showing that what counts as access has always been produced through political struggle rather than neutral technical progress [74, 75]. Bennett et al.’s biographical prototypes push against savior narratives in design by recognizing the prototyping disabled people already do in their own lives [12]. Shew names the broader cultural failure *technoableism*: the belief that technology will “fix” disabled people, held mostly by people who never asked disabled people what they want [167]. And within visualization specifically, Hsueh et al. bring crip technoscience to bear on data experiences, articulating how *access friction* arises when one design that includes some people excludes others [92].

We read this literature first as a set of constraints on what this thesis may claim. It is why we reject techno-solutionist framing outright: no tool in this thesis “solves” accessibility, because access is negotiated among people, institutions, and policy, and tools do not meaningfully redistribute power entirely on their own. It is also why access friction appears throughout this thesis as a design reality rather than an edge case, and why personalization is treated in Chapter 8 as an ethical question and not merely an intellectual problem to solve. And this body of work imposes a boundary we want to name early: a thesis about practitioner tooling mostly serves practitioners building *for* disabled users. The later chapters, Cross-perception and Softerware, begin to shift who holds the power to shape our interfaces. The critical literature is the measure of how far there is still to go.

We also read this literature as relatively divorced from accessibility work in practice. Critical scholarship has even been critical of “access” as a framing and its shortcomings [74, 75, 76, 142, 201]. Yet, practitioners are building systems all over the world that must be made accessible.

And rather than argue that people with disabilities should not be included in the design and development of new projects, instead we argue that, at least in some cases, the empirical record of what people with disabilities want and need is already enough imperative for builders and designers to begin working. For something new and unexplored? Yes, people with disabilities should be included. But what if we have a sufficient breadth and depth of guidance from people with disabilities already? Aren't we then ready to take action and begin practical application? *Chartability* in Chapter 4 takes this stance: people with disabilities have already said enough for us to build out comprehensive guidelines and an evaluation framework. Between standards, research, and community practice, we have an authoritative set of criteria we can test against. This should be our starting point; the bare minimum. And *Data Navigator* in Chapter 5 didn't seek to build a novel interaction but rather build a better system than what currently exists, following existing empirical work that laid the foundation for rich navigation as a necessity [213]. And in Chapter 5 we *do* demonstrate that our system enables novel interaction designs and how practitioners can approach co-design in a simple, effective, and thoughtful way with partners with disabilities. Chapter 6 sought to investigate the barriers that sighted practitioners (without disabilities) face, which is outside of the scope of critical disability literature, and Chapters 7 and 8 then centered on the experiences of blind and low vision people, and how they might have interactive and innovative power over their experiences.

2.4 Why Navigation, Interaction, and Personalization

If the gap were uniform, any contribution would do. It is not uniform. Looking across the empirical record, the prototypes, and the practitioner studies, the places where practitioners most consistently lack the means to act cluster in three domains, and each domain fails for a different kind of reason.

It is worth being explicit about why these three and not others, because the field's other major needs are of a different kind. Description is the best-served domain: practitioners have models, guidance, and concrete heuristics for what to write [103, 116], and what remains is largely a matter of doing it. Evaluation is hard, but it is hard as a knowledge problem, a matter of gathering, judging, and applying scattered guidance, and a well-built resource can carry a practitioner a long way; Chapter 4 contributes exactly such a resource and uses it to map the territory. *Chartability*, from Chapter 4, ultimately ends up framing the rest of this thesis. As we did work alongside our co-designers, clients, and industry partners over the years, applying *Chartability* to their work, it became clear that three major barriers emerged as the most pernicious (or least-resourced) for practitioners to address.

Navigation, interaction, and personalization are different in kind: in each of them, a practitioner who knows precisely what good access requires still cannot build it, because the means do not exist in their tools. Guidance cannot substitute for means. These areas then serve as opportunities to innovate, to build better, more general, or more specific tooling.

We unpack these three tooling gaps more below.

2.4.1 Navigation

The empirical case for structured navigation is now strong. Zong et al. showed through co-design that screen reader users want to traverse a visualization’s structure, not only hear a summary: structure, navigation operations, and description together determine the quality of the experience [213]. Systems work has begun to deliver this for specific stacks, with Chart Reader designing navigation experiences with screen reader users [186] and Olli adapting existing charting libraries into traversable structure [17]. What practitioners still lack is twofold. They lack low-level building blocks: a way to give *any* data interface, whatever its renderer or framework, a navigable structure with arbitrary shape, rather than the fixed hierarchies particular systems assume. And they lack any visible, manipulable representation of navigation structure in their workflows: a sighted practitioner can see a color palette and act on it with existing tooling but cannot see a navigation graph and act on it, which makes navigation design incredibly burdensome to iterate on and wholly downstream from other visual work. Chapter 5 (*Data Navigator*) contributes the structural substrate, and Chapter 6 (*Skeleton*) contributes the authoring representation; their chapter-level related work sections cover visualization toolkits and authoring systems more in depth.

2.4.2 Interaction

History shows that input design, not just output conversion, changes what is possible. Kane et al.’s *Slide Rule* contributed multi-touch techniques that let blind users operate touchscreens, outperforming a button-based baseline and prefiguring techniques that later appeared in commercial screen readers [104]. That simple lesson has not yet reached data analysis. Accessible visualization interaction today is overwhelmingly what we call *access-oriented*: it exposes information that is already there, through navigation, sonification, or description. These methods are *interactive* as existing work as argued [213], but we argue that the interaction is focused on perception of the representation and not enabling analytical interaction with and manipulation of the information that lies beneath the representation. Analytical work of the kind sighted analysts perform with cross-filtered, coordinated views depends on rapid, continuous loops of input and perception [82, 114], and a screen reader’s serial, query-at-a-time interaction may be structurally unable to sustain such loops. Fast, complex manipulation of a data space is presently made inaccessible by the primary assistive tool used by blind analysts. Rethinking input for blind analytical interaction, beyond an *access-oriented* framing, is nearly unexplored territory. Chapter 7 (*Cross-perception*) takes this on, formalizing a design framework and contributing a hardware prototype; that chapter’s related work covers tactile and haptic input devices in more detail.

2.4.3 Personalization

Accessibility research has long argued that systems, not users, should do the adapting. Put simply, this means that systems must be capable of adaptation. Arguably, this is the “soft” part of *software*. Yet, this softness can create conflict in system design, especially when an entire field of work must refactor their dominant tooling in order to enable system adaptation by end users. Visualization faces this exact challenge.

Existing work has demonstrated that interfaces adjusted to fit a person’s abilities and preferences measurably improved speed, accuracy, and satisfaction for users with motor impairments compared to default interfaces [57], and ability-based design generalized the stance into principles [206]. Good systems are made to fit a user’s abilities and remove barriers that they face. It could be argued, in broad strokes, that the field of HCI centers on the goal of *fit* between a user and a computational system.

Yet, visualization toolkits are largely not made to be personalized by end users. They are rigid and *hard*; lacking that softness required for end-user manipulation. Visualizations are typically built with a *universal design* [75] approach, when being made accessible (if being made accessible at all): one design is intended to fit as many people as possible. In visualization, recent work shows screen reader users benefit from customizing even small things, like the textual tokens spoken during navigation [101]. Yet a significant gap exists that enables visualization systems to adapt to personalization at scale and for many different people. How should these systems be built and what would they require? Which dimensions and features and functionality of data visualizations *should* be manipulable and customizable to fit end users? How should systems be constructed in order to make personalization effective and useful? And ultimately: how do we make visualization toolkits *softer*?

Some existing work on customizable accessibility settings identifies what drives people to adopt or abandon personalization features [208], but not all of this work neatly translates into visualization system design and development. Additionally, and fundamentally, we simply lack an empirical record of what exactly different people with disabilities would even want or need to manipulate. Presently, no mainstream visualization library or system offers practitioners infrastructure for end-user preferences over data representations: no shared option vocabulary, no persistence, no way to reconcile preferences with authorial defaults. Most visualizations become completely static in their design. Those with manipulable elements are only manipulable through hacking or manual overrides, placing a significant burden of access on end users, exactly where ability-based design says it should not [206]. Chapter 8 (*Towards Softerware*) engages this domain in collaboration with a production charting library, considered by the empirical record to be the most-accessible toolkit according to present evaluation methods by a wide margin [111]; Chapter 8’s related work section covers adaptive systems and personalization research in more depth.

2.5 What to Take Away

Five things from this chapter frame everything that follows. First, tools direct the work and the workers who use them, which makes tool-making a rich potential site for accessibility intervention and research into better understanding practitioner barriers when making and using tools. Second, the conversation between accessibility and data visualization is old, rich, and increasingly self-critical. Third, the field’s output is lopsided: empirical breadth, exemplary prototypes, and guidance exist in abundance, while production tooling and practitioner action barely exist. The broader field’s critical voices impose real constraints: tools do not solve access, and this thesis will not claim they do. Instead, critical work implores us to bridge the gap between research and practice. People with disabilities have been saying enough, it’s time to get to work

and try to apply what we know at scale. Fourth, the gap is concentrated where practitioners' means run out in three specific ways: navigation lacks a structural substrate, interaction lacks input techniques for analysis, and personalization lacks infrastructure. Fifth, the first step in our thesis work should be to collect the empirical record and synthesize it for applied accessibility work in the context of interactive data visualization. This is Chapter 4's *Chartability* project. Given where the field stands, *navigation*, *input*, and *personalization* are the categories of work that follow Chapter 4; the chapters ahead carry this out as the body of this thesis, and our concluding chapters return to what tool-making accomplished and what it cannot accomplish, on the other side.

Chapter 3

Overview of Contributions

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Skeleton, Cross-perception, and Softerware, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework, contributed first and mapped the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softerware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosystems.

We engage each domain below with the questions: “Why does this work matter?”, “Why is it hard?”, and “What has tool-making within this domain showed us?”

The four domains of work that we engage in this thesis start first with **evaluation**. In work contexts where someone is designing and developing interactive data experiences, the practitioner must have the knowledge, tools, and resources available to systematically identify how their interfaces produce barriers for people with disabilities. A significant portion of professional accessibility work (arguably most, if not all) is founded on auditing and evaluating barriers to access. This work is pre-dominantly done through a standards-based approach [189], although in more robust evaluation work, people with disabilities are actively involved in the process [142].

Evaluation is difficult work because much of it is contextually defined by the author themselves, and most tasks at this intersection require careful, non-automated processes and meth-

ods [38, 142]. To make matters more difficult, no comprehensive guidelines, tests, and tools exist in any singular location. Practitioners often must gather these resources themselves, which tend to be situated towards accessibility in general or are high-level and provide minimal usefulness in practice. Additionally, practitioners themselves often have little knowledge about the veracity or quality of any given bit of information they gather [102, 165], and often do the work themselves to synthesize this disparate space of information into a usable format they can apply to their own evaluation work.

To engage this, our first main chapter focuses on *Chartability*, a heuristic framework that enables designers, developers, and auditors to systematically evaluate data visualizations and interfaces for a wide range of accessibility barriers, considering people with visual, motor, vestibular, neurological, and cognitive disabilities. In this project, we did the hard work for other practitioners and contributed our collection of synthesized accessibility resources in a single workbook. We had practitioners try out our resource in real environments, in-situ, in order to learn more about the challenges and barriers they themselves faced in evaluation work. With *Chartability*, practitioners, especially those with limited accessibility expertise, gained more confidence and clarity in assessing and improving their work. Additionally, *Chartability* has since become widely applied as a framework that isn't just used for evaluation but also as design guidance in many contexts, internationally, including policy organizations, governmental groups, and more than 100 companies and businesses.

Chartability then opened up a significant landscape of new projects and research directions. From a combination of my existing expertise as a visualization designer and engineer, in addition to our continued application of *Chartability* in the wild, we began to identify the trickiest and most-difficult domains of work for practitioners. *Chartability* has 50 total heuristics, or tests, each organized under one of seven principles. These three domains did not simply “emerge” on their own; they were evidenced by our applied work. Alongside the research activities that comprise this thesis, in a freelancing role we applied *Chartability* by partnering with industry, government, and non-profit organizations, conducting over 150,000 tests (both manual and automated) and working closely with dozens of designers and engineers. From that body of work, three larger domains stood out as the areas where the most severe and dramatic accessibility barriers remained unaddressed: data *navigation*, analytical *interaction*, and interface *personalization*.

So the next section of this thesis engages the first of these three: **navigation**. Navigation is a fundamental type of interaction that is leveraged by modern software-based assistive technologies. Screen readers, the primary tool used by people who are blind to interact with computers, navigate content. Additionally, many other assistive technologies, such as a sip and puff device (like the “POSSUM” from as far back as ’63 [121]) also navigate. Navigational technologies are leveraged by people with a significant array of disabilities, yet tend to be entirely ignored by existing data visualization tools, which are pre-dominantly built to support direct input (using a computer mouse).

Empirical work has already demonstrated that structural navigation is actually good [213], even when regular alternative text (image descriptions) exist. This is both because people who are blind can gain both a high level understanding (from the description) as well as lower-level sense of the data's structure and arrangement, in addition to the fact that discrete, structural navigation exposes interactivity that may exist on any visualization elements (such as they can be hovered or clicked with a mouse in order to perform some action). So, if good empirical work exists: *why*

haven't practitioners put this research into action? What makes this work hard to do?

We first built *Data Navigator* to provide the building blocks we needed in order to address the technical and conceptual gaps that were required to make any visualization or visualization tool provide a navigable, interactive structure. *Data Navigator* is a low-level toolkit which can be used to construct accessible navigation structures such as lists, trees, and diagrams from an underlying graph structure. We leveraged graph theory for an applied HCI problem: nodes and edges represent any relationships within the data as a structure, which then supports rich expressiveness of data navigation experiences. Users can navigate discrete marks in a visualization, clusters, groupings, and more. In addition to its structural scaffolding, *Data Navigator* also supports a wide array of input modalities leveraged by people with disabilities (screen readers, keyboards, speech, and gestures).

Data Navigator provided a substrate, but this contribution alone wasn't enough to engage the question *why is navigation so hard, in practice?*. So the chapter following *Data Navigator* introduces *Skeleton*, a data navigation authoring tool built on top of *Data Navigator*. Our novel approach in *Skeleton* involves visualizing and making manipulable the nodes, edges, and textual data that comprise non-visual end user experiences. *Skeleton* visualizes the building blocks that comprise *Data Navigator*. Additionally, *Skeleton* provides expressive, rapid scaffolding capabilities that leverage data visualization rendering engines. This scaffolding engine helps practitioners quickly create common configurations for non-visual data navigation structures that retain visual congruence to the underlying structure.

But most importantly, *Skeleton* serves as a framework that shapes designerly consideration. Our conjecture was that because sighted practitioners cannot *see* navigation building blocks, they will not treat those elements as iterable design materials. We conjectured: Navigation is hard in practice because sighted designers face barriers to iteration and understanding. We conducted an empirical study with sighted practitioners and found that making non-visual elements visual helped practitioners shift from treating accessibility as a compliance task to treating it as a design problem, re-iterating on the visual aspects of their design, and engaging in the complex and nuanced components that comprise data navigation experiences.

Now, **interaction** becomes the next area we want to engage. Existing accessible data interaction for people who are blind, including our previous work on navigation (which is a form of interaction), predominantly seeks to expose information. This is what we call *access-oriented interaction*. In terms of low-level analytical tasks, most are then made feasible through navigation, sonification, or summarization-based and question-answering approaches: retrieving values, filtering, computing derived values, sorting, determining ranges, clustering, and finding outliers. What remains are analytical tasks that, despite being “low level” (understood as *unable to be reduced into more fundamental tasks*), are cognitively highly complex: finding correlations and characterizing distributions [5]. These tasks require complex hypothesization and exploration, rather than a system that simply encourages surfacing what is known or what is already present in the data: it requires combining, remixing, restructuring, and dividing data.

But blind *analytical interaction* isn't just important to engage because it is understudied, it is important to engage because many of interactive information visualization's most impactful tools for data science enable it [55, 82, 134, 194]. In visualization, *cross-filtering* is one example of an interaction that enables a user to filter one visual space while seeing a coordinated change in another visual space simultaneously and near-instantaneously. The speed of input interaction and

perception of output also matters: even a small bit of latency changes the quality of a user’s data exploration activities [114]. We conjectured that a screen reader, the most-used tool leveraged by blind people when interacting with computers, may be insufficient for engaging this task.

To engage this, this thesis introduces *Cross-perception*, an approach for building analytical interactions that support perception in one space of input interaction with simultaneous, non-competing perception of output in another space of data representation. We first formalized a design framework for producing *cross-perception* experiences and then built a novel prototype device, the *cross-feelter*, that enables blind *cross-perception* of a cross-filtering data exploration interface. In an empirical study with blind users (with and without existing data expertise), we found *cross-perception* speeds up analytical exploration by 90% and helps blind users consider vastly more questions of their dataset (+188% computational queries, +54% spoken aloud) compared to a screen reader-driven interaction.

Beyond performance, we found that our input modality itself shaped the character of analytical engagement: participants didn’t just work faster, they considered more dimensions of their data and asked qualitatively different questions. The *cross-feelter* also reduced anxiety and substantially increased enjoyment, particularly for participants without prior data expertise, suggesting that the barriers blind practitioners face in data work are not only functional but affective. Additionally, we had our blind practitioners imagine new interaction possibilities that *cross-perception* could enable including and beyond our *cross-feelter* device.

Our final domain of work is the most difficult for visualization practitioners to engage: **personalization**. While *navigation* demands better software tooling and visual support and *interaction* requires new hardware, *personalization* completely re-orientes how software authoring takes place. Personalization matters because of *access friction*, which is a design challenge where one design or interface configuration that might be accessible for one person or group of people turns out to create barriers for someone else [74, 92]. In existing work on personalization and accessibility, studies have demonstrated that end-user control is great to have and can alleviate friction [101, 208], but little work has been done to explore what personalization looks like for an existing data visualization library and how practitioners should build and maintain their existing systems to support it.

Our final chapter introduces *Software* to address the tension between standardized accessible design and the diverse needs of real users with disabilities. In the wild, *access friction* exists in every design that reaches a public audience; it is inevitable. This tension ultimately means that some users have a worse experience, and may even face exclusion, with any particular design configuration. So for this work, we conducted our research in-situ with visualization software engineers and designers and worked to build a scalable, flexible software system dubbed “softerware” that enables end users to manipulate the appearance and functionality of the charts and graphs they encounter according to their own preferences. We conducted empirical research to inform our collaborators, as well as other visualization system authors, with guidelines and considerations for building *softerware* systems. In our study, no two participants chose the same preference configuration and participants with the same diagnosed condition sometimes needed opposite design treatments. Practitioners, meanwhile, immediately raised ethical concerns about whether personalization would let designers off the hook for poor defaults. These findings revealed that the real barriers to personalization are not at the level of any individual chart but at the level of system infrastructure: without persistence, cross-system interoperability, and shared

standards, the effort required of end users exceeds the value they receive.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part III

Navigation: Making Data Structures Traversable

Chapter 6

Skeleton: Visual Authoring of Non-visual Data Experiences

6.1 Preface

When sighted practitioners author accessible data visualizations, they build navigation structures (the nodes, edges, and input bindings that govern how assistive technologies traverse an interface) entirely in code, with no visual representation. This invisibility makes navigation structures difficult to inspect, debug, and iterate on. To sighted practitioners, every other aspect of a visualization is iterated on because it is visible; navigation structure ships as a first draft, if at all, because it is not. Without a representation to react to, practitioners cannot develop judgment about what makes navigation good or bad, and the quality ceiling of non-visual experiences is set by the absence of a feedback loop. We address this problem through longitudinal co-design with practitioners across cartography, design systems, and open-source visualization, and make three contributions.

First, we introduce technical advancements for making the properties of accessible navigation structure visible and directly manipulable during authoring, grounded in two foundational pieces of infrastructure produced by our co-design work: an *Inspector* that renders navigation graphs as interactive node-link diagrams, and a *Dimensions API* that expresses navigation in terms of data dimensions rather than explicit graph construction.

Second, building on these, we present *Skeleton*, a direct-manipulation authoring environment in which the properties of an accessible navigation structure are translated into visual representations authors can observe and manipulate. Key techniques include a dual-view editor that simultaneously shows the system’s navigation model and the end user’s spatial experience, a scaffolding engine that automates spatial node placement by re-purposing a visualization rendering pipeline, a live label-template editor with real-time screen-reader-output preview, and a testing mode that makes traversal sequence visually trackable.

Third, we evaluate *Skeleton* through an in-situ study with 8 practitioners across visualization design, engineering, and research. Making navigation structure visible changed how practitioners engaged with accessible design: they reconsidered the architecture of their own visualizations, attended to a broader range of input modalities, and shifted from treating accessibility as a compliance task to treating it as a design problem.

6.1.1 Context

Data Navigator was successful enough to help us acquire practitioner-centered funding to apply it in real-world contexts. We worked with three different communities as a researcher-designer-engineer alongside the teams attempting to apply *Data Navigator* to their work. These projects, while ultimately responsible for *Skeleton* (spoiler alert), also quickly motivated a new hypothesis: that making data navigation experiences isn’t just *hard* for some generic reasons but that it may

be hard because it involves perceptual accessibility barriers for sighted practitioners (or in other words: we conjectured that sighted practitioners face barriers when working with non-visual design materials).

Data Navigator evolved and grew and found a place in real-world work after its release. However, *Data Navigator* eventually also became a vehicle to open up new research directions. Notably, we made improvements to the system from a technical standpoint and also identified new barriers that practitioners faced when applying *Data Navigator*, both of which we discuss further in this chapter.

The following work was proposed as the culmination of this thesis, and chronologically was the final work we completed for this body of work.

This chapter has also, notably, been modified from the submitted version to IEEE VIS, for the sake of this thesis’ overall narrative cohesion as well as the fact that this thesis can afford more space to expand in several areas beyond the page limits that VIS allows.

It is worth stating plainly, before the chapter proceeds, how we intend each artifact to be read. The two pieces of infrastructure described in later subsections, the *Inspector* and the *Dimensions API*, are engineering contributions in the ordinary sense: they are released, reusable, and already used in production-facing collaborations beyond this research. The integrated authoring environment we call *Skeleton* is something different. We built and studied it as a design probe [73, 96], an instrument whose purpose is to make a specific thing observable rather than to ship a finished workflow. What the probe is designed to make visible, up front, is twofold: first, the navigation structure itself, rendered as a manipulable visual object so that sighted practitioners can perceive and react to it; and second, the practitioners’ own relationship to that structure, what they notice, question, and revise once it stops being invisible. The first is a property of the artifact; the second is what the study is built to surface.

This framing has consequences for what we claim and how we evaluate. A production authoring tool would be judged on throughput, robustness, export fidelity, and task completion time. A probe is judged on what it reveals. Accordingly, our study (Section 6.4 onward) reports qualitative shifts in how practitioners engage with accessible design, not task performance against a baseline, and the prototype deliberately leaves production concerns such as deployable export out of scope (see the Limitations and Future Work section). Treating *Skeleton* as a probe is also why we study it first with sighted practitioners alone: the probe is positioned to test whether making structure visible changes sighted authors’ reasoning, which is a precondition for, not a substitute for, evaluation with the disabled users those structures ultimately serve. We return to the boundary between what the probe makes *designable* and what only end-user evaluation can establish as *good* in the Limitations and Future Work section.

This chapter was adapted from our paper, currently conditionally accepted with IEEE VIS:

F. Elavsky, C. Nnadozie, L. Nadolskis, P. Carrington, and D. Moritz, ‘*Skeleton: Visual Authoring of Non-visual Data Experiences*’, *IEEE Transactions on Visualization and Computer Graphics*, 2026.

6.2 Overview

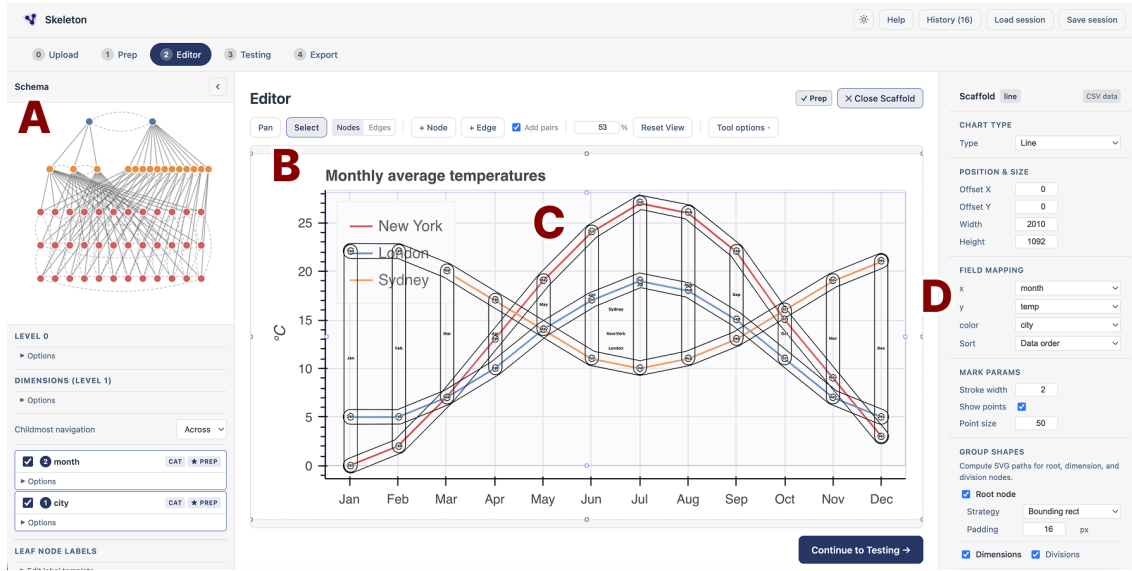


Figure 6.1: *Skeleton* being used to build a navigation structure over a line chart. **A**: a dual tree visualization of nodes and edges built for a line chart with 3 lines, spanning over 12 months. **B**: a direct-manipulation canvas that resides over a static png image of a line chart. **C**: a skeleton-like structure of nodes, edges, and group outlines that correspond to the schema shown in (A), overlaid on the visual space of the chart in (B). **D**: controls for rapidly scaffolding the nodes, edges, and their shapes shown in (C), using Vega’s visualization engine for automatically generating spatial properties.

We start this work with a provocation: How might the discipline of visualization help the discipline of accessibility?

Visualization has spent decades developing techniques for a specific class of problem: representing and interacting with information visually. Grammars of visual encoding [128, 154, 196, 199, 214], direct-interaction interfaces [46, 95, 169], and iterative feedback loops between representation and understanding [53, 81, 114] are all methods that enable abstraction and manipulation of the information that underlies the visual representation. We argue these methods have a direct application within the discipline’s own accessibility challenges.

Sighted practitioners who build accessible data visualizations face an unusual authoring problem. The non-visual navigation structures they construct (the nodes, edges, focus states, input bindings, and semantics that govern how assistive technologies traverse a chart) exist only as code. A practitioner can write a navigation hierarchy, but cannot see it, click on a node to inspect what will be announced, observe the spatial relationship between a navigation path and the chart it overlays, and then manipulate its properties through direct interaction. Every other aspect of a visualization has a visible, inspectable representation during authoring: the visual encodings are visible, the layout is visible, the interaction states are visible. And yet, navigation structure is not.

This invisibility has practical consequences. Without a way to see what they are building, sighted practitioners cannot easily catch structural errors, compare design alternatives, and iterate. Accessibility becomes downstream of every other design choice, not because practitioners choose to deprioritize it, but because the authoring conditions do not support anything else [102, 165]. The floor and ceiling of non-visual data experiences are constrained by what sighted authors can perceive of their own work.

We engage this space with the following research questions:

R1 (Qualitative, Exploratory): What challenges do sighted practitioners face when designing and engineering navigation structures for accessible visualizations?

R2 (Qualitative, Exploratory): How do sighted authors reason about the non-visual experiences that accompany their visualizations?

R3 (System, Design): How can we make the properties of accessible navigation structure visible and directly manipulable during authoring?

R4 (Qualitative): In what ways does a directly manipulable visual representation of navigation structure change how practitioners find errors and improve upon their designs?

We address these questions through longitudinal co-design with practitioners across cartography, design systems, and open-source visualization, following an action research orientation [80] in which the research team was embedded in each community’s active development work. This paper makes three contributions:

First, **we introduce technical advancements** for making the properties of accessible navigation structure visible and directly manipulable during authoring. These techniques are grounded in our co-design collaborations, which produced two foundational pieces of infrastructure: an *Inspector* that renders any navigation graph as an interactive node-link diagram, and a *Dimensions API* that formalizes a declarative grammar for expressing navigation in terms of data dimensions rather than explicit graph construction (**R1, R2, R3**).

Second, building on this infrastructure, **we present *Skeleton*, a direct-manipulation authoring environment** in which the topology, spatial mapping, semantics, and input logic of an accessible navigation structure are made visible and directly manipulable. *Skeleton* is built on Data Navigator [45], a code-based library for constructing interactive data navigation structures (**R3**). Our intention with *Skeleton* is to continue to develop it towards a usable, practical system beyond the scope of this research.

Third, **we contribute findings from an in-situ interview study** with 8 practitioners across visualization design, engineering, and research. We evaluate *Skeleton* as a design probe [73, 96] rather than a deployable system, focusing on qualitative shifts in practitioner engagement rather

than task performance. We find that making navigation structure visible shifted how participants engaged with accessible design: they reconsidered the architecture of their own visualizations, attended to a broader range of input modalities, and shifted from treating accessibility as a compliance task to treating it as a design problem (R1, R2, R4).

6.3 Related Work

6.3.1 Visualizing Non-visual Structure

Long before accessibility, computing established a practice of taking structures that are not inherently visual and giving them a visual form that can be seen and acted upon directly. Sutherland’s Sketchpad turned the constraints and topology of a drawing into a manipulable graphical object rather than coordinates in memory [180]; visual programming environments rendered computation as arrangements of on-screen objects [172], syntax-directed editors treated a program as a hierarchical structure edited through its visible derivation tree [183], and structured flowcharts gave control flow a fixed spatial form [137]. Myers later named the distinction this lineage had been drawing, separating *visual programming* from *program visualization*, depicting otherwise non-visual programs graphically [136]. A consequence is easy to overlook: many structures we now treat as inherently visual, including trees, graphs, flowcharts, and the document object model, are abstractions whose visual rendering has become so dominant that the representation now stands in for the structure.

Direct manipulation interfaces, in turn, provide a continuous representation of the objects of interest and give rapid, incremental feedback, letting people work with a structure they would otherwise hold in their heads or reach only through code [95, 168]; grammars of interactive graphics extend the same move to authoring [19, 154]. Yet across this tradition, one structure has remained conspicuously without a visual form to manipulate: the UI structure that a screen-reader traverses. This lack of visual representation is especially surprising because screen readers evolved in parallel with the emergence of graphical interfaces [185].

6.3.2 Non-visual Data Experiences

Blind people who rely on assistive technologies interact with data in fundamentally different ways than sighted users who use a direct pointer, like a mouse [110, 125, 163]. A substantial body of research has documented what these experiences look like across modalities, and what it takes to make them meaningful. In the context of “data experiences,” this paper focuses specifically on interactive navigation structures, but we briefly survey adjacent modalities to situate our contribution.

Alternative text and natural language. A dominant strand of this work concerns the generation and evaluation of textual descriptions of visualizations [103, 107, 111, 116]. More recent LLM-driven systems and Q/A approaches can caption charts with varying degrees of semantic depth, some at a risk of producing bias [43, 51, 109]. While alt text makes a visualization’s message available without sight, it is by nature static: a description conveys what a visualization says, but not how a user might explore or interact with it.

Sonification, haptics, and tactile rendering. Non-visual data experiences extend well beyond text. Sonification encodes data as sound [23, 84, 123], with declarative grammars emerging for authoring these experiences [108]. Haptic and tactile representations offer another channel through refreshable displays, 3D-printed graphics, and multimodal touchscreen interactions [27, 89, 126, 146]. Recent systems integrate multiple non-visual representations of the same data [30, 90, 160].

Interactive navigation structures. The primary focus of our work centers on the state of research related to structured navigation: the traversal of data points, groupings, and interface elements through assistive technologies and keyboard input [45, 173, 192, 213]. Existing systems and interfaces have demonstrated that going beyond static descriptions to support hierarchical, traversable data structures meaningfully improves how blind users can explore and reason about charts [186, 213]. Giving users control over the textual tokens surfaced during navigation improves comprehension and agency [101]. And more recent work has found that perceptually congruent navigation structures for charts and diagrams can improve goal-driven exploration [130].

6.3.3 Authoring Non-visual Data Experiences

Accessible visualization has historically centered the experiences of disabled users, but a parallel body of work examines the experiences of visualization practitioners working on accessibility.

Practitioner challenges. Research consistently finds that sighted visualization practitioners struggle with accessibility [50, 102, 165]. Most visualizations in the wild are inaccessible and designers themselves report lacking guidance, especially for complex and interactive graphics [102, 165]. And screen reader users experience the downstream effects of these gaps: inconsistent structure, poor keyboard support, and information that is present visually but absent in the accessibility tree [50, 163]. The pattern across this work is consistent: the practitioners who build visualizations lack the tools and feedback mechanisms to make non-visual experiences effective, useful, and good.

Across practitioner-centered literature, a recurring finding is that non-visual experiences are treated as downstream of visual design choices, added after the visual representation is finished rather than designed in parallel [102, 117, 165, 215]. This sequencing has consequences: what is navigable and how it is structured is constrained by whatever visual decisions came first.

Authoring-oriented systems. A distinct line of work has focused on building authoring tools and libraries that give practitioners more tractable paths to accessible output. Few visualization tools support the kinds of interactive navigation structures that assistive technology users most benefit from [111]. Of those that do, most rely on code-based approaches [17, 45, 162, 164]. *Umwelt* [215] takes a different and notable approach: it is a structured editing environment where authors specify representations *across* modalities (sonification and visualization) in an integrated interface, where navigation is made available over the visualization using *Olli* [17]. The latest work in this space is *Arboretum*, a tool that provides automatic conversion of diagrams to a tabular structure, navigable structure, and tactile representation [204]. In *Arboretum*, the input visual is treated as the ground truth for the navigation structure, which is treated as downstream output.

Communicating visually, authoring invisibly. There is a revealing irony across this body of related work: research about navigation structures almost invariably communicates those

structures visually. Papers such as “rich screen reader experiences” [213], *ChartReader* [186], *Benthic* [130], and *Data Navigator* [45] each use visual node-link diagrams and hierarchical schematics to explain navigation paths to their readers. The same pattern holds in adjacent domains that involve structuring data for navigation, such as PDF remediation [135].

Yet, most other non-visual modalities are not only communicated but *authored* in visual and non-visual modalities. Sonification is designed in editors that are both visual and auditory [108]. Tactile graphics are composed by transcribers on a graphical canvas before being physically embossed or printed [18].

Inspecting accessibility structure. In authoring-oriented systems, none provide a visual interface through which practitioners can interactively inspect and manipulate navigation structures as a first-class design material. Structure output is either downstream of code or static visuals. Structure is always *derived* or *specified*; indirectly manipulated. Additionally, verification of the structure across all of these systems requires developers to manually navigate using a screen reader after the structure has been authored and rendered, before returning to the upstream visual design space or code. Navigation structure is thus understood and communicated visually by sighted researchers and designers, yet built entirely without visual feedback by developers; this is the gap we address.

6.4 Co-design Foundations

Our research follows an *action research* orientation [80], in which knowledge is generated by engaging with a community to solve a real problem in-situ, alongside them rather than studying it from the outside. These collaborations started from a shared motivation: practitioners needed to make their existing systems accessible to navigational assistive technologies, using *Data Navigator* [45] as a foundation. Across three projects, we worked with 12 individuals outside our research team. Three blind co-designers shaped the work throughout: CD1 and CD2 (anonymous) and a co-author on this paper, [REDACTED]. CD1’s contributions are noted in [Section 6.4.1](#), CD2’s are noted in [Section 6.5.3](#). [REDACTED] contributed to early ideation, problem formation, and framing for the project as a whole, helping define the authoring challenges that *Skeleton* addresses, in addition to feedback on study design.

The co-design literature on accessibility has centered people with disabilities as primary design partners [35, 36, 143, 186, 212, 213], and rightly so. Our co-design takes sighted practitioners as primary partners because the authoring challenges we address happen on the sighted side of the process: it is sighted authors who cannot see what they are building.

Two themes converged across all three engagements, and we present them here before describing the individual projects that surfaced them. First, **practitioners communicated and reasoned about accessible navigation using visual representations**: while designing for language, sound, and structure, our collaborators drew on paper, built wireframes of nodes and edges, and reflected on the design space using visual artifacts. Even when collaborating with blind co-designers, a visual medium was the first language of sighted authors. Second, **development that followed visual design work faced severe iteration barriers**: verifying a navigation experience required building a working code prototype and manually navigating it with a screen reader. The gap between visual design and code-based scaffolding with manual testing produced

repeated mistakes, misinterpretations, and abandoned prototypes.

These themes did not arrive as abstractions; each project below surfaced them as concrete friction, and together the three engagements produced five design goals that *Skeleton* was built to satisfy. We state them here and, in the subsections that follow, mark where each emerged:

DG1 Make structure visible and inspectable. Render the navigation structure as a manipulable object during authoring, not only as code, so that practitioners can perceive and react to it.

DG2 Specify navigation at the level of data. Provide a higher-level, data-driven abstraction so practitioners describe traversal in terms of data dimensions rather than hand-wiring every node and edge.

DG3 Shorten the iteration loop. Let practitioners verify a design without a full code build or a manual screen-reader pass.

DG4 Support image-only and bespoke visualizations. Do not require a single underlying dataset, format, or recognizable chart type, since the cases that most need accessible tooling often have neither.

DG5 Reach beyond keyboard input. Accommodate input modalities other than assuming keyboard navigation, including touch and text.

6.4.1 Geologic Map

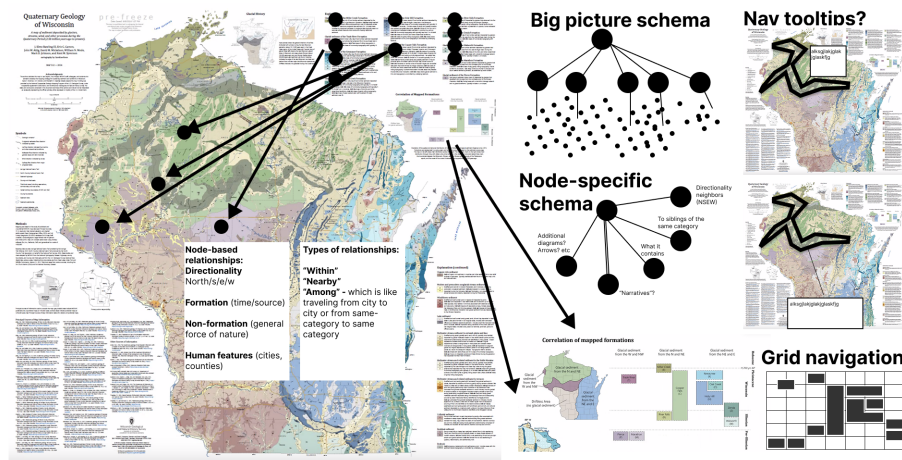


Figure 6.2: Our visual design work in Figma over a static geologic infographic map of Wisconsin. We use visual forms and illustrations over and beside the map to communicate flows, structure, navigation styles, and interaction patterns.

Our longest engagement spanned just over two years, with a cartographer, accessibility consultant, and blind co-designer (CD1) building an interactive version of a quaternary geologic map of Wisconsin [144]. The map combined dozens of irregular geographic regions organized categorically by a legend.

Early design sessions took place in Figma (Figure 6.2), where we laid out node-link diagrams of how a screen reader user might traverse the map, legend, and peripheral information. We

annotated each node with text a screen reader would announce and connected them to relevant regions. Drawing the structure made it possible to discuss design choices, catch dead ends, and debate traversal strategies. We were informally doing what *Skeleton* later formalized.

The friction began when we moved from design to implementation. We had no way to verify that our designed structure would be good or ideal, and scaffolding the project into Data Navigator was arduous: the translation from even simple navigational designs to functional code was too complex to hand off or meaningfully iterate on. The collaboration also surfaced a design-space boundary: for within-map spatial navigation, an egocentric audio-game approach [15] fit better than a node-link graph, a distinction CD1 helped surface. Reaching this boundary was only possible because we could see the structure we were trying to build (DG1). And the map’s lack of a single tabular dataset made clear that a useful navigation tool should not assume structured data or a grammar of graphics (DG4).

6.4.2 Design System Library

Our second engagement, spanning 7 months, was with 6 engineers and designers on Adobe’s React Spectrum Charts library, an open-source chart component system. We worked in their codebase and in Miro on navigation design for bar charts, clustered and stacked bars, line charts, and related types. Because we needed generalized, reusable patterns, our design problems differed from the geologic map: we needed to account for use, re-use, and edge cases across chart types.

Our Miro sessions produced two kinds of artifacts: diagrams of navigation structure for specific chart *instances* and *schema* diagrams capturing generalized patterns in dimensional terms. The concept of a “dimension” emerged naturally: common transformations on Data Navigator’s graph structure corresponded to properties within a dataset, such as a “categorical” dimension or a “numerical” dimension, where traversal took place within collections of grouped siblings. This dimensional thinking directly foreshadowed the Dimensions API (Section 6.4.4).

The dominant friction was iteration speed. We could sketch and converge on a schema in Miro in an hour, but verifying the design in a functional example required embedding Data Navigator into a large codebase, implementing changes, rebuilding, and manually testing with a screen reader. The collaboration also surfaced a limitation: mobile screen reader navigation uses swipe gestures rather than keyboard input, and our keyboard interaction model did not account for it. Each of these frictions pointed somewhere: the dimension concept toward a higher-level abstraction (DG2), the cost of verifying each change toward a faster loop that does not require a full build or screen reader (DG3), and the mobile-gesture gap beyond keyboard input (DG5).

6.4.3 Open Source Visualization Library

Our third collaboration, spanning nearly two years, was with Quansight Labs and contributors to Bokeh, a Python-based open-source visualization library. We performed an accessibility audit [48] based on Chartability [44] and identified that Bokeh visualizations with interactive chart elements needed to be navigable by assistive technologies.

Unlike Adobe, Bokeh has no native chart types. Its API operates at the level of glyphs, renderers, and data sources, which users assemble freely. There was no standard unit around

which to anchor a navigation pattern, and any grammar or tooling would need to accommodate an open-ended range of encoding combinations. The iteration gap from Adobe was present in a more severe form: making library-wide contributions to a fully open-source project required incremental tooling for the most common cases. For Bokeh, we needed to test functional, data-driven navigation abstractions without fully embedding Data Navigator into the library. A library with no native chart types required the abstraction to be data-driven and chart-type-agnostic rather than tied to fixed templates, sharpening **DG2** into a concrete constraint.

6.4.4 Infrastructure from Practice

The three collaborations converged on the design goals above. Two of them demanded new infrastructure before *Skeleton* could exist: a way to visually render and inspect navigation structures (**DG1**), and a higher-level, data-driven abstraction for specifying navigation without hand-wiring every node and edge (**DG2**). We built two pieces of infrastructure to address these, described next; the remaining goals (**DG3**, **DG4**, **DG5**) are met by *Skeleton*'s authoring environment (Section 6.5). These infrastructural improvements as well as *Skeleton* are open source on [our GitHub repo](#).

6.4.4.1 An Inspector Gadget

We built an *Inspector* (`@data-navigator/inspector`) to render any Data Navigator structure as an interactive node-link graph using D3, with an accompanying console for debugging. Hierarchical structures are colored by level; edges are drawn as directed links; the entry point is visually marked. The *Inspector*'s graph can itself be navigated using Data Navigator, with visual focus tracking during navigation, allowing practitioners to manually verify structure and reachability.

This made structural verification immediate: a practitioner could generate a structure and check at a glance whether the hierarchy had the right levels, whether circular extents produced expected wrap-around edges, or whether a particular path was reachable. The interactive console logs API information and underlying data when nodes are activated, and hovering or focusing logged information highlights the corresponding node in the graph.

The *Inspector* remains a developer tool, however. It requires code familiarity to attach and renders structure as an abstract graph with no connection to the spatial layout of the underlying visualization. It shows topology but not instantiated geometry: a practitioner can see that two nodes are connected but not where their focus indicators will appear on-screen. This gap between navigable structure and its spatial instantiation over a rendered chart motivated *Skeleton*.

6.4.4.2 Alternative Dimensions

The original Data Navigator library requires explicit graph construction: practitioners specify nodes, edges, and navigation rules by hand. This is general but scales poorly. Our *Dimensions API* addresses this.

Our new vocabulary mirrors how practitioners already think about their data, much as a grammar of graphics lets a designer write a visualization at the level of data fields rather than primitive

marks. Rather than specifying a graph directly, a practitioner describes the *dimensions* of their data, the meaningful axes along which a user might want to navigate, and a single build step constructs the full node-link structure automatically. Notably, this enables systems to programmatically construct navigable structures as long as dimensions are configurable. Each dimension declares a type (categorical or numerical) and a `dimensionKey` naming the field it traverses, together with a `behavior` block that governs traversal (how navigation behaves at group boundaries, and whether leaf nodes are reachable laterally across divisions) and an `operations` block that governs how the data is shaped into the hierarchy (sorting, filtering, and binning numerical ranges). [Section 6.4.4.3](#) gives a deeper look into the API’s parameters.

This data-dimensions framing both aligns with and departs from *Olli* [17], which similarly lets authors express a navigable structure over a chart from a higher-level specification rather than hand-built nodes. Both share the premise that fields, not graph primitives, are the right authoring unit. The difference is what the specification produces and where it lives: *Olli* compiles a chart specification into a traversable tree that an author does not subsequently manipulate as a graph, whereas the *Dimensions API* generates an explicit Data Navigator node-link structure whose every node, edge, and rule remains addressable, so that the declarative output is itself the object a practitioner can then inspect in the *Inspector* and edit directly in *Skeleton*.

6.4.4.3 Dimensions API Reference

Each dimension in the *Dimensions API* ([Section 6.4.4](#)) declares a type (categorical or numerical, inferred from the data when omitted) and a `dimensionKey` naming the data field it traverses. Two further groups of properties govern behavior. A `behavior` block controls traversal: `extends` sets boundary behavior (`terminal` stops at the edges of a group, `circular` wraps around, and `bridgedCousins` carries focus across to the nearest element of an adjacent group rather than stopping or wrapping), and `childmostNavigation` determines whether leaf-level nodes are reachable laterally across a dimension’s divisions (`across`) or only within a single division before returning to a parent (`within`, the default). An `operations` block controls how the data is shaped into the hierarchy: a `sortFunction` orders divisions and leaves, a `filterFunction` selects which data participate, and `createNumericalSubdivisions` bins a numerical dimension into a chosen number of ranges.

The generated structure is a multi-level hierarchy: each dimension produces a root node, below which division nodes group the data, below which leaf nodes represent individual data points. Multiple dimensions over the same dataset share leaf nodes, so users navigating via different dimensions reach the same data through different paths.

To make the input and output concrete, the examples below use this small four-row dataset of average temperatures, with an explicit `id` on each row:

```
const data = [
  { id: '_0', month: 'Jan', city: 'NYC', temp: 0 },
  { id: '_1', month: 'Jul', city: 'NYC', temp: 25 },
  { id: '_2', month: 'Jan', city: 'Sydney', temp: 22 },
  { id: '_3', month: 'Jul', city: 'Sydney', temp: 8 }
];
```

With the Dimensions API, bar chart navigation that would otherwise require constructing every node and edge by hand is expressed as a single dimension declaration:

```
dimensions: {
  values: [
    {
      dimensionKey: 'month',
      type: 'categorical',
      behavior: { extents: 'circular' }
    }
  ]
}
```

From this declaration, the build step does three things automatically. It assembles the nodes and the parent-child and sibling edges that connect them, including, when multiple dimensions are declared, the cross-dimension edges that let a user move between hierarchies over shared leaves. It applies the boundary and childmost rules to decide what happens at each edge of the structure. And it assigns keyboard navigation rules to each dimension, drawing from a pool of directional key pairs; because that pool is finite, the API recommends keeping a structure to six dimensions or fewer and asks for explicit navigation rules beyond that point.

This bounds the structures the abstraction expresses by default: it generates the regular, dimensionally-organized hierarchies that recur across common chart types, and structures that do not decompose into a small set of data dimensions (or that exceed the directional-key budget) fall back to explicit graph construction or manual editing in the editor.

The abstraction is otherwise chart-type-agnostic: bar charts, scatter plots, line charts, and layered charts all use the same vocabulary, with different combinations producing different navigation topologies. This directly addressed Bokeh's problem, where no native chart types exist to anchor a pattern: the API mirrors data fields and encoding choices as a set of dimensions, so a thin wrapper can map an arbitrary glyph assembly onto a handful of recurring dimensional shapes ([Section 6.4.3](#)).

The vocabulary composes to richer topologies simply by adding dimensions. Declaring two categorical dimensions over the same dataset, for instance a temperature series recorded by month and by city, produces a dual hierarchy in which left and right traverse one axis and up and down traverse the other, with both sharing the same leaf nodes; here the month dimension wraps at its boundaries while the city dimension stops:

```
dimensions: {
  values: [
    {
      dimensionKey: 'month',
      type: 'categorical',
      behavior: {
        extents: 'circular',
        childmostNavigation: 'across'
      }
    },
  ],
}
```

```

    {
      dimensionKey: 'city',
      type: 'categorical',
      behavior: {
        extents: 'terminal',
        childmostNavigation: 'across'
      }
    }
  ]
}

```

This two-dimension declaration assigns a keyboard control to each axis. [Table 6.1](#) lists the controls Data Navigator generates for it; the arrow keys move between siblings along each dimension, Enter drills toward the data, and each dimension receives its own drill-out key from the default key pool (here W and J).

Table 6.1: Keyboard controls generated for the two-dimension (month, city) structure over the sample dataset. Movement along month wraps (circular); movement along city stops at the ends (terminal).

Key	Action
← / →	Move to previous / next month (wraps Dec–Jan)
↑ / ↓	Move to previous / next city (stops at ends)
Enter	Drill in (dimension → division → data)
W	Drill out along the month grouping
J	Drill out along the city grouping
Esc	Exit the structure

Finally, to show what the build step emits, the two-dimension declaration above, applied to the four-row dataset, expands into the nodes and edges below. Each dimension produces a dimension node carrying its `dimensionKey`, a set of division nodes (one per distinct value, each tagged with the field it was derived from), and the shared leaf nodes that carry the original rows. A node’s edges list the edges it participates in; each edge names a source, a target, and the `navigationRules` that traverse it. Crucially, a single edge is bidirectional: the left/right pair on one edge means pressing left moves toward its source and right toward its target, so the month and city axes at the leaf level each collapse to one edge with two rules (abbreviated here for space):

```

// nodes (excerpt: one division + one leaf per axis)
{
  month: {
    id: 'month',
    data: { dimensionKey: 'month' },
    edges: ['month-Jan', 'any-exit']
  },

```

```

city: {
  id: 'city',
  data: { dimensionKey: 'city' },
  edges: ['city-NYC', 'any-exit']
},
'Jan': {
  id: 'Jan',
  derivedNode: 'month',
  data: { month: 'Jan' },
  edges: [
    'Jan-Jul',
    'month-Jan',
    'Jan-_0'
  ]
},
'NYC': {
  id: 'NYC',
  derivedNode: 'city',
  data: { city: 'NYC' },
  edges: [
    'NYC-Sydney',
    'city-NYC',
    'NYC-_0'
  ]
},
_0: {
  id: '_0',
  data: { month: 'Jan', city: 'NYC', temp: 0 },
  edges: [
    'leaf-_0-_1',      // month axis (l/r)
    'leaf-_0-_2',      // city axis  (u/d)
    'Jan-_0', 'NYC-_0',
    'any-exit'
  ]
}
// Jul, Sydney, and _1/_2/_3
// follow the same pattern
}

// edges
{
  'month-Jan': {
    source: 'month',
    target: 'Jan',

```

```

    navigationRules: ['drill-out_month', 'drill-in']
  },
  'city-NYC': {
    source: 'city',
    target: 'NYC',
    navigationRules: ['drill-out_city', 'drill-in']
  },
  'Jan-Jul': {
    source: 'Jan',
    target: 'Jul',
    navigationRules: ['left', 'right']
  },
  'NYC-Sydney': {
    source: 'NYC',
    target: 'Sydney',
    navigationRules: ['up', 'down'] },
  'leaf-_0-_1': {
    source: '_0',
    target: '_1',
    navigationRules: ['left', 'right']
  },
  'leaf-_0-_2': {
    source: '_0',
    target: '_2',
    navigationRules: ['up', 'down']
  },
  // ...etc
}

// navigationRules
{
  left: { key: 'ArrowLeft', direction: 'source' },
  right: { key: 'ArrowRight', direction: 'target' },
  up: { key: 'ArrowUp', direction: 'source' },
  down: { key: 'ArrowDown', direction: 'target' },
  'drill-in': { key: 'Enter', direction: 'target' },
  'drill-out_month': {
    key: 'KeyW',
    direction: 'source'
  },
  'drill-out_city': {
    key: 'KeyJ',
    direction: 'source'
  }
}

```

}

Every node, edge, and rule in this output remains individually addressable: this is the structure the *Inspector* renders and that practitioners select and edit directly in *Skeleton*, rather than a compiled artifact they cannot reopen.

6.5 *Skeleton*: System Design

With a visual structure renderer (the *Inspector*) and a declarative abstraction for producing navigation structures (the *Dimensions API*), the remaining problem was to bring these capabilities into an integrated, direct-manipulation authoring environment where practitioners could not only see structure but manipulate, test, and iterate on it with immediate feedback. *Skeleton* is that environment. It integrates both and extends them with a guided preparation phase, a spatial rendering canvas, and a live testing mode, reversing the *Inspector*'s code-first direction: practitioners design a navigation structure visually and inspect the code representation as a consequence of that design. Each of its authoring techniques makes visible a specific property of non-visual interaction that sighted practitioners otherwise cannot see during authoring: the topology of what is navigable, the spatial mapping of where navigation lives over the chart, the semantics of what is announced, and the temporal sequence in which a user encounters nodes.

A principle runs through the environment: everything *Skeleton* generates is an editable default, not a fixed output. The *Dimensions API* proposes a topology, the scaffold proposes spatial positions and group outlines, and the preparation wizard proposes labels and rules, but each is materialized as concrete nodes, edges, positions, and text that a practitioner can then select and override by direct manipulation, or bypass entirely by constructing nodes and edges manually over an image. The declarative and automated layers lower the cost of a reasonable starting point; they do not constrain what the final structure can be.

6.5.1 Staging: Input and Preparation for Authoring

The authoring workflow proceeds through four stages: upload, prepare, edit, and test. Making a visualization accessible involves decisions about what is navigable, how navigation is triggered, and where focus indicators appear in space. These decisions are related but distinct, and collapsing them into a single undifferentiated interface, as code-only workflows effectively do, makes each one harder to reason about. The stage architecture surfaces them as separate concerns. It also preserves a direct correspondence between what practitioners see in the interface and the structure of the Data Navigator API: each stage maps onto a distinct layer of the API, lowering the barrier to moving from visual authoring into code when production deployment requires it.

Upload. The upload phase is deliberately permissive. Practitioners can bring a dataset, a chart image, both, or neither. When a dataset is present, *Skeleton* parses its fields and infers dimension types automatically, producing a default, starting configuration for the *Prepare* step. When only an image is present, practitioners proceed directly to editing and construct nodes and edges manually over the image. Because dimensions are derived from whatever fields a dataset actually carries rather than from a fixed, pre-registered schema, the approach does not require the data to be known in advance: a different dataset simply yields a different set of inferred

dimensions, and a visualization with no tabular dataset at all is handled through the image-only path. This was motivated by our geologic map co-design work: many bespoke visualizations do not have a single underlying dataset, and any tool that requires structured data as a precondition excludes the cases that need it most. *Skeleton* can be applied to any 2D image surface, not only to visualizations in the conventional sense.

Prepare. The prepare stage addresses a hard authoring bottleneck: not placing nodes, but deciding what structure to build at all. A practitioner who has never designed a navigation structure faces an open configuration space with no obvious entry point. The prep stage presents a four-chapter Q&A wizard that moves through authoring decisions sequentially: (1) whether the chart should have a root node and what it should announce, (2) which data fields should be navigable dimensions and how those dimensions should behave at their boundaries, (3) which keyboard interactions each dimension should be assigned to, and (4) what text labels each level of the hierarchy should produce. Each chapter is accompanied by an illustrative schematic diagram of which part of the hierarchy is being edited as well as a diagram showing examples of what these decisions look like when showing on a chart. The wizard’s output populates a configuration in the editor that practitioners can then inspect, refine, and revise.

6.5.2 Edit: Interacting with Topology, Layout, and Semantics

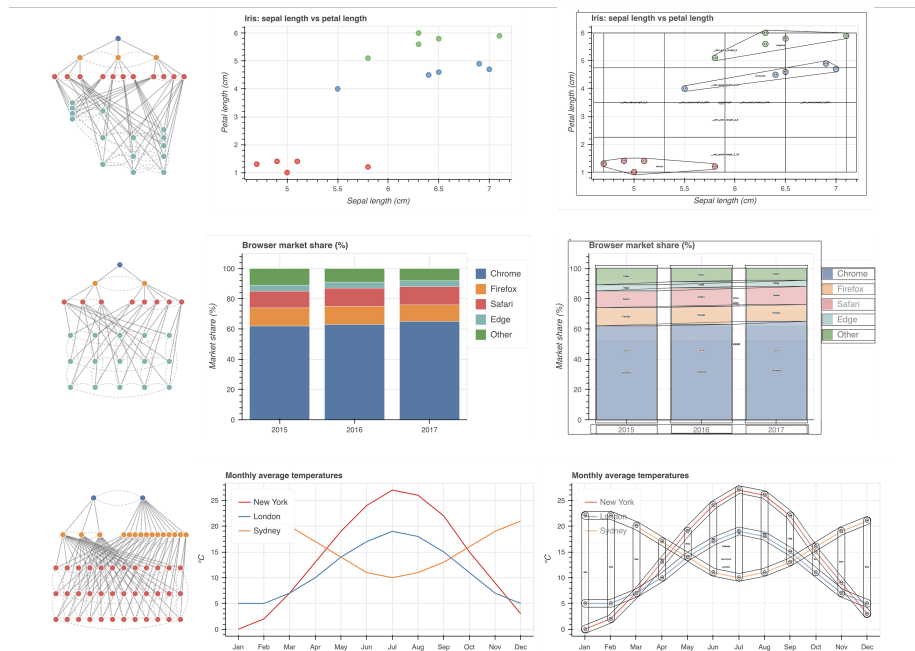


Figure 6.3: Input data transformed into a navigable structure using the *Dimensions API* and visualized with our *Inspector* gadget (left). The input chart (middle). The navigable structure is transformed and drawn over the chart using the *Scaffolding Engine* (right).

Seeing the system, seeing the experience. The editor is *Skeleton*’s primary authoring environment (Figure 7.1) and presents two interlinked representations of the same navigation structure. A schema panel shows the structure as an abstract hierarchical tree layout that makes levels

and parent-child relationships immediately readable. A graph canvas shows the same structure rendered as geometric elements positioned over the uploaded chart image, representing what an end user would encounter spatially. These two representations are bidirectionally linked: selecting a node in either view propagates the selection to the other, so practitioners can simultaneously hold in mind both the abstract topology of what is navigable and the spatial instantiation of where that navigation will live. The dual-view design makes visible a real conceptual divide between the system’s model of navigation and a user’s experience of it, one that practitioners recognize once they can see it, even without prior vocabulary for it.

Leveraging visualization as a scaffolding engine. Manually positioning leaf nodes over each data mark is the most mechanical step in the authoring workflow, and it scales poorly with dataset size. In our early pilot sessions, actually placing nodes in the canvas space was the slowest and most tedious part of the process. To address this, *Skeleton* includes a scaffold tool (Figure 6.3) that automates spatial placement by repurposing Vega [154] as a layout engine.

The scaffold renders a Vega chart specification to a hidden, off-screen container and extracts node positions via the library’s internal scale and view APIs, using the rendering engine purely as a coordinate computation step: no actual chart is ever shown to the practitioner. Coincidentally, none of our co-designers were crafting visualizations using Vega or Vega-Lite (or derivatives), yet the Vega rendering engine could faithfully reconstruct every necessary mark position over the underlying, inaccessible data visualization provided to *Skeleton*.

Additionally, positions and outline strategies for category-level group nodes are computed from the leaf positions using geometric algorithms: a union of child node paths, a convex hull, a grid over numerical bounds, or a bounding rectangle. These designs were consistently produced as ideal treatments during co-design, so we chose them for our starting set of outline strategies.

The scaffold is optional and works by generating synthetic placeholder positions that practitioners can adjust manually. It dramatically improved authoring speed: in light pilot tests, a research team member completed a three-dimension structure without scaffolding in 8 minutes 22 seconds and with scaffolding in 56 seconds. A co-designer completed the task incorrectly (failing to account for one dimension’s division nodes) in 13 minutes 7 seconds without scaffolding, and correctly in 2 minutes 44 seconds with it.

Specifying token patterns, editing instances. Selecting any node populates a properties panel with spatial properties (position, size, shape) and semantic properties (ARIA role, description, and a label template editor; Figure 6.4). The label template allows practitioners to assemble the text a screen reader will announce at that node from tokens drawn from data fields, including aggregate statistics at group-level nodes and precise data values at leaf nodes [101]. Editing labels can apply to all nodes of the same type or to a single instance.

At the bottom of the semantic section, a live preview displays the full assembled announcement string in the exact form a screen reader would produce: role, semantics, group membership, and label combined into a single rendered output that updates in real time. Prior to this preview, understanding what would be announced at a given node required running a screen reader and navigating to it sequentially. In deeply nested structures, this meant spending several seconds listening to labels and drilling in. This was consistently the main point where bugs were produced and missed during our co-design work.

The preview makes text announcements inspectable as visible, editable objects. This surfaces a class of highly specific, low-level problems that code-only workflows leave invisible: redun-

AGGREGATE SUMMARIES FROM CHILDREN

☐ Count of children

☐ Min and Max of temp

☐ Sum of temp

☒ Average of temp

Trend direction and R²

X variable

Y variable

month
temp

☒ Trend direction
 ☐ R² value

Label template

{value:"city"}, trend for {key:"temp": {trend:"
Clear

LABEL

PREVIEW:

New York, trend for temp: flat, average temp: 13.78

City 3 of 8.

Figure 6.4: Group label pattern builder, including an array of aggregate summary options, template formatter field, and preview.

dancies in announced text, missing contextual framing, and label ordering and punctuation that affects comprehension and reading speed. Of all the authoring decisions *Skeleton* exposes, label templates involve the most degrees of freedom and have the most direct bearing on the quality of the non-visual experience. Getting the structure right ensures navigability; getting the labels right determines whether navigation communicates anything meaningful.

6.5.3 Test: Debugging Interaction Interactively

The testing stage allows practitioners to navigate the structure they have built using the same keyboard input and navigation rules that assistive technologies would use, without leaving the tool. When a practitioner enters testing, the three Data Navigator modules are instantiated in sequence: the structure module rebuilds the navigation graph from the current configuration, the rendering module creates an HTML layer positioned over the chart image at each node's spatial coordinates, and the input module registers keyboard listeners for all navigation rules. The result is a live, keyboard-navigable structure. An event log records navigation events in order, letting practitioners verify that all nodes are reachable and that label sequences make sense when encountered serially rather than read simultaneously.

The abstract graph continues to display during testing. As the practitioner navigates, the focused node is highlighted simultaneously in the canvas (showing its spatial position over the chart image) and in the abstract graph (showing its structural position in the hierarchy). Practitioners can verify at a glance both where focus is and what role it occupies. A mobile-friendly text-chat

mode is also available, in which practitioners navigate by typing natural language commands, motivated by the Adobe collaboration ([Section 6.4.2](#)).

Practice-based validation. The testing stage also served, during development, as the primary debugging interface for *Skeleton*’s own data pipeline. Additionally, CD2, a subject matter expert who professionally evaluates interfaces for screen reader access and is familiar with other visualization navigation systems, used *Skeleton*’s testing stage to evaluate navigation output across several chart types and dimension configurations: line charts (3 configurations), bar charts (2), stacked bar charts (3), and scatterplots (4). This evaluation followed a manual, systematic approach combining standards-based criteria with expert screen reader testing. Scatterplots required the most iteration, surfacing bugs in Data Navigator’s core library that were then fixed. CD2 also recommended that we use list-based navigation while in the editor (not in testing) in case users build themselves into a keyboard trap.

6.5.4 System Implementation: Technical Details

This appendix details the three technical components behind *Skeleton*’s spatial authoring ([Section 6.5](#)): how the scaffold repurposes a visualization rendering engine to approximate mark positions, how group outlines are computed from those positions, and how a lightweight, non-ML computer vision pass aligns the scaffold to an uploaded chart image. We intentionally chose deterministic math approaches at any opportunity, for speed and inspectability benefits.

6.5.4.1 Vega as a Deterministic Layout Engine

The scaffold ([Section 6.5](#), “Leveraging visualization as a scaffolding engine”) treats a visualization grammar as a coordinate oracle rather than a rendering target. From the active configuration, which carries a chart type, field-to-channel mappings (`xField`, `yField`, and an optional `colorField`), mark parameters, and an outer plot box, *Skeleton* assembles a *Vega-Lite* [154] specification and renders it through `vega-embed` into a hidden, off-screen container using the SVG renderer. The container is never attached to the visible document; it exists only so that *Vega* will compile a full scenegraph and, with it, the scale functions that map data values to pixels.

Positions are then read directly from the compiled view rather than parsed from the rendered output. For each leaf node, *Skeleton* evaluates the view’s `x` and `y` scales on the corresponding data values: a band scale supplies a bar’s horizontal position and, through its `bandwidth`, its width, while a linear scale maps a quantitative value to a pixel height with the baseline taken at the scale’s zero. Stacked bars accumulate per-segment offsets in data space before scaling; clustered bars add the sub-band offset; scatter and line marks evaluate both scales at each point. Because the scale functions are pure, this path needs no DOM measurement and is fully deterministic: identical configurations yield identical coordinates. A secondary path, used when scale evaluation does not apply, parses the hidden SVG and reads mark bounding boxes instead.

The configuration distinguishes the inner mark area from the outer box. *Vega-Lite*’s `width` and `height` describe the inner area in which marks are drawn, while a `padding` object reserves space for axes and labels; *Skeleton* treats `plotWidth` and `plotHeight` as the outer, border-box dimensions and derives the inner area by subtracting padding. A mark at normalized position f within the inner area therefore lands at image coordinate $o + p + f w_{\text{in}}$, where o is the plot

offset, p the leading padding, and w_{in} the inner extent. This relation is what later makes padding recoverable.

A practical subtlety motivates the rest of this appendix. The engine that produced the uploaded image is generally unknown and is rarely *Vega*; our co-designers authored charts in other tools entirely (Section 6.4). What transfers across engines is the *relative* geometry of the marks, the spacing of bars or the placement of points within the data rectangle, not the absolute pixel offsets, which depend on how much room each particular chart reserved for its axes, ticks, labels, and title. The scaffold reconstructs the relative geometry faithfully; reconciling it with the absolute placement in the image is the task taken up by the padding estimation described below.

6.5.4.2 Computing Group Outlines

Once leaf positions exist, the group nodes above them (divisions, dimensions, and the root) receive a visible outline computed purely from the geometry of their descendants. Each leaf contributes a rectangle (for bars) or a circle (for points), and a chosen strategy turns the collection into a single SVG path string. These functions live in the Data Navigator core (`geometry.ts`) and are side-effect free, taking arrays of rectangles or circles and an optional padding term and returning a path `d` attribute, so they can be reused outside *Skeleton* and packaged for export.

Four strategies cover the cases that recurred during co-design (Section 6.4). A *bounding rectangle* encloses all descendants in the single axis-aligned box given by their combined extent. A *union* emits one rectangular subpath per mark and relies on the SVG fill rule to render disconnected regions, which suits groups whose marks are spatially separated. A *convex hull* collects the corner points of every rectangle (or sixteen sampled points around every circle) and runs a Graham scan [66], optionally pushing each hull vertex radially outward from the centroid to add padding; this gives a tight, convex boundary appropriate to scatter clusters and bar groups. For line series, an *offset path* traces the polyline of points at a fixed perpendicular distance on each side, using per-vertex averaged normals for miter joins and semicircular arc caps at the ends, producing a closed band that follows the line. Numerical dimensions additionally support a grid drawn over the dimension’s value bounds.

Because every outline derives from the same leaf coordinates the scaffold computed, changing a mark position, a chart type, or the padding propagates automatically to the group shapes, and authors edit outline strategy and padding as ordinary node properties rather than drawing shapes by hand.

6.5.4.3 Estimating Plot Padding from the Image

The scaffold’s layout is internally consistent but free floating: it knows where marks sit relative to one another, not how the uploaded image split its canvas between the data rectangle and the axis space, which authors originally recovered by dragging the four padding values until the scaffold marks settled onto the real marks. Our most recent work automates it with a small, fast, dependency-free computer vision pass that runs entirely in the browser, in the spirit of chart reverse-engineering that recovers marks from bitmap images [141, 155] but reduced to coarse mark localization, since the scaffold already supplies the data.

Detection proceeds on a downscaled copy of the plot region. The image is drawn into an off-screen canvas with its longer side capped near one thousand pixels, and a binary foreground mask is built by thresholding the pixel data [139]. The key observation is that data marks are almost always rendered in color while chart chrome (axis lines, gridlines, tick labels, titles, and legend text) is grayscale. A saturation threshold therefore isolates the marks and prevents a dark axis line from bridging adjacent bars into a single blob; for the rare monochrome chart the pass falls back to a dark-ink mask. Connected-component labeling [148] with eight-connectivity groups the mask into candidate marks, each carrying a bounding box, centroid, area, and mean color. Two filters clean the result: components far smaller than the median are discarded as legend swatches, and for line and area charts, where the mark is one long stroke rather than discrete glyphs, the endpoints of the dominant component are used in place of separate marks. The pass returns the extreme detected marks along each axis (leftmost and rightmost, topmost and bottommost) as points in image space.

Padding then follows in closed form. Recall that a scaffold mark at normalized inner position f sits at $o + p + f w_{\text{in}}$, and that f is invariant to padding because it is fixed by the data and the scale. Taking the two extreme scaffold marks on an axis, with inner positions f_{lo} and f_{hi} , and the two detected image positions d_{lo} and d_{hi} they should align to, the inner extent and the two padding values follow while the outer box (o, W) is held fixed:

$$w'_{\text{in}} = \frac{d_{\text{hi}} - d_{\text{lo}}}{f_{\text{hi}} - f_{\text{lo}}}, \quad p'_{\text{lo}} = d_{\text{lo}} - o - f_{\text{lo}} w'_{\text{in}}, \quad p'_{\text{hi}} = W - w'_{\text{in}} - p'_{\text{lo}}.$$

The two axes are solved independently. Several guards keep the result trustworthy: an axis whose two marks are nearly coincident in f is ill-conditioned and is skipped, negative padding is clamped, and a solved inner extent larger than the outer box signals that the box itself is too small and is reported rather than forced. The pass runs once automatically when an image, a chart type, field mappings, and at least two scaffold marks are all present, and on demand thereafter from a control in the scaffold panel; after applying the padding it re-measures the residual between detected and scaffold marks as a one-shot verification.

In practice this recovers padding consistent with the original chart across the four canonical types we tested (bar, stacked bar, line, and scatter), shown in Figure 6.5. On the bar example, for instance, the solver recovers a left padding close to the width of the chart’s y -axis label gutter and a small right padding, with no manual adjustment. The approach is deliberately heuristic and assumes colored marks over light chrome; its single entry point makes a heavier detector (for example an `OpenCV` build) substitutable without changing the padding solver. We regard this as promising but early, and a fuller evaluation across more diverse chart images remains future work.

6.6 User Study

Our co-design work was motivated by the problems of the communities the research team was embedded in, and our co-designers were deeply familiar with the problem space, having spent months or years working on accessible navigation. We still needed to understand what happens when practitioners who are *not* embedded in this process design, author, and debug navigation

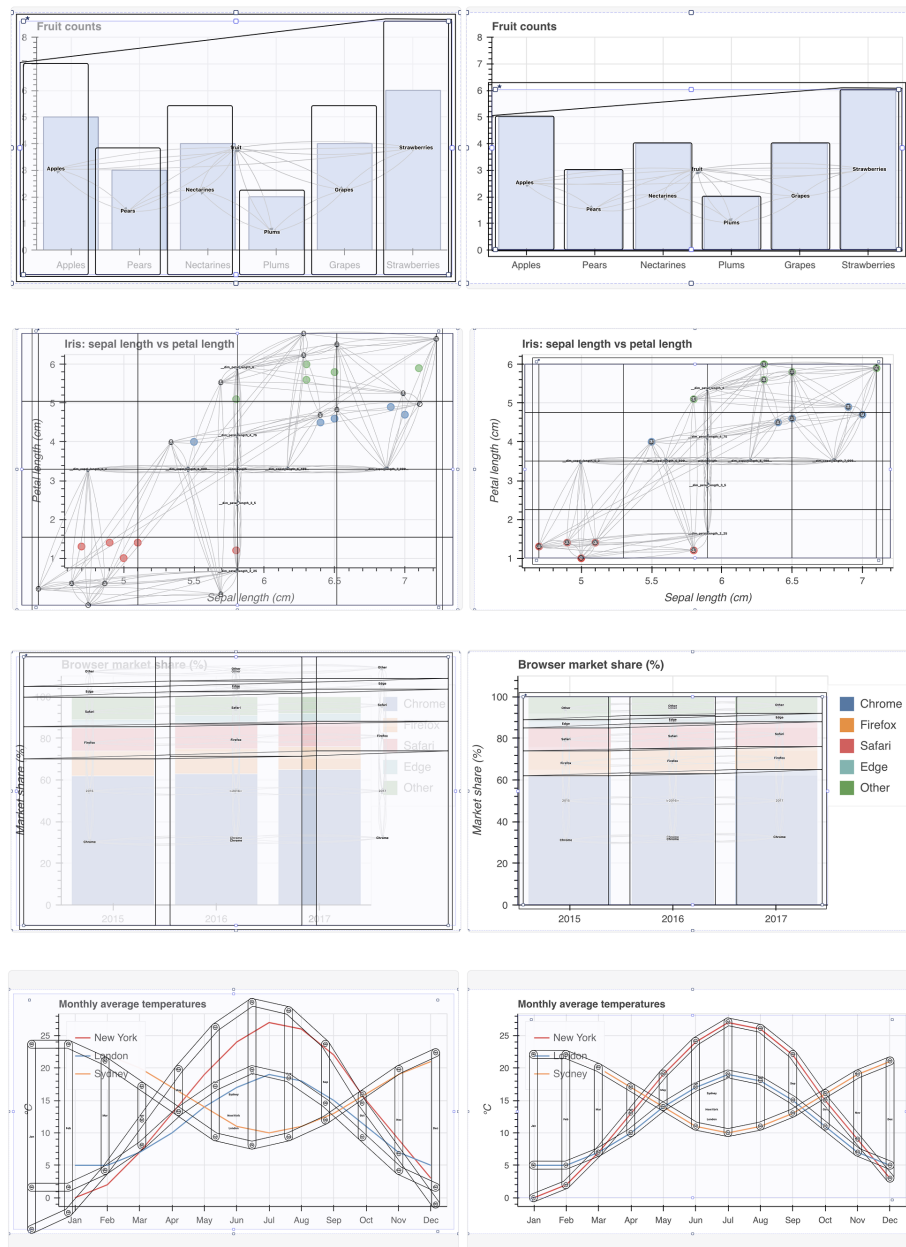


Figure 6.5: Computed layout for marks via Vega’s visualization engine plus group outlines (left) and with padding estimation using our non-ML computer vision approach (right) for various types of common visualizations.

structure visually: whether the representations we built are legible to them, whether the techniques change how they reason about accessible design, and what new questions or problems emerge when navigation structure becomes visible. These are empirical questions that required a study.

To evaluate how *Skeleton* influences the way practitioners engage with accessible design, we conducted an in-situ interview study with 8 participants across visualization design, engineering, and research. The study was conducted remotely over video call, took approximately 45–60 minutes per session, and was approved by our IRB. Participants provided verbal consent at the start of each session. Video and audio were not recorded, however some participants consented to share the data/image they brought to the study as well as screenshots of their workflow; data collection was note-based throughout.

6.6.1 Participants

Participants were recruited through snowball sampling within the visualization and accessibility community, and through referrals from co-designers involved in our earlier collaborations. We asked each participant to self-report their primary work role (engineering, research, design, or student) and their existing level of accessibility expertise on a 1–5 Likert scale. We also asked whether the visualizations they build are ever bespoke, that is, custom rather than instances of a recognizable, standard chart type. This distinction mattered because bespoke visualizations represent an especially underserved case in accessible design tooling: no library pattern applies, and every navigation structure must be designed from scratch. Participants were not compensated.

6.6.2 Procedure

Each 45-minute session proceeded in four phases. Before the session, all participants were asked to prepare a chart image they were currently working on or had recently built, for use in the third phase.

Phase 1: Introduction and demographics (5 minutes). After obtaining verbal consent and recording a pseudonym, we collected self-reported role, accessibility expertise level, and whether the participant regularly builds bespoke visualizations.

Phase 2: Generic chart think-aloud [4] (10 minutes). Participants used *Skeleton* on a provided bar chart and dataset of fruit counts (Apples, Pears, Nectarines, Plums, Grapes), asked to design an accessible navigation experience for a screen reader user with no instructions on how the tool worked. The tool loaded with a default structure having both a categorical dimension (`fruit`) and a numerical dimension (`count`) active. This default was intentionally problematic for two reasons: numerical navigation sorts by count value and groups data into subdivided ranges, producing a traversal order different than the visual layout an additional, largely unhelpful level in the hierarchy for such a simple chart. We observed how participants reasoned about what they saw and whether and how they noticed this extra dimension.

Phase 3: Own-chart think-aloud (15 minutes). Participants loaded their own chart image and attempted the same task, except they were also asked to explain their graphic to the research team (purpose, role, data, and domain). This phase was open-ended: charts ranged from standard

types to bespoke visualizations, and the goal was to observe how participants reasoned about navigation structure when the context was their own work.

Phase 4: Reflective interview (15 minutes). We conducted a semi-structured interview in which participants reflected on their decisions in Phase 2 versus Phase 3, their experience with the generic versus their own chart, and their assessment of the tool’s capabilities and limitations. We asked what felt possible or impossible, what they wanted to do that they could not, and what they found themselves thinking about that they had not considered in Phase 2.

6.6.3 Analysis

Notes from each session were compiled into a shared document. Participant quotes reported in the results are reconstructed from these researcher notes, not verbatim transcripts. We analyzed the data using a combination of thematic analysis [21] and affinity diagramming [78], iterating across both methods to surface recurring patterns while preserving the specificity of individual participant experiences. Analysis attended particularly to differences in how participants engaged with accessible design before and after using the tool, the range of input modalities and user scenarios they considered, and moments when participants reconsidered or wished to redesign their own visualizations.

6.7 Results

We organize our findings into five themes that emerged from thematic analysis and affinity diagramming across all eight sessions. Each theme captures a qualitative pattern in how practitioners engaged with accessible navigation design when its structure was made visible and manipulable. We report these findings descriptively and ground them in specific participant moments; interpretation follows in the Discussion.

6.7.1 Seeing Navigation Made Structural Problems Legible as Design Problems

The generic bar chart in Phase 2 loaded with an intentionally problematic default: both a categorical dimension (`fruit`) and a numerical dimension (`count`) were active, producing overlapping navigation structures with different traversal orders over the same data. This configuration is a poor design choice, arguably a design failure, but one that would be difficult to detect in code alone (R1, R4).

Participants varied widely in how quickly they recognized the problem. P1 turned off the numerical dimension within seconds of seeing the editor, without commenting on it. Most participants, however, initially struggled to understand what they were seeing, remarking on the unfamiliar structure: “what is this? what are these?” when encountering the numerical divisions for the first time. P8 spent time trying to guess what the extra divisions represented but did not remove them during Phase 2, only realizing during Phase 3 that the additional dimension was “probably bad.” P2 and P5 expressed suspicion early: P5 asked, “Is this too much data? This

seems like way too much to just navigate through,” and P2 noted, “I feel like a lot of data points would be bad, yeah? Like, too many at once is bad?”

The testing stage (Section 6.5.3) proved critical for resolution. P4, P5, and P7 each removed the extra dimension only after navigating the structure with keyboard input in testing mode, where the traversal sequence made the redundancy experientially apparent. In total, five of eight participants resolved the problem during the session: P1 and P2 during editing, and P4, P5, and P7 after testing.

Beyond the intentional default, participants identified other problems through visual inspection. P8 reacted to a generated node name: “Okay `dim_fruit` node...that is horrible, what is that?” During Phase 3, P7 looked at the edges of their multi-line chart and asked, “are all these bad? Is it bad that I don’t even really know what the takeaway of this [structure] is?” P3, seeing a full hierarchy for their own simple six-item bar chart (during Phase 3), concluded “I should just skip the root and grouping and go straight to the data. This seems like too many steps.” In each case, the visual representation of their navigation structure motivated judgment about potential negative design qualities.

6.7.2 Practitioners Developed a Designerly Interest in What Constitutes Good Navigation

The most pervasive pattern across sessions was that participants began asking design questions about navigation quality, unprompted by any instruction or guidance from the research team (R2). These questions went beyond identifying errors: participants wanted to know what *good* navigation would be for their charts.

P2, working through the bar chart in Phase 2, deliberated over boundary behavior: “Loop back or stop? I don’t think there is a right way. I will just pick *fruit* for now and *loop* and see what this does.” P6, who brought a bespoke flower visualization, wondered how to translate the affective quality of their chart: “I think my visualization should be more about the vibes, but I don’t know how to make the alt text have good vibes. What is *fun*?” P4 asked fundamental questions about the interaction model itself: “Why do screen readers and keyboards have to work this way? Do people like that?... why do we navigate?” And later after testing, P4 concluded, “I bet we should make this *faster*” before cutting out the additional numerical dimension in Phase 2.

Several participants engaged with the concept of narrative and flow. P8 articulated this as a question about the goal of the visualization: “sometimes I want a big picture, not precision. I may want to drill down a little...” and observed that “we think too much in terms of components... sometimes accuracy isn’t the actual goal, it’s getting a general sense of something.” P4, upon discovering that *Skeleton* supports text-based input, asked, “How do you make that good, though? Like chatGPT, or do people want to, like, interact with the chart [elements]?”

During the interview, several participants explicitly requested guidance. P1, P2, P3, and P6 wanted to see examples of well-designed navigation experiences. P1, P3, P4, P5, P6, and P7 wanted embedded guidelines within the tool. P2 and P4 actually used web search to look for “chart navigation for accessibility guidelines” (P2) and “accessible viz screen reader design” (P4). P3 was interested in automation and heuristics that could suggest reasonable defaults.

These requests are consistent with the pattern (R2): practitioners who could see the design space wanted orientation within it.

6.7.3 Iteration Was Substantive, Self-directed, and Concentrated on Semantics

Every participant iterated on their navigation designs, and this iteration was neither perfunctory nor prompted by the research team (R4). Participants revisited decisions, revised configurations, and in some cases restructured their entire approach after encountering their design in a new stage of the tool.

The most sustained iteration concerned labels. Every participant spent the largest share of their authoring time in the label template editor (Figure 6.4), editing the text that a screen reader would announce at each node. This editing was granular: participants wrote full sentences, rearranged the order of data tokens, debated whether to include field keys alongside values or values alone, experimented with how to name groups and individual elements, and considered the length and density of the resulting announcements. At the division and dimension levels, some participants added multiple aggregate statistics (count, sum, average, range, trend) and then returned to trim them. P2, for instance, edited data point labels, left them, and returned to revise them two additional times: “This label is way too complicated, I think.”

A second, distinct pattern of iteration emerged around testing. The editor displays all navigation nodes simultaneously, showing the full structure at a glance. The testing stage (Section 6.5.3), by contrast, shows only the currently focused node, highlighting it in the canvas one at a time as the practitioner navigates. This difference consistently prompted participants to revise. Several returned to the editor after testing to adjust group node outlines, because outline strategies that looked distinct when displayed simultaneously (such as bounding rectangles vs. convex hulls) became harder to differentiate when encountered one at a time. Others adjusted their dimension configurations: P3 and P5 restructured their dimensions after testing, and P4, P6, and P8 experimented with different key bindings and navigation rules. P2 used the testing stage specifically to identify labels that needed revision at the dimension and division levels.

The most extensive iteration came from P1, who returned to the preparation wizard after reaching the testing stage and re-took the entire wizard from scratch. Seeing the full structure in testing motivated them to re-evaluate their earliest decisions. They reduced their navigation from three dimensions to one, producing a navigation experience they described as deliberately minimal. In the reflective interview, P1 explained that they had been thinking about trimming excess, joking that they were “working on a data-to-word ratio.”

The scaffolding tool shaped the pace and character of these iterations. Participants who used the scaffold (Figure 6.3) generally required only minor manual adjustment of node positions, spending one to two minutes refining placements in-aggregate before moving on. Iteration on spatial layout was primarily about the appearance of group-level outlines rather than individual node positions: because the scaffold computed leaf positions from the chart’s encoding, the main remaining spatial decision was how division and dimension nodes should visually indicate their grouping.

6.7.4 Seeing Navigation Prompted Practitioners to Reconsider the Architecture of Their Own Charts

An unexpected finding was that working with *Skeleton* prompted several participants to question the design of the visualization they had brought, not just the navigation structure they were building over it (R4). Five participants (P4, P5, P6, P7, P8) expressed a desire to simplify their own charts after seeing what the corresponding navigation structure required. Three (P1, P5, P6) considered whether a different chart type might serve the same communicative goal with a simpler navigational architecture. And in 2 cases (P1, P6), participants questioned whether a chart was the right medium at all.

P5, who brought a scatter plot with over two thousand data points, initially tried to build a navigation structure over the full dataset, recognized that the result was unmanageable, and then asked: “do I even need visualization?” They went on to wonder whether the information could be communicated as “a few sentences or like, some data [users] can prompt” (referring to using a large-language model). P6, who brought a bespoke flower visualization in which each petal encoded a different variable, initially tried to express the chart’s full complexity in navigation (grouping by flower and then navigating by petal) but then reversed course: “okay, what if we actually just treat this like a bar chart?” Using the scaffold tool, they produced a simple list-style navigation that worked well for their data, and reflected: “my visualization has a lot, but actually, this could be pretty easy to navigate, I think.” P6 also wondered whether there was a non-chart way to “tell their story,” and in further discussion described something closer to a scrollytelling article or guided walkthrough than a single interactive chart.

P8’s case was the most striking. They brought a voronoi pie chart (Figure 6.6) used to communicate to students that 26 assignments make up 45% of their grade, the largest slice of the pie. Working through *Skeleton*, P8 first considered making all 26 voronoi cells navigable, then considered making only the three main slices of the pie navigable, and then questioned whether navigation was needed at all. The “point of this,” as they described it, was to communicate a single ratio; students did not need to traverse individual cells. P8 concluded that well-written alternative text was sufficient for the screen reader experience and chose not to build any structure. They also reflected on the design of the visualization itself, concluding that the voronoi treatment served a visual purpose (it is visually striking) that was separable from the informational purpose (communicating a grade breakdown). They kept the visual design and simplified the non-visual experience accordingly.

6.7.5 Experiencing Keyboard Navigation Surfaced a Broader Range of Users and Input Technologies

The testing stage (Section 6.5.3) provided what was, for most participants, their first experience of keyboard navigation through a data visualization. Only three participants (P1, P2, P8) reported prior experience using a screen reader, and only two (P1, P8) had previously navigated a visualization using keyboard input alone. Yet during the study, every participant navigated using a keyboard.

Several participants responded to this experience with genuine engagement. P4 reacted with enthusiasm: “Woah, this is incredible. Wait, what other visualizations can do this?” and later,

Editor

Draw nodes and edges to define the navigation graph.



Click and drag to pan the canvas — Escape to exit

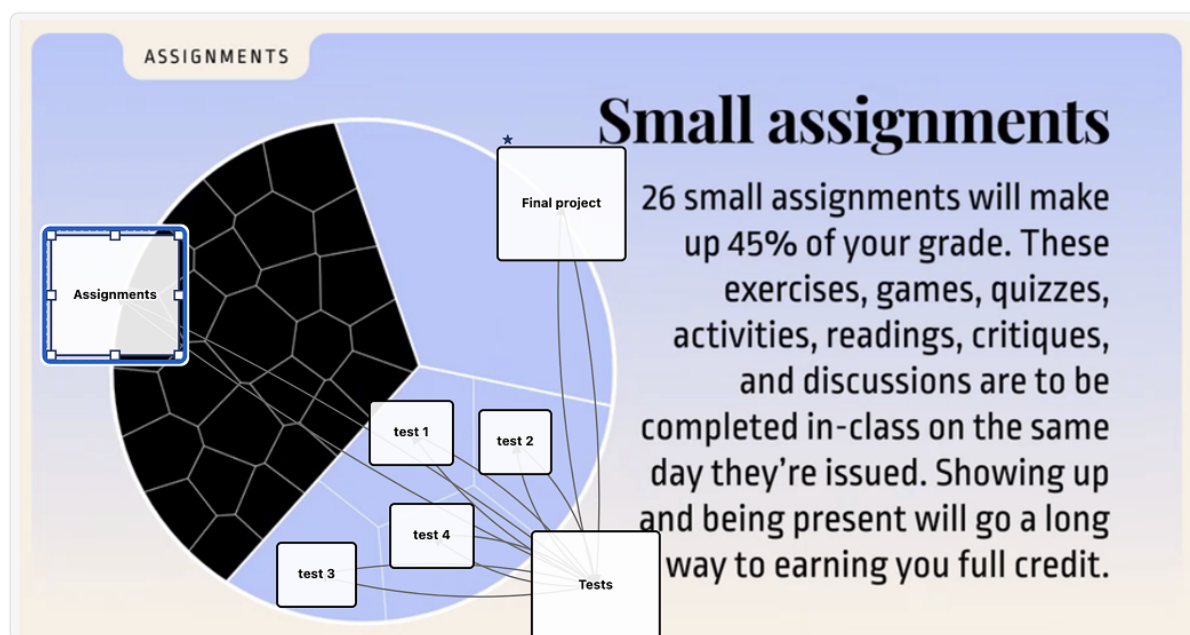


Figure 6.6: Re-creation of P8’s moment of realization, placing nodes manually: not every element in their voronoi pie chart *should* be navigable.

“Maybe you could make a visualization game, where you can move around the data?” P7 said, “I love this. I want to make charts with this.” P5, who had not previously encountered keyboard-navigable charts, said, “I didn’t know you could do this.”

The experience also expanded participants’ sense of who these structures serve (R2, R4). P8 adjusted a node’s spatial dimensions and said, “Let’s make the hitbox as big as possible here... for my neighbor with Parkinson’s to click these,” treating the navigation structure as relevant to motor accessibility, not only screen reader access. P3, observing the visual focus indicators that appeared during scaffold-based placement, remarked that “this is for more people than [someone] blind,” recognizing that sighted keyboard users and users with partial vision also rely on visible focus states. During the interview, P4 asked sustained questions about what kinds of technologies different people with disabilities use, and P4 and P7 both asked why focus indication was designed to be visible rather than invisible.

P4’s curiosity extended to alternative input modalities. Upon learning that *Skeleton* supports text-based navigation and (due to leveraging Data Navigator) also supports a wide array of other input modalities, they asked, “Can you talk at it, too? Is that what some people do?” and followed up with, “How cool is that? How do you make that good, though?” P8 raised a concern about discoverability in the current interaction model: “I’ve never used j or w to drill out, always shift arrow or option arrow...” and worried that a user might not know how to exit a nested level. These observations reflect a broad mental model of the user, from an abstract “screen reader

user” to a person with multiple capabilities, preferences, and interaction patterns (R2).

6.8 Discussion

6.8.1 Visibility as a Precondition for Iteration

The central finding of this work is not that *Skeleton* taught sighted practitioners how to design accessible navigation, but that it gave them something to react to. When navigation structure was invisible, our co-designers would only seek to verify whether navigation followed our design documentation. Once visible, questions shifted to whether it was good, why, how it could be improved, and what else it could do. Visibility encouraged iteration, and iteration is enabled continuous design improvements.

This mechanism is consistent with Schön’s account of reflective practice [159]: designers iterate by externalizing a representation, perceiving its properties, and responding to what they see. The representation talks back, and the designer adjusts. But this loop requires a representation. In our co-design work, we assumed that the gap was hand-off between design and development, and the slow process of manual verification. Instead, the gap is that development was not participating in, and encouraging, further design reflection. In a developer’s code-only workflow, navigation structure has no externalization that supports this kind of perceptual engagement. A practitioner can read the code that specifies a navigation graph, but they cannot see the graph, cannot perceive its topology at a glance, cannot notice that a label is redundant or that a hierarchy is too deep by looking at it. The reflective loop is disrupted and occluded at the first step.

Skeleton encourages this loop by rendering navigation structure in a form that supports the same kind of perceptual judgment that visualization practitioners already apply to every other aspect of their work. The results show what happened when this loop was available: every participant iterated, substantively and self-directedly (Section 6.7.3). They revised labels repeatedly, restructured dimensions after testing, adjusted group outlines when sequential traversal revealed problems that simultaneous display had hidden. P1 restarted the entire preparation wizard after seeing their structure in testing mode. These are not the behaviors of practitioners following a specification; they are the behaviors of practitioners negotiating with a design material.

The co-design work (Section 6.4) provides complementary evidence: in each collaboration, sighted practitioners reached for visual representations when reasoning about navigation, through node-link diagrams in Figma, schema sketches in Miro, and annotated wireframes on paper. They were already thinking visually about non-visual structure; the development tooling simply had not caught up. The practical implication is broad: if sighted authors depend on visibility, then any authoring workflow that keeps navigation structure invisible limits design iteration. Making structure visible does not guarantee good design, but it is a precondition for the kind of sustained, judgment-driven refinement that good design requires.

6.8.2 From Compliance to Design

A consistent pattern across the accessibility literature is that practitioners frame accessibility as a compliance problem: a set of requirements to satisfy, a checklist to complete, a legal or insti-

tutional obligation to meet [44, 102, 165]. The framing matters because compliance and design orient practitioners toward fundamentally different activities. Compliance asks: “does this pass?” Design asks: “is this good?” Our results suggest that *Skeleton* produced a partial shift from the first orientation to the second. Participants’ initial instincts clustered around compliance-oriented responses: provide alternative text, follow guidelines, ask an expert (Section 6.7.2). But alongside that desire for guidance, they began doing something compliance framing does not typically produce: they encountered complexity and then *iterated* (Section 6.7.3). This sustained, self-directed refinement is the behavioral signature of design, not compliance. As P4 put it: “I’d love to have someone blind actually just with me while I make this, but I also understand that I should learn what makes a good experience too.”

The shift extended beyond the non-visual experience itself. As reported in Section 6.7.4, five participants reconsidered the design of the visualization they had brought, not just the navigation structure. Making the accessibility consequences of visual design choices visible prompted practitioners to question whether a different chart type, a simpler encoding, or a non-chart medium might better serve their communicative goals. This interrelation between non-visual and visual design suggests that the widespread treatment of accessibility as a compliance activity may be partly a consequence of tooling that offers no legible, manipulable design surface. Auditing frameworks are valuable, but they are *evaluative* tools, not *authoring* tools. The field needs both.

6.8.3 Bespoke Visualizations as an Unaddressed Accessibility Research Problem

Diagrams, infographics, and data-driven illustrations are often one-off, custom designs with bespoke symbols, layouts, and visual languages. These representations are increasingly common in journalism, scientific communication, personal projects, art, and public-facing data work, and they represent the cases where accessibility-focused tools are needed most and available least. *Skeleton*’s image-based workflow (upload any 2D image, place nodes manually) provides a starting point, but the study made clear that bespoke visualizations need more than node placement. They need support for reasoning about what navigational structure (if any) is appropriate when no template applies, a research problem that remains largely unexplored. It is worth being precise about where generality holds and where it stops. Because the *Dimensions API* derives structure from whatever fields a dataset carries rather than from a fixed schema, the data side of the approach already extends to datasets not known in advance; what does not yet generalize is the prior, harder question of which dimensions, if any, a bespoke visual should expose when its visual form was never organized around a tabular structure to begin with. Generalizing the abstraction is therefore not primarily a matter of supporting more schemas but of supporting the design judgment that precedes choosing a schema at all.

6.8.4 What Visualization Owes Accessibility

Our approach, to make visual non-visual experiences, should not be limited to data visualization’s own accessibility challenges. Navigation structure is a foundational component of accessible experience across domains: PDF and document reading order, web page structures, and

software application layouts. In each of these areas, sighted practitioners author non-visual experiences without visual feedback, and in each, the same gap between design intent and verifiable outcome constrains quality. Testing is slow, error prone, and requires expertise in assistive technology use. Visual tooling for authoring, inspecting, and debugging non-visual structure (**R3**) is a tractable and high-value problem across application domains.

There is also a deeper question about what visibility can accomplish in principle. Work on multi-modal authoring environments has argued for de-centering visual representation and treating modalities as equal partners in the design process [215], an important ethical commitment. *Skeleton* does not do this: it re-centers visual representation as the medium through which sighted practitioners engage with non-visual structure. We believe this is justified pragmatically, because sighted authors need to articulate navigation design in their own perceptual language before they can reason about it at all, and this paper provides evidence that they do. But articulating a design in one’s own language is not the same as understanding how it will be experienced in someone else’s. The structures participants built during our study were never evaluated by blind users, and visibility alone cannot substitute for that evaluation. The risk we want to name is that making non-visual structure visible to sighted practitioners could be mistaken for making it *understood*, when in fact it makes it *designable*, a real but bounded gain. The fuller design process requires collaboration with disabled users, not as an occasional supplement but as a regular practice. *Skeleton* can make that collaboration more productive by giving both parties a shared representation or space of translation between representations, but it cannot replace it.

What visualization owes accessibility, then, is not simply better output but authoring tools that better stimulate reasoning, both individually and collaborative, about design.

6.9 Limitations and Future Work

Skeleton surfaces design questions but does not answer them: several participants asked what constitutes good navigation, and the tool had nothing authoritative to offer. As [Section 7.8](#) argues, our study evaluated whether making structure visible stimulated design consideration, not whether the resulting designs were good for the people who will use them, which is a distinct question that remains open. CD2’s expert screen reader evaluation of *Skeleton*’s navigation output ([Section 6.5.3](#)) provided practice-based validation for several common chart types and surfaced concrete bugs, but this evaluation was neither comprehensive nor controlled: many configurations remain untested, and expert review is not a substitute for evaluation with a broader population of end users.

Our approach using *Skeleton* as a design probe only with sighted participants is a limitation and, we believe, the right sequencing: *Skeleton* improves the intentionality and iterability of what sighted practitioners produce, which is a necessary precondition for a subsequent evaluation or future collaborative design work between sighted and blind authors. Further research and evaluation should close this loop, ideally to engage how mixed-ability teams co-design multi-modal data experiences.

A further boundary concerns the kind of visualization *Skeleton* targets. Highly interactive or dynamic visualizations, where selecting an element reloads the data, where marks animate, or where the navigable structure itself changes in response to user interaction, are not yet addressed.

We see this as an important future direction rather than a fundamental obstacle: the same argument for making structure visible applies with more force when there are several structures to coordinate, and an inspectable representation may be precisely what makes a dynamic navigation experience tractable to author.

Skeleton is also a prototype with substantial work remaining. Not within the scope of the paper, but our participants provided ample feedback on the functionality of the prototype itself. The most urgent gap is export functionality: practitioners can design and inspect navigation structures in the tool but cannot yet produce deployable output.

6.10 Conclusion

Accessible navigation structure has long occupied an awkward position in visualization practice: known to matter, difficult to design, and invisible to the people responsible for building it. Without a way to see what they were making, sighted practitioners could not catch errors, could not iterate, and could not develop the kind of considered judgment that good design requires; accessibility remained downstream of every other decision not because of any single failure, but because the authoring conditions did not support anything else. *Skeleton* demonstrates that those conditions can be changed. Making navigation structure visible and manipulable (as an interactive graph rendered over the spatial layout of a real visualization, with live label previews and testable traversal) shifted how practitioners engaged with and reasoned about accessible design, prompting them to ask whether the design of their structure was *good* rather than merely whether it passed.

The implication generalizes beyond our tool: wherever a non-visual structure or user experience is authored without a visible representation, building one the author can react to and manipulate invites them to design. What the paper leaves open is more than what it closes. We used *Skeleton* as a design probe with sighted practitioners; we did not evaluate the structures they produced with the end users those structures are meant to serve, and the broader disciplinary conversation about what visualization research owes accessibility, and what methods might transfer between them, has more questions than answers. We offer *Skeleton* not as a solution to these problems but as evidence that engaging them directly, with the full weight of visualization's methodological tradition, is both possible and fruitful.

6.11 Postface

While this chapter did not take the liberty of making a strong position statement, it is clear to us now that when viewed through the social model of disability (where disability is produced between human interaction with systems and environments) practitioners who are not traditionally considered “people with disabilities” in the medical sense may still face barriers to access when using an interface or system. *Systems can disable any user.*

As another note worth making, we also attempted to “vibe-code” the prototype for this work as well, and encountered (ironically) many barriers while developing, both that we personally faced and that were unfortunately also produced for end users. Remediating our prototype has taken a non-trivial amount of manual and agentic effort.

Large-language models are not necessarily skilled enough out of the box, at the time of this writing, to make highly complex research prototypes that focus on cutting-edge accessibility functionality. The overall work spent vibe-coding now vastly outpaces whatever work we would have invested if we hadn’t vibe-coded at all, because we will need to virtually rebuild this entire prototype from scratch before deploying it live for real-world use. It is, in this sense, a throwaway research prototype in its current condition. But at the same time, we did learn that vibe-coding in contexts with significant system complexity also takes from its user an opportunity to solve the engineering problems themselves, which is a challenge that we enjoy and believe helps us to better understand our own work.

Perhaps in the future, we will submit an experience report or write a workshop paper on the *self* and *social* barriers that vibe-coding introduced into our accessibility-centered prototyping work.

Part IV

Interaction: Exploring New Possibilities for Blind Data Science

Chapter 7

Cross-perception: Rethinking Input Design Towards Blind Analytical Interaction

7.1 Preface

Interactive visualization research has produced a rich repertoire of interaction techniques (brushing, linking, and cross-filtering chief among them) that accelerate exploratory data analysis for sighted users. And accessible visualization research has made substantial progress on making data perceivable and navigable for blind users.

Yet what remains largely unaddressed is the *analytical* tier of blind interaction: the capacity to actively filter, conjecture, and test hypotheses through direct data manipulation. In existing systems that accomplish interactive, analytical tasks, users predominantly query text or sequentially navigate across discrete units of information, paradigms which focus the user on one piece of information at a time. These experiences limit a user’s ability to rapidly explore and interrogate multiple dimensions within a dataset simultaneously.

To target this analytical gap, we introduce *cross-perception*, an interaction technique that adapts cross-filtering for non-visual, tactile interaction. *Cross-perception* enables coordinated tactile brushing over one data space to produce perceptible computational output in a separate, linked space, without requiring serial navigation between them.

We instantiate *cross-perception* in the *cross-feelter*, a prototype hardware device with dual motorized faders, and evaluate it against screen reader methods in a within-subjects study with 15 blind participants, 7 of whom have professional data expertise. *Cross-feelter* substantially improves exploration speed (+90%), query generation (+188% computational, +54% spoken), enjoyment, and reduces anxiety (particularly for participants without prior data expertise). We discuss implications for interaction design in accessible visualization and propose *cross-perception* as a framework for a broader class of non-visual, analytically-oriented data interaction techniques.

7.1.1 Context

This project has been a pet project, tandem to the main thread of the others in this thesis. It stands out a bit: centered exclusively on *blind* practitioners and people working with data, as opposed to sighted practitioners who are making things for people with disabilities.

We unpack this in the chapter, but our prior co-design work with a blind colleague eventually led to this project. And if anything, the greatest takeaway from this chapter is that, in true scientific fashion, simply observing something with a fresh perspective and conjecturing on possibilities can lead to fantastic places.

This project was also borne out of a tiny resistance to the then-rapid rise of large-language models. All of a sudden, certain problems that were very hard in accessible visualization (such as generating descriptions for visualizations with dynamic and changing data) were orders of

magnitude less difficult. While we argue (outside of the scope of this thesis) that these new technologies exacerbate existing issues of bias, trust, and validation for people who are blind [43], generally speaking they are undeniably revolutionary.

And yet, I (Frank) had often spoken about my core motivation for leaving a well-paying, highly-supported, staff-level engineering role to start a PhD in HCI: because to me, the greatest challenge was figuring out how to motivate people to think more deeply about their work. At Visa, we automated much of our accessibility by default. And when we realized designers and developers rarely deviated from the defaults, we added validation and checks in the console that would log or throw errors to motivate them to stop and think about how they were using our charts. This ultimately made me want to study human behavior and how tools can shape practitioners towards better outcomes.

But to me, I had always assumed that *practitioners would do the work*. Automation takes away the thinking! And my mission was to elicit deeper consideration and better design. I had assumed that humans *want* knowledge and expertise. But agentic tooling and LLMs began to challenge those assumptions. Perhaps *convenience* is what people really want. People seem to now call on LLMs for whatever labor they value the least. And accessibility is now a common project for LLM-based intervention.

So *cross-perception* was motivated by looking where language models have little influence: in novel hardware development. As of writing, it is still required for a human to solder boards, design circuitry, model, 3D print, and assemble hardware prototypes. So this project was a way to look in a direction that few others are paying any attention. While there seems to be a gold rush to move to higher and higher levels of software abstraction, this project is a return to lower levels of hardware components, tactors, and interaction design.

This project was also immensely difficult to recruit for. On the one hand, I had several blind data practitioners, researchers, and scientists who were colleagues, friends, and mentors supporting me. But few of them had the time or bandwidth to participate in our study. Recruiting locally was quite hard, so we decided to recruit non-data practitioners locally and, if we managed to get lucky, pull off running our studies “in the location of our participant’s choosing, should they consent to assume risk” (roughly paraphrased, as our IRB allowed). This verbiage meant that our participants could opt not to participate in our lab space on campus but rather choose their own location (with certain conditions). This ultimately enabled us to recruit blind data practitioners at conferences (ASSETS and CSUN, in particular), which made the study for this project stretch nearly a year and a half (but helped us to see the finish line).

But it was worth it. Our participants were fantastic. And not to spoil the chapter’s findings, but we discovered that hardware and input design might be the key to a whole new spectrum of studies where data interaction is framed as a fun and generative activity, and not just something dry, boring, or professionalized as data work often is, in practice.

This chapter was adapted from our paper, soon to be submitted to TVCG:

F. Elavsky, Y. Li, J. Jang, L. Nadolskis, P. Carrington, and D. Moritz, ‘*Cross-Perception: Rethinking Input Design Towards Blind Analytical Interaction*’, *IEEE Transactions on Visualization and Computer Graphics*, 2027.

7.2 Overview

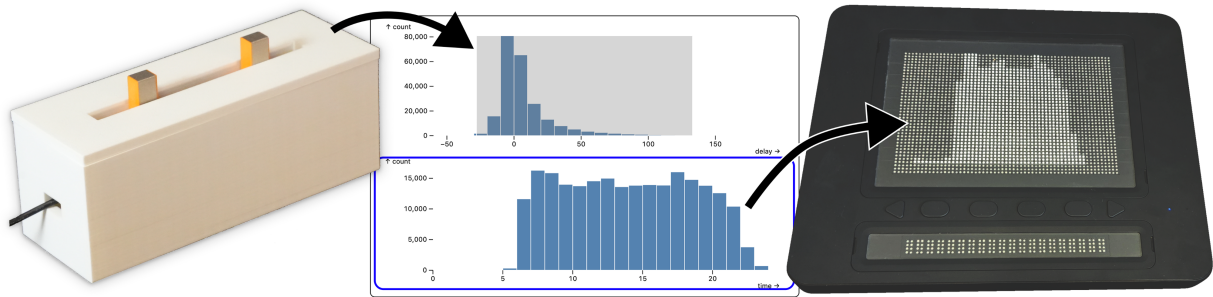


Figure 7.1: Interaction and perception in one space while being able to perceive output in a separate space is the cornerstone of cross-filtering. Our prototype *cross-feelter* (left) can manipulate a visual cross-filter on one visualization (middle) in order to produce output in a separate visualization, as a tactile graphic (right).

The visualization community has spent decades formalizing interaction techniques for sighted users: brushing scatterplots [10], linking coordinated views [25], and cross-filtering aggregated data to rapidly surface relationships across dimensions [1, 194]. These techniques are not only conveniences. Empirical work has shown that fast, direct manipulation of data fundamentally changes the speed and character of hypothesis generation, and that even small increases in interaction latency measurably reduce the quality of exploratory analysis [53, 114].

Accessible visualization research has made substantial and genuine progress in recent years. Blind users can access data through rich text descriptions [103, 116], sonification [90, 108, 215], and tactile representations [27, 89, 125, 126, 171, 179]. Increasingly, this work extends beyond rendering to *interaction*: screen readers support structured traversal of chart elements and their relationships at varying levels of granularity [45, 130, 213], and multimodal systems pair conversational agents with tactile displays to support rich data exploration [146].

What existing work addresses is, broadly, the question: *what is here?* A blind user can navigate a data representation to understand its structure, locate specific values, extract summary statistics, ask descriptive questions, and build a mental model of what the data contains. This is *access-oriented* interaction, and current tools support it with increasing sophistication.

What remains largely absent from accessible visualization is a different kind of interaction entirely: the *analysis-oriented* kind, in which a user not only reads a dataset but actively operates on it: filtering subsets, conjecturing relationships, testing hypotheses, and generating new

knowledge through iterative manipulation. These are the fundamental activities of interactive data science, and they have a different character from access-oriented interaction. Questions shift from “*what is here?*” to “*what happens to Y when I constrain X?*” According to taxonomies of analytical tasks, existing work has largely not addressed “characterizing distributions” and “correlating” tasks, which are fundamental to hypothesis-construction [5].

We propose *cross-perception*, an interaction technique that adapts cross-filtering for non-visual contexts while preserving the simultaneity that makes it analytically useful. Cross-perception is defined by two properties. First, *coordinated tactile brushing*: the user manipulates a persistent tactile input in one interaction space, selecting a range across a data dimension. Second, *linked perceptual output*: the computational result of that manipulation is immediately perceivable in a separate space, without requiring serial navigation between them. Together, these properties support not only access to data but active analytical engagement with it: the capacity to ask iterative questions through direct manipulation.

To evaluate cross-perception and explore its design space, we built the *cross-feelter*: a prototype device with two motorized linear faders that maintain persistent, absolute tactile position for brush handles, paired with a refreshable braille display for tactile output. We conducted a within-subjects study with 15 blind participants, comparing the cross-feelter against a screen reader baseline on structured and open-ended data exploration tasks.

In our work, we address four research questions:

- R1 (Exploratory):** How might we formalize a tactile interaction approach that enables coordinated, non-visual interaction and perception across multiple spaces?
- R2 (Quantitative: Objective):** In what measurable ways can we observe improvements to blind data interaction using our approach, in terms of speed, accuracy, precision, and quantity of data queries?
- R3 (Quantitative: Subjective):** What are the self-reported effects on stress, cognitive load, anxiety, and enjoyment when using our approach?
- R4 (Qualitative):** In what ways do blind data practitioners imagine extending our approach and its instantiating hardware to new contexts?

This paper makes three contributions. First, we formalize *cross-perception* as an interaction technique for non-visual *analytical* data interaction; one that equips blind users to actively filter, conjecture, and test hypotheses through direct tactile manipulation, not only to access and navigate data (Section 7.4). Second, we provide an empirical evaluation demonstrating that cross-perception yields substantial improvements in exploration speed, query generation, and user experience relative to screen reader interaction, while maintaining comparable accuracy and precision (Section 7.5–7.7). Third, we argue that accessible visualization has made significant progress at the access-oriented tier of interaction, and that our field’s frontier has shifted: the work now should be to give blind users *analytical agency*: the ability to interrogate data, not only to read it (Section 7.8).

7.3 Related Work

This paper sits at the intersection of two bodies of work: research on cross-filtering and analytical interaction in data visualization, and research on making data accessible to blind users. We review each in turn, then characterize the specific gap that cross-perception addresses.

7.3.1 Cross-Filtering and Interactive Visual Analysis

A note on terminology. *Brushing* is the direct manipulation gesture of selecting a contiguous visual range across a data dimension [10]. *Linking* propagates that selection across coordinated views [25]. *Cross-filtering* applies the linked operation as a filter on shared underlying data, constraining what is visible across views rather than merely annotating selections [1, 194].

What makes cross-filtering analytically valuable is not that users can manipulate the interface, but that the gesture and its computational consequence are perceived simultaneously. Even 500ms of additional latency measurably reduces the rate at which users generate observations and formulate hypotheses [114], and providing partial results incrementally reduces but does not eliminate this cost [53]. Temporal coupling between input and output is the mechanism by which direct manipulation supports hypothesis generation, not an implementation detail [81]. Recent work on scalable cross-filtering has continued to optimize this coupling through improved architectures [82], including pre-computed aggregates [134]. This distinction, between analytical interaction that constructs new information through manipulation and navigational interaction that locates information already present, motivates the core design challenge of cross-perception.

Previous work establishes that cross-filtering supports *analytical* interaction, in which the user constructs and tests hypotheses through rapid, iterative, direct manipulation, rather than navigating toward information that is already present. The value of this analytical interaction is that the user, not a system, gains new insights, questions, and understanding [169].

7.3.2 Non-Visual Data Representations

Accessible visualization has produced many systems for making data perceivable to blind users across text, audio, and tactile modalities [47, 64, 103, 108, 116, 143, 179]. We position cross-perception as complementing this work: many rendering and representation problems have made significant advancements in recent years; but analytical interaction problems, comparatively, have not.

Sonification. Sonification, mapping data variables to auditory properties such as pitch or timbre, is a distinct and sophisticated field with its own tooling, theory, and evaluation methodologies [84]. Recent work has formalized sonification as a first-class rendering approach with grammar-level expressiveness, enabling mappings comparable in richness to visual encodings [108]. Some work has explored data-centric approaches that treat sonification and visualization as parallel co-equal representations [215].

A note on auditory perception is warranted: the human auditory system is capable of parsing concurrent streams and segregating them into distinct perceptual objects [22]. However, the relevant constraint for blind and low vision users working with data is more specific: in task-driven contexts, synthesized speech from a screen reader competes with data-representing audio

in the same channel, increasing cognitive load of the user and providing complex challenges for authors when designing speech audio and sonification interfaces [60, 108, 186]. Dual-task and multi-task operations become difficult or impossible, which is a design challenge for cross-filtering specifically, where the filter state and its output must be perceived simultaneously.

Tactile and haptic output. Research on tactile data representation spans several decades, from foundational work on tactual perception [158] to recent studies formalizing tactile alternatives to common chart types [47, 143]. More recent work has explored simultaneous tactile-and-audio output [27], slide-tone and tilt-tone haptic encodings for shape characteristics of graphs [49], comparative studies on tactile vs audio output [32], and design guidelines for multimodal touchscreen-based graphics that inform both resolution and tactile landmark placement [64].

7.3.3 Access-Oriented Data Interaction

In parallel with rendering advances, accessible visualization has developed increasingly sophisticated interaction techniques for blind users, such as serial and multi-directional navigation across data structures using a screen reader [45, 213]. However, screen reader traversal is sequential and discrete, which impacts a blind user’s ability to extract information from visualizations, impacting accuracy by 61% and costing 210% more time doing so [163]. Empirical work has engaged “questions” that everyday blind folks have of data representations [109], and most recently, conversational agents have been paired with refreshable tactile displays to support question-based exploration combining voice input with tactile output [146].

These systems share a common character: they support *access-oriented interaction*, attempting to bridge the gaps that exist between sighted and blind access to information. However, the activities that remain unsupported require constructing information through manipulation rather than locating it through navigation; the work of professional data science. In analysis-oriented interaction, the question is not merely lookup or query tasks such as *what is the value at this position* but along the lines of *what is the joint distribution within this subrange*, an answer that does not exist until a filter is applied (and a question that may not even exist unless within a system that supports direct interaction and iteration with the data).

7.3.4 Tactile and Haptic Input

Tactile and haptic input for data interaction has been explored in immersive contexts (not related to accessibility or blind interaction). For example, Cordeil and colleagues’ *embodied axes* allow manipulation of data axes as physical objects in augmented reality [34], and the *MADE-Axis* extends this to motorized dual-axis faders for AR/VR data environments, demonstrating a cross-filtering use case [170]. Both leverage tactile input and motorized output, though both presuppose sighted users in visual contexts.

Within accessibility-specific work, a touchpad-based brush interface showed 1D filter selection feasibility but provides no queryable positional state when the user’s hand is lifted [166]. And a non-motorized slider within an accessible smartphone interface shows the principle of persistent tactile state [210] but cannot be programmatically positioned, which cross-filtering requires for two-way synchronization.

The gap that remains is specific: persistent physical state and simultaneity of input and output perception are both necessary for analytical interaction, and no existing accessible system combines them.

Cross-perception is our attempt to address this gap. Our design is informed by the principles reviewed above: the temporal coupling of cross-filtering, the persistent-state affordances of motorized tactile input, and the insights from prior work on non-visual data interaction regarding the importance of simultaneity, spatial memory, and multi-channel perception. The following section formalizes these into a set of design goals for the interaction technique.

7.4 Formalizing Cross-Perception

Cross-perception is a tactile interaction technique defined by two properties. First, *coordinated tactile brushing*: the user manipulates a persistent tactile input in one interaction space, selecting a range across a data dimension. Second, *linked perceptual output*: the computational result of that manipulation is immediately perceivable in a separate space, without requiring serial navigation between them. These properties directly instantiate the simultaneity requirement established in [Section 7.3](#): brush state and its consequence are perceived together, in the same moment, supporting the rapid inference cycles of analysis-oriented interaction.

In this section, we formalize cross-perception by deriving six design goals from the principles reviewed in the related work. We begin with a design vignette with our co-designer and co-author Nadolskis, that illustrates how the same cognitive mechanisms that underpin our goals appear in naturalistic blind interaction with physical documents.

7.4.1 A Design Vignette: Fuzzy Search and Dual-Task Comparison in Naturalistic Blind Reading

The following vignette, shared with permission by our blind co-designer and co-author Lucas Nadolskis, illustrates two tactile interaction strategies that correspond directly to our design requirements formalized below. We present this vignette as both grounded in the cognitive science and visualization literature reviewed in [Section 7.3](#) as well as evidence that the principles we derive from that literature describe real, expert-level behavior in the wild.

Nadolskis, a blind neuroscience engineer, was reading an embossed academic paper and explaining a particular equation to the primary author as the equation recurred and evolved across multiple pages. The embossed format provides high-resolution tactile rendering at up to 100 dots per inch, and Nadolskis interacted with it using two strategies we have named *fuzzy tactile search* and *dual-task comparison*.

Fuzzy tactile search is a spatial indexing strategy. Nadolskis picked up the stack of pages and thumbed to an approximate location based on tactile position in the stack, touched near the top of the page, immediately recognized from the local texture pattern that he was one page off, flipped to the next page, and navigated directly to the spatial region where the equation began. He then repeated this process to locate the equation’s recurrence later in the paper. The strategy relies on two cognitive functions reviewed in [Section 7.3](#): spatial memory, encoding the

approximate location of content within a document’s physical structure, [24, 33, 37] and *object-location memory*, the ability to retain and retrieve the position of objects without continuous visual or tactile contact [58, 113, 115]. Both functions require that objects *have* a stable spatial position to encode: they cannot operate on a system where position is reassigned or meaningless.

Dual-task comparison is a simultaneous spatial reasoning strategy. Having located the same equation in two places, Nadolskis placed one hand on each and skimmed both simultaneously, stopping when his hands detected the point of divergence between them. He explained the evolution of the equation by reference to that divergence. This strategy demonstrates exactly the interaction character that cross-filtering supports for sighted users: two data regions perceived in parallel, with relational structure emerging from simultaneous contact rather than sequential reading. It also demonstrates the additional requirement of *simultaneous multi-location tactile interaction*: a user resting a pinky on a known reference point while exploring relative to it with another finger, a pattern documented in prior empirical work on tactile interactive systems [64].

Together, these strategies illustrate a key observation: skilled blind readers of physical documents are not limited to sequential access. The constraint is architectural: the medium must maintain a persistent, meaningful spatial state so that object-location memory can operate, and it must allow simultaneous multi-point contact so that comparison can occur. Our design goals operationalize these requirements for computational, analytical interaction with tactile interfaces.

7.4.2 Design Goals

We derive six design goals from the principles established in [Section 7.3](#) and illustrated by the vignette above.

G1: Maintain and reflect system state through tactile location. Object-location memory requires objects with stable, queryable positions [58, 115]. Tactile input elements must therefore have absolute, persistent positional state: the knob must remain where the user or system last placed it and be readable by touch without performing an action. This also enables motorized two-way synchronization, where the system sets knob position as output and the user sets it as input.

G2: Allow for multiple simultaneous channels of perception. Synthesized speech competes with data-representing audio under cognitive load [60]. Cross-perception routes data manipulation to a tactile input channel, eliminating this competition. Audio output via ARIA-live announcements remains available for precise positional feedback but is kept serial and non-competing.

G3: Correspond tactile-to-computational functional semantics. Interface elements should have names and behaviors that describe what they do [44, 190]. The visual cross-filtering metaphor involves dragging handles along a spatial dimension; cross-perception preserves this by mapping a physical rail sliding to setting a filter boundary, reducing translation between user action and analytical action.

G4: Provide linked interaction across multiple data dimensions. Cross-filtering requires that a filter on one dimension immediately propagates to constrain coordinated views of others that share the same underlying dataset [1, 25]. The filter state set by tactile input must immediately determine the output displayed in linked views; without this, the technique only accomplishes single-view selection.

G5: Support fast, dynamic, and reversible interaction. The analytical value of cross-filtering depends on rapid hypothesis iteration [53, 114]. Filter adjustments must be continuous and incremental, system response must be low-latency, and any state must be immediately reversible. These motivate motorized faders with 1024 units of precision and a pre-computed data cube backend [82].

G6: Facilitate user-driven exploration. Cross-filtering is a direct manipulation paradigm in which the user, not an algorithm, controls the analytical flow [1, 169]. The user’s physical manipulation of the filter handles is the direct and sole determinant of what the tactile output displays, with no intermediary statistical, language model, or non-deterministic interpretation.

7.5 Prototype: The Cross-Feelter

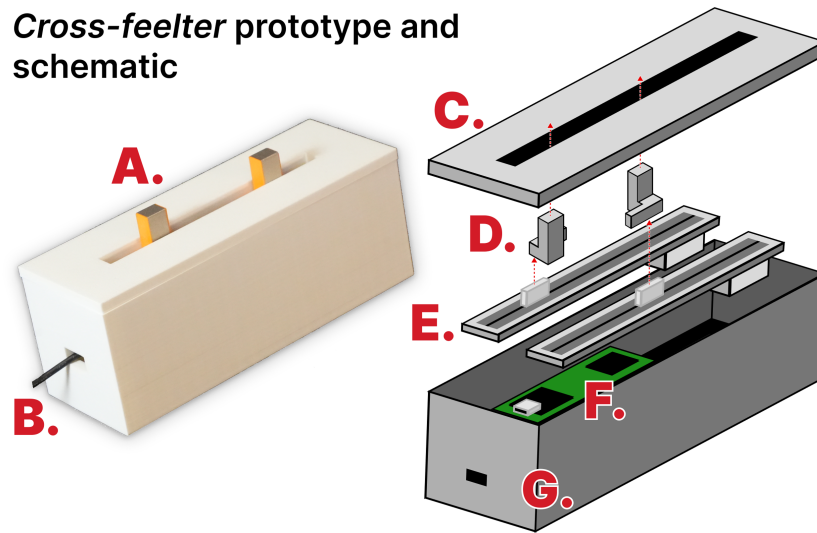


Figure 7.2: A. Our prototype *cross-feelter*. B. USB output (to computer). C. 3D printed rail cover. D. 3D printed knobs that consolidate rails into one. E. 2 motorized linear potentiometers. F. Arduino board. G. 3D printed casing.

We instantiate cross-perception in a hardware prototype we call the *cross-feelter*: a device with two motorized linear faders that provide persistent, absolute tactile input state for brush handles, paired with a refreshable braille display for coordinated tactile output. All hardware schematics, firmware, analysis, and interface code are [archived on OSF](#); 3D-printable STL files for the housing components are available in our [supplemental materials on OSF](#). The total cost of hardware components is \$30–50 USD, which we designed intentionally to enable replication by the DIY-AT and maker communities [94, 171].

7.5.1 Interaction Flow

We begin with an idealized walkthrough of what a blind analyst actually does when using the *cross-feelter* to perform a cross-filtering task. This is the ground truth from which the hardware

design choices derive; we describe the hardware in [Section 7.5.2](#) with reference to the steps below.

Consider an analyst exploring a dataset of local rent prices and transit access across city neighborhoods. Three linked histograms are rendered in sequence on the refreshable braille display. The analyst wants to ask: *do neighborhoods with the highest rents also have better transit access?* At startup, the left knob sits at 0% and the right at 100%, representing an unfiltered view of the full distribution.

Step 1: Feel and filter. The analyst feels the starting state of the tactile data representations on the refreshable braille display, navigating between each chart and hearing supplemental screen reader descriptions alongside the touchable shapes formed from the pins. The analyst moves their hand or hands to the feeler device and slides the left knob rightward, narrowing the filter from below. As they move, an ARIA-live announcement reports the current percentage and knob name via the screen reader, providing precise serial feedback for positioning ([G2](#)). The analyst moves the right knob slightly leftward, hearing another announcement. The two knobs now define a range, say 75% to 90% of the rent distribution.

Step 2: Perceive the consequence. During filtering, the analytical environment updates linked histograms immediately on each knob movement. The analyst navigates the braille display to the transit histogram without touching the knobs. Because the knobs are motorized and maintain position, the filter state is preserved and readable by touch at any time ([G1](#)). They feel the updated distribution and compare it to the shape they perceived in the unfiltered state.

Step 3: Iterate. The analyst slides the left knob further right, narrowing to the top 10% of rents. The display updates. They feel the new shape. The pattern sharpens or changes, and they iterate again, building toward a hypothesis in the same cycle that sighted analysts perform visually. Each knob adjustment is a computational query: a gesture that constructs a question and immediately exposes its answer.

7.5.2 Hardware Design

The cross-feelter uses two motorized linear potentiometers (commonly called motorized faders in sound design hardware) mounted side by side in a 3D-printed housing (see [Figure 7.2](#)). Each fader provides a physical knob on a bounded rail.

Motorized, absolute-position faders ([G1](#), [G5](#)). We chose rail-based faders over alternatives (touchpad, rotary encoder, joystick) for two reasons grounded in the [G1](#) requirement. First, the fader is *absolute*: the knob’s physical position on the rail directly corresponds to the filter boundary value, with no drift or accumulation. Second, the fader is *motorized*: the system can programmatically set knob position as well as read it. This bidirectionality is essential for cross-filtering: the system must be able to reset filter state on condition changes, counterbalance, and initialization, not only receive input. Prior work on brush-based accessible interfaces [[166](#)] used a touchpad, which provides no queryable positional state when the hand is lifted; prior work using non-motorized sliders [[210](#)] cannot be programmatically positioned. The motorized fader solves both problems.

Each rail terminates in a tactile boundary, a strong edge inset 4mm from the housing wall per tactile design guidelines [[64](#)], so the analyst can feel both the knob position *and* its position relative to each end of the scale ([G1](#)). The API reads and writes fader position every 5ms,

providing up to 1024 units of precision, which maps well to histogram bin resolutions in practical cross-filtering datasets (G6).

Merged-rail housing (G3). We designed two housing configurations: one exposing the rails independently, and one using angled knobs to place both rails along a single track (Figure 7.2, part D). The merged-rail design, used for all data tasks in the study, ensures the left knob remains to the left of the right knob at all times (they cannot cross over). This preserves the functional semantics of a spatial filter range: left boundary is always left, right boundary is always right, matching the computational filter and the visual histogram (G3). The independent-rail configuration was used for the video interaction training task (Section 7.6.4), where the two rails served different functions (volume and position).

Our dual-fader design is similar in form to motorized fader-based systems proposed for axis interaction in AR/VR data environments [34, 170]; however, those systems presuppose sighted users operating in 3D visual contexts. Our design differs in three ways: (a) the housing constrains the two faders to a single interaction dimension to support filter semantics; (b) the motorized positioning enables two-way synchronization with the data interface; and (c) the entire system is designed for eyes-free operation as the primary use mode.

ARIA-live auditory feedback (G2). When a knob is moved, the interface emits an ARIA-live announcement to the screen reader reporting the new filter percentage. ARIA-live is set to `polite` mode, meaning announcements queue behind any active screen reader speech rather than interrupting it. This provides precise, serial auditory feedback for filter positioning without competing with other auditory streams in the environment (G2). Tactile feedback (the knob’s physical position) and auditory feedback (the ARIA announcement) thus serve complementary roles: tactile provides continuous, queryable approximate state; auditory provides precise on-demand readout triggered by movement.

Electronics and firmware. Figure 7.3 provides the full wiring of the device, sufficient for a reader to replicate it. Each motorized fader (a Behringer MF100T: a 100mm fader with a 10k Ω linear resistive track, a capacitive touch-sense track, and a small DC motor) exposes six terminals: the two fixed ends of the resistive track, the position wiper, the touch line, and the two motor terminals. A Teensy 4.0 microcontroller, running Arduino-style firmware, reads and drives both faders. The track ends connect across the Teensy’s 3.3V supply and ground, so each wiper reports absolute knob position as a 10-bit analog value (0 to 1023) on pin A0 (rail 0) or A2 (rail 1). The touch lines connect to pins A1 and A3. The firmware samples them every 5ms and counts readings at the extremes of the analog range (below 50 or above 973) over a rolling ten-sample window; four or more extreme samples within the window register as a touch. This works because a finger on the conductive knob couples body capacitance and ambient electrical noise into the otherwise stable touch track, swinging successive samples toward the rails. Touch state gates the motors: any pending motor target is cancelled the moment a knob is touched, so the device never fights the analyst’s hand (G6).

The motors are driven through a DRV8833 dual H-bridge motor driver: Teensy digital pins D2 and D3 control rail 0 (up and down respectively), pins D4 and D5 control rail 1, and the driver’s output pairs connect to the two motors. The driver’s motor supply takes 5V from the Teensy’s USB power pad, so the entire device is powered and controlled over a single USB cable. The firmware runs a 1ms control loop that drives an actively commanded motor toward its target until the wiper reads within a small tolerance band, and a 50ms reporting loop that emits

Cross-feelter wiring schematic

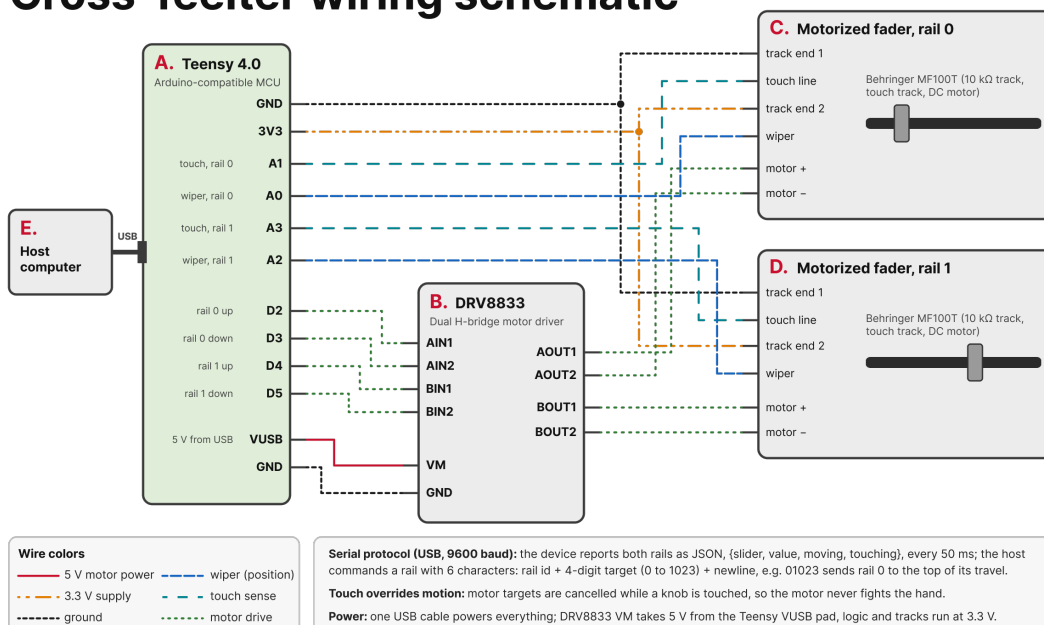


Figure 7.3: Wiring schematic of the cross-feelter. A. Teensy 4.0 microcontroller. B. DRV8833 dual H-bridge motor driver. C. Motorized fader, rail 0. D. Motorized fader, rail 1. E. Host computer, connected over USB. Wire colors distinguish 5V motor power, the 3.3V supply, ground, wiper position lines, touch-sense lines, and motor drive lines.

the state of both rails (position, moving, and touching) as a compact JSON message over a 9600 baud serial connection. Host-to-device commands are six characters: a rail identifier, a four-digit zero-padded target position, and a newline. For example, 01023 sends rail 0 to the top of its travel.

7.5.3 Analytical Environment

The cross-feelter operates within a web-based analytical environment purpose-built for this study (Figure 7.4). The environment presents three linked histograms derived from different variables in a shared dataset. Each histogram is rendered both visually (for debugging and sighted collaborators) and as a 60×40 pin array sent to the Dot Pad refreshable braille display.

Cross-filtering backend (G4, G5). We use Mosaic [82] to manage the cross-filtering logic. Mosaic pre-aggregates filter results across binned input values, enabling fast, low-latency responses to filter changes even over larger datasets (G5). It also provides a straightforward API for specifying a brush-filter interaction layer over a histogram dimension, including programmatic control of filter state and undo actions (G5). When the user moves a fader knob, the new filter boundaries are sent to Mosaic, which immediately updates the linked histograms, which are then re-rendered on the braille display. The render cycle from fader movement to updated tactile output is 500–1200ms on the Dot Pad hardware, fast enough to support iterative exploration,

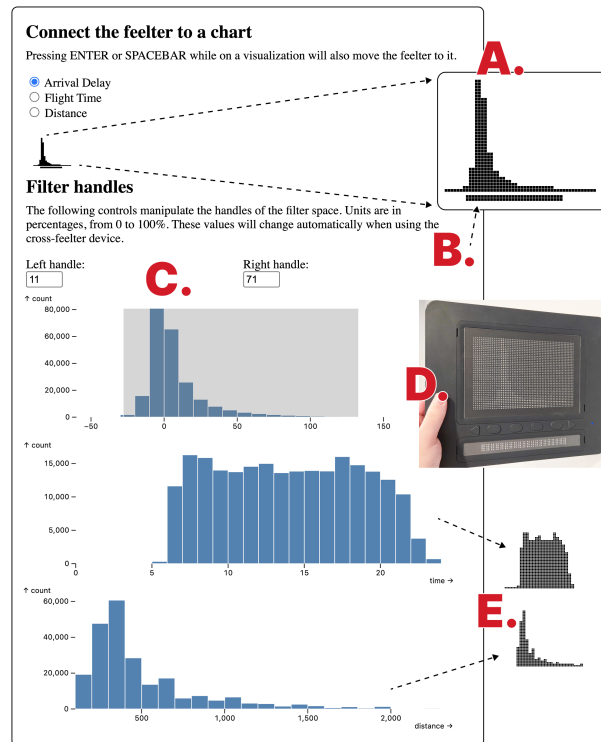


Figure 7.4: Our analytical environment. A. a 1 to 1 translation between visual and tactile that represents a 60x40 pixel-to-pin array. B. A tactile version of the filter location (if on the chart being focused). C. Cross-filtering controls, including text inputs for screen-reader only manipulation. D. Refreshable tactile display output. E. Example renderings of the other 2 charts.

though slower than visual cross-filtering systems and a recognized limitation ([Section 7.10](#)).

Software architecture. The analytical environment runs entirely in the browser. It connects to the cross-feeler through the Web Serial API, filtering on the microcontroller’s USB vendor identifier so that no driver installation is required, and polls the device’s serial reports every 25ms. When a report shows a knob that is both moving and touched, the interface maps the raw fader value through a clamped linear scale onto the pixel range of the corresponding Mosaic brush handle and applies it by synthesizing native pointer events on the handle itself. The fader therefore drives exactly the same interaction pathway as a mouse drag on the visual brush, which keeps the tactile and visual input methods behaviorally identical by construction. Mosaic propagates the updated selection as a cross-filter, DuckDB-WASM re-aggregates the data in the browser, and all three linked histograms re-render. On each update, the focused histogram is redrawn onto an internal 60×40 canvas, including a low horizontal bar marking the current brush extent, packed into the Dot Pad’s cell-based packet format, and sent over Web Bluetooth; unchanged rows are diffed out so that only modified lines of pins are refreshed. Programmatic motor control flows in the other direction: when the analyst attaches the feeler to a different chart, the interface writes positioning commands over the same serial connection to drive both knobs to the ends of their

rails, resetting the physical filter state to match the new chart’s unfiltered view.

Braille display output (G2). The Dot Pad provides a 60×40 binary pin array (pins raise or lower on command). We render each histogram as a bitmap at this resolution, trading fidelity for speed relative to a high-resolution braille embosser (which takes 3–4 minutes per page). At any time, the braille display shows one chart; the analyst navigates between charts using the screen reader. The visual representation of the target pin array is visible to sighted observers and collaborators in the environment (Figure 7.4, part A), illustrating the resolution gap that future advances in refreshable display technology could close.

Screen reader compatibility. The analytical environment is a standard web interface, accessible via any JAWS, NVDA, or VoiceOver-compatible screen reader. Cross-filter input controls are exposed as labeled range inputs so they can be operated by screen reader alone (providing the baseline condition for our study, Section 7.6). The ARIA-live region for fader announcements is separate from these controls and does not interfere with standard screen reader navigation of the interface.

7.6 Data Exploration Study

We conducted a within-subjects study comparing cross-feelter-based cross-filtering (CF condition) against screen-reader-based cross-filtering (SR condition) across structured and open-ended data exploration tasks. Sessions were 90 minutes, conducted in person, and approved by our IRB.

7.6.1 Study Design

Overview. Our study employed a 2×2 Latin square design, fully counterbalanced across two factors: (1) condition order (CF-first vs. SR-first) and (2) dataset order (flight data vs. neighborhood data), producing four participant groups (Table 7.1).

Table 7.1: 2×2 Latin square study design. CF = cross-feelter condition; SR = screen reader condition.

Group	Condition order	Dataset order
A (n=4)	CF $\rightarrow$ SR	Flight $\rightarrow$ Neighborhood
B (n=4)	SR $\rightarrow$ CF	Flight $\rightarrow$ Neighborhood
C (n=4)	CF $\rightarrow$ SR	Neighborhood $\rightarrow$ Flight
D (n=3)	SR $\rightarrow$ CF	Neighborhood $\rightarrow$ Flight

Within each condition, participants performed two tasks on the same dataset: a 3-minute structured data-relationship task followed by a 10-minute open-ended data exploration. Counterbalancing both orders controlled for dataset familiarity, learning effects, and dataset-specific task difficulty.

Conditions. In the **CF condition**, participants used the cross-feelter (merged-rail housing; Section 7.5.2) to manipulate cross-filter bounds, with the Dot Pad displaying linked histograms. In the **SR condition**, participants used their screen reader to operate labeled text input fields

exposing the same filter controls, with the same Dot Pad display. The analytical environment was identical across conditions; only the input method differed.

Session procedure. Each session followed this order regardless of group:

1. Tech setup, baseline interview, introduction to tactile histograms and cross-filtering, baseline anxiety Likert question (*20 min*)
2. Device familiarization: video interaction task and brief semi-structured interview (*15 min*)
3. First data condition: structured task, open-ended exploration, Likert-scale interview (*20 min*)
4. Second data condition: structured task, open-ended exploration, Likert-scale interview (*20 min*)
5. Closing semi-structured interview (*15 min*)

7.6.2 Participants

We conducted our study with 15 blind participants: 8 without prior professional data expertise and 7 with it (data analysts, scientists, or engineers currently or recently employed in data-related roles). 7 participants identified as male, 8 as female.

Participants without data expertise, plus 2 with it, were recruited locally through community outreach. The remaining 5 participants with data expertise were recruited through professional networks and coordinated at mutual academic conferences, due to difficulty recruiting blind data scientists locally. All participants were screened for eligibility (blind, screen reader proficient, 18 or older, US citizen or permanent resident), informed of risks and voluntary participation, and compensated with a \$50 gift card.

7.6.3 Datasets

Study tasks used two datasets selected through a community data-preference workshop. Pilot sessions revealed that participants were not engaged by our initial generic dataset, so we held a workshop at a local braille and talking book library ($n=21$ community members). Participants voted on datasets from topics including local politics, housing, and community life. The two highest-voted were: (1) rent prices across city neighborhoods, and (2) fish fry locations across city neighborhoods (a dataset reflecting a regional Catholic Lent food tradition observed broadly across denominations). Both are publicly available. A generic flight dataset was included as the second counterbalanced dataset.

Each dataset was configured with three histogram variables from a shared source, with cross-filter tasks designed to target a 75%–100% filter range producing a clear, unambiguous relationship between variables.

7.6.4 Device Familiarization: Video Interaction Task

Rationale. Before data tasks, participants completed a device familiarization activity using the cross-feelter to control a video feed rather than a data chart.

The goal was to build fluency with the motorized fader mechanism (absolute-position input, motorized output, the relationship between physical knob position and system state) without pre-exposing participants to the cross-filtering metaphor, data tasks, or the study datasets. A practice task using data histograms would have constituted condition-specific training, introducing learning effects that could confound the within-subjects comparison. Video scrubbing and volume control are familiar and intuitive interaction goals for all of our users, which served as a way for all of them to understand how to operate the device.

Task. Participants were asked to find the name of the person in the video recording, announced early in the clip. This simple information-retrieval goal required genuine device use while keeping cognitive demand low before the main study.

7.6.5 Introduction to Tactile Histograms and Cross-Filtering

At session start, after consent and the baseline interview, we provided a scripted introduction to tactile histograms and cross-filtering. We explained histogram structure, introduced the concept of linked histograms sharing a source dataset, and demonstrated cross-filtering with three histograms from a fruit example dataset, using two sheets of paper held over one histogram to physically simulate a spatial filter.

We framed filtering explicitly as an analytical activity: “filtering is a way to ask questions more than to find answers, to look for possible relationships that might exist in the data.” Two follow-up questions gauged comprehension and interest before proceeding.

7.6.6 Data Tasks

7.6.6.1 Structured data-relationship task

Participants cross-filtered the dataset and answered a three-option multiple-choice question within three minutes. The question and options were available via screen reader in the interface; the facilitator restated them only if asked and did not otherwise intervene.

For the flight dataset: “*What is the relationship between filtering for longer flight distances and the arrival time of flights?*” (Correct: longer flights tend to arrive earlier compared to all flights.)

For the neighborhood dataset: “*What is the relationship between filtering for a higher number of fish fry locations and the rent price of a neighborhood?*” (Correct: neighborhoods with more fish fry locations tend to have higher rent prices.)

We measured task completion rate (binary), time to completion (seconds), filter placement precision (distance of handles from target positions of 75% and 100%, scored in $\pm 4\%$ increments awarding 0.5/0.3/0.1 points per handle tier), and answer accuracy (binary).

7.6.6.2 Open-ended data exploration

Participants then performed a 10-minute open-ended exploration of the same dataset using a prompted, concurrent think-aloud method [4]. Before beginning, we framed the activity: “the

purpose is not to *answer* questions but to *come up with* questions. This is how many data scientists explore unfamiliar datasets, by looking for possible relationships.”

We measured: (1) *computational queries*: any filter manipulation followed by braille display inspection of one or more linked charts (filter changes without output inspection did not count); and (2) *spoken queries*: any data-related question voiced aloud, regardless of formality or answerability via cross-filtering.

7.6.7 Measures and Analysis

Quantitative analysis. Objective measures (completion rate, time, precision, accuracy, query counts) and subjective measures (5-point Likert scales for stress, cognitive load, future-task anxiety, and enjoyment) were collected per condition. A pre-study baseline anxiety item provided a pre-condition reference.

Analysis was conducted in Python (code and de-identified data in supplemental materials). Objective continuous measures were compared using paired *t*-tests. Likert-scale differences were evaluated using Wilcoxon Signed-Rank tests given the ordinal nature of the data. We report means, standard deviations, test statistics, and *p*-values; group-level results (data experts vs. non-experts) are reported separately where they diverge.

Qualitative analysis. Qualitative data comprised session audio/video recordings, facilitator notes, and closing semi-structured interview transcripts. The interview covered three topics: overall experience and condition comparison, interface and environment feedback, and future application ideas.

Analysis proceeded in two stages. In the first stage, two members of the research team independently reviewed all recordings, notes, and transcripts, generating initial codes at the utterance level. Codes captured themes across four domains: device usability (physical interaction, feedback quality, learning curve); condition comparison (speed, cognitive load, perceived control); analytical engagement (question generation, hypothesis articulation, confidence with data); and future use speculation. The two coders then resolved all disagreements through discussion, reaching consensus before proceeding.

In the second stage, codes were organized using affinity diagramming [78], conducted collaboratively by members of the research team working from the reconciled code set. Resulting themes inform the qualitative findings in [Section 7.7](#) as well as our future use cases in [Section 7.9](#).

7.7 Results

Our analysis compared 15 blind participants performing cross-filtering tasks under two conditions: using the cross-feelter (CF) versus using a screen reader alone (SR). We report objective and subjective measures in turn. Full analysis code and de-identified data are available in our supplemental materials.

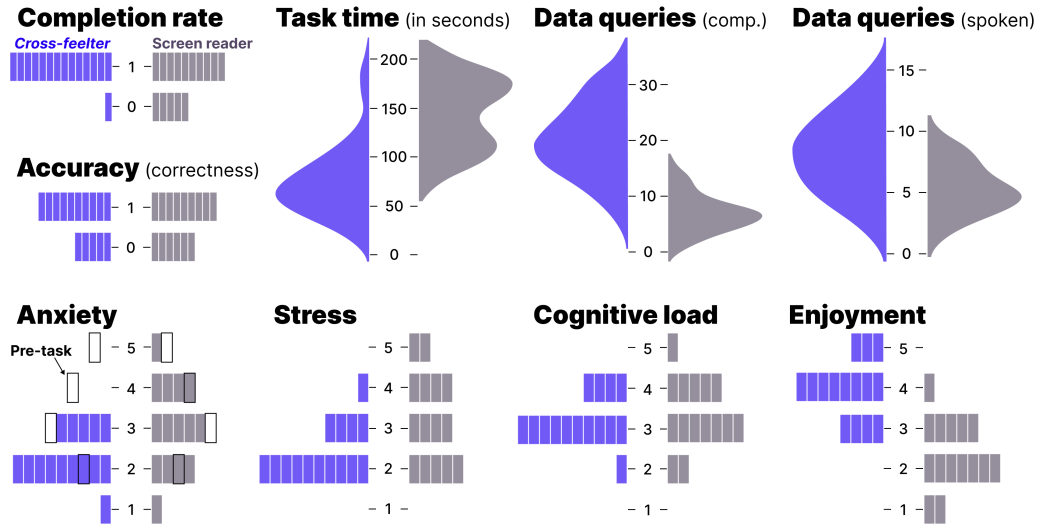


Figure 7.5: Results. Completion rate and accuracy are counts. Task time, data queries (computational), and data queries (spoken) are objective, observational measures. Anxiety, stress, cognitive load, and enjoyment are Likert-scale results from 1 (“very little”) to 5 (“quite a lot”).

7.7.1 Objective Results

7.7.1.1 Substantially improved task completion rate and speed

For the structured data-relationship task, participants achieved a significantly higher completion rate with CF (mean = 0.93, SD = 0.26) compared to SR (mean = 0.60, SD = 0.51; paired t -test: $t = 2.646$, $p = 0.019$). Task completion time was significantly lower in the CF condition (mean = 75.53s, SD = 36.14) than in the SR condition (mean = 143.93s, SD = 34.65; $t = -6.876$, $p < 0.001$), a 90% improvement in speed.

7.7.1.2 Comparable accuracy and precision

Accuracy (correct multiple-choice answer) was higher in the CF condition (mean = 0.67, SD = 0.49) than SR (mean = 0.53, SD = 0.52), but this difference was not statistically significant ($t = 1.468$, $p = 0.164$). Notably, when filtering for only completed tasks, SR users showed marginally higher accuracy (SR: mean = 0.78; CF: mean = 0.71), though again non-significant ($t = 0.555$, $p = 0.594$). We interpret this in [Section 7.8.2](#).

Filter placement precision (scored by proximity to target handle positions of 75% and 100%) was marginally higher in CF (mean = 0.66, SD = 0.37) than SR (mean = 0.59, SD = 0.34), with no significant difference ($t = 0.617$, $p = 0.547$).

7.7.1.3 Substantially increased quantity of data queries

During the open-ended exploration task, CF elicited substantially more computational queries (mean = 20.13, SD = 6.45) than SR (mean = 7.00, SD = 3.18; $t = 9.567$, $p < 0.001$), a 188%

increase. Spoken queries were also significantly higher in CF (mean = 8.20, SD = 3.14) than SR (mean = 5.33, SD = 1.99; $t = 3.765$, $p = 0.002$), a 54% increase.

7.7.2 Subjective Results

Subjective measures varied between participants with and without professional data expertise; we report group-level differences where they diverge meaningfully.

7.7.2.1 Reduced stress

Stress ratings were significantly lower in the CF condition (mean = 2.40, median = 2.0) than SR (mean = 3.20, median = 3.0; Wilcoxon $W = 4.000$, $p = 0.013$). The effect was driven primarily by participants without data expertise (CF: mean = 2.62, SR: mean = 3.75; $W = 2.500$, $p = 0.047$). Data experts reported similarly low stress across both conditions (CF: mean = 2.14, SR: mean = 2.57; $W = 0.000$, $p = 0.083$).

7.7.2.2 Comparable cognitive load

Cognitive load showed an observed difference in means (CF: mean = 3.13, SR: mean = 3.47) but was just outside statistical significance ($W = 4.000$, $p = 0.059$). We interpret both conditions as producing comparable cognitive load during task performance.

7.7.2.3 Lower future-task anxiety

Participants in the CF condition reported significantly lower anxiety about performing a future timed data task using the same tools (CF: mean = 2.27, SR: mean = 3.00; $W = 10.000$, $p = 0.017$). CF anxiety was also significantly lower than participants' baseline anxiety measured before any tasks ($W = 5.000$, $p = 0.003$), while SR anxiety was not ($W = 8.000$, $p = 0.132$), suggesting the cross-feelter reduced anticipatory anxiety relative to participants' pre-study state, while the screen reader did not.

7.7.2.4 Substantially improved enjoyment

Enjoyment was substantially higher in CF (mean = 3.93, median = 4.0) than SR (mean = 2.33, median = 2.0; $W = 0.000$, $p = 0.001$). This effect was strongest for participants without data expertise (CF: mean = 3.875, SR: mean = 2.000; $W = 0.000$, $p = 0.015$). One participant's unprompted remark captures the qualitative texture of this finding: "*Sighted people get to have this much fun when they work with data?*"

7.8 Discussion

Our findings suggest that cross-perception shows meaningful promise as an approach to accessible analytical data interaction. We discuss our results in relation to our four research questions, then develop the broader implications for the field.

7.8.1 Research Question Responses

R1 (Exploratory): How might we envision and formalize a tactile interaction approach that enables coordinated, non-visual interaction across multiple spaces? Cross-perception, as formalized in [Section 7.4](#), provides one answer: an interaction technique grounded in principles from visual cross-filtering, spatial cognition, and blind information interaction, instantiated in a device with persistent-state motorized input and a linked tactile display.

The six design goals ([G1–G6](#)) constitute the formalization, and the design vignette demonstrates that the underlying cognitive mechanisms are ecologically grounded in expert blind reading practice. We intend this formalization to be extensible and generative; we demonstrate this extensibility in our future use cases in [Section 7.9](#).

R2 (Quantitative: Objective): In what measurable ways can we observe improvements to blind data interaction using our tactile interaction approach? Cross-perception produced significant improvements in task completion rate (+55% relative), task completion speed (90% faster), and quantity of data queries during open exploration (+188% computational, +54% spoken). Accuracy and precision were statistically comparable between conditions, though with a nuance discussed in [Section 7.8.2](#) below.

R3 (Quantitative: Subjective): What are the self-reported benefits, opportunities, and challenges that blind people experience with our tactile interaction approach? Participants using the cross-feelter reported significantly lower stress, lower future-task anxiety, and substantially higher enjoyment than in the screen reader condition. These improvements were particularly pronounced for participants without data expertise, suggesting cross-perception may lower affective barriers to data work for blind users who are new to or learning data analysis.

R4 (Qualitative): In what ways do blind data scientists imagine extending our approach? Participants, especially those with professional data expertise, proposed substantive extensions including: a pre-rendered “interaction cube” for environments without a refreshable display; feelter-only navigation of data arrays and sequences; sonification paired with rail brushing; and multi-dimensional panning, zooming, and scatterplot brushing using multiple feelter units. These proposals are detailed in [Section 7.9](#).

7.8.2 Accuracy and Precision Considerations

The comparable accuracy and precision results merit careful interpretation. Participants maintained similar correctness across conditions while performing substantially faster in CF. However, when filtering for completed tasks only, SR users showed marginally higher accuracy. One interpretation is that the additional time the serial screen reader interface imposed allowed for more deliberate verification before committing to an answer, while the cross-feelter’s speed encouraged more rapid, and occasionally less careful, decision-making.

This is not necessarily a weakness of cross-perception. In genuine exploratory analysis, speed of iteration is often more valuable than precision on any individual query, since the point is to generate and test many hypotheses rather than verify a single one carefully. Some precision can be a little “fuzzy” if the gained effect has analytical benefit overall [99]. This difference in interaction patterns warrants further investigation and suggests the two approaches may support different analytical strategies rather than being simply faster and slower versions of the same

strategy.

Future iterations should explore mechanisms that preserve the speed benefits of direct tactile manipulation while supporting careful verification when the task demands it (for example, a confirmation gesture or a “hold” mode that locks the filter and prompts output inspection before advancing).

7.8.3 The Case for Analytical Interaction as a Primary Research Site

The result most significant for the field is not the speed improvement but the query count. 20 computational queries in 10 minutes compared to 7 is not merely a faster version of the same activity. This increase in speed also reflects a qualitative shift in what participants were doing. In the CF condition, participants iterated hypotheses through direct manipulation; in the SR condition, they navigated toward output values that were structurally separate from their space of interaction. This maps directly to the distinction between analysis-oriented and access-oriented interaction: the screen reader, however capable at navigation, kept participants in a mode that structurally separated interaction and information; a descriptive mode. The cross-feelter enabled a *generative* mode where interaction and information are coupled.

The enjoyment gap is legible through the same lens. One participant’s unsolicited remark, “*Sighted people get to have this much fun when they work with data?*” describes the experience of analytical agency. This is what visualization researchers have understood for decades to be the core value of interactive exploratory analysis: an experience of power over and with data interaction. This agency has not been consistently available to blind data practitioners, and the affective signature of its absence is measurable in our stress and anxiety results.

The implication is that analytical interaction accessibility constitutes a distinct research problem, not an extension of the accessible rendering/description problem. The accessible visualization community has made substantial progress on helping blind users read and navigate data. What our results suggest is that reading data and analyzing data are sufficiently different activities that the techniques developed for one cannot simply be adapted for the other: they require different interaction architectures, different evaluation approaches, different design primitives, and sometimes even different hardware. We encourage researchers working at this intersection to treat analytical interaction as a primary site of inquiry: studying the existing practices of blind data analysts during open-ended exploration, identifying what analytical operations (filtering, outlier detection, hypothesis iteration, assumption testing) current tools support or foreclose, whether existing tools support direct interaction with data representations in these tasks, and designing interaction techniques grounded in these blind-centered findings, rather than in adaptations of what sighted-user interfaces already provide.

7.8.4 Toward Hardware and Physicalization in Accessible Data Interaction

Building on that previous point, a recurring practical constraint in accessible data interaction research is the emphasis on solutions that work within existing consumer device ecosystems: screen readers, keyboards, web browsers, and computationally synthesized audio. This is rea-

sonable: it maximizes immediate reach and avoids asking blind users to acquire specialized equipment if the capabilities are already realized on existing computer and mobile hardware. But it also implicitly constrains the design space to what those platforms can support, and some of the interaction properties that matter most for analytical work (such as persistent physical state, absolute spatial position, simultaneous bilateral manipulation) are not well-approximated by software alone on current consumer hardware. Existing work (mentioned in [Section 7.3](#)) hasn't ignored this problem, but this existing work is a small minority. Priorities in research have predominantly focused on leveraging *existing* software, hardware, and computational models towards problems and barriers blind people face, rather than innovation towards *novel* hardware and physicalization systems.

The cross-feelter's component cost (\$30–50 USD) and the availability of all schematics under an open license suggest that the hardware barrier is lower than it might appear. The more substantive barrier is that the visualization and accessibility communities have not yet developed strong shared practices around hardware prototyping and physical interaction design. Work from the tangible and physicalization communities is directly relevant here, including the insight that physical materials afford interaction properties that screens do not, among them permanence, resistance, and the kind of spatial memory that our design vignette illustrates. Bringing these ideas into contact with accessible data interaction research is, we believe, a productive direction.

The future use cases our participants proposed, discussed in detail in [Section 7.9](#), illustrate the range of what becomes thinkable when the design space is extended beyond the screen: pre-rendered interaction outputs for environments without powered displays, motor-driven data array navigation, multi-axis spatial exploration using multiple feeler units, and sonification navigated through physical rail position. These proposals emerged from a structured speculative futuring exercise [41] with participants who are themselves blind data practitioners. They are worth treating as design starting points, not only as participant feedback.

The field has successfully applied careful empirical and design methods to the challenge of making non-visual data representations perceivable. Extending that rigor to the challenge of making data manipulable through the hands, with all the hardware and prototyping work that entails, is the direction we are advocating.

7.9 Future Use Cases

The following use cases were developed through the speculative futuring exercise [41] conducted as part of our closing interview, with contributions from participants and the research team. We present them as design starting points grounded in the cross-perception framework, each addressing a limitation of the current prototype or extending the approach to new contexts. [Figure 7.6](#) illustrates all four cases.

7.9.1 Cross-Perception Without a Powered Display: The Pre-Rendered Interaction Cube

Refreshable braille displays remain expensive (\$8k–\$20k USD) and are not universally available. The current prototype depends on the Dot Pad to render cross-filter output in real time, which

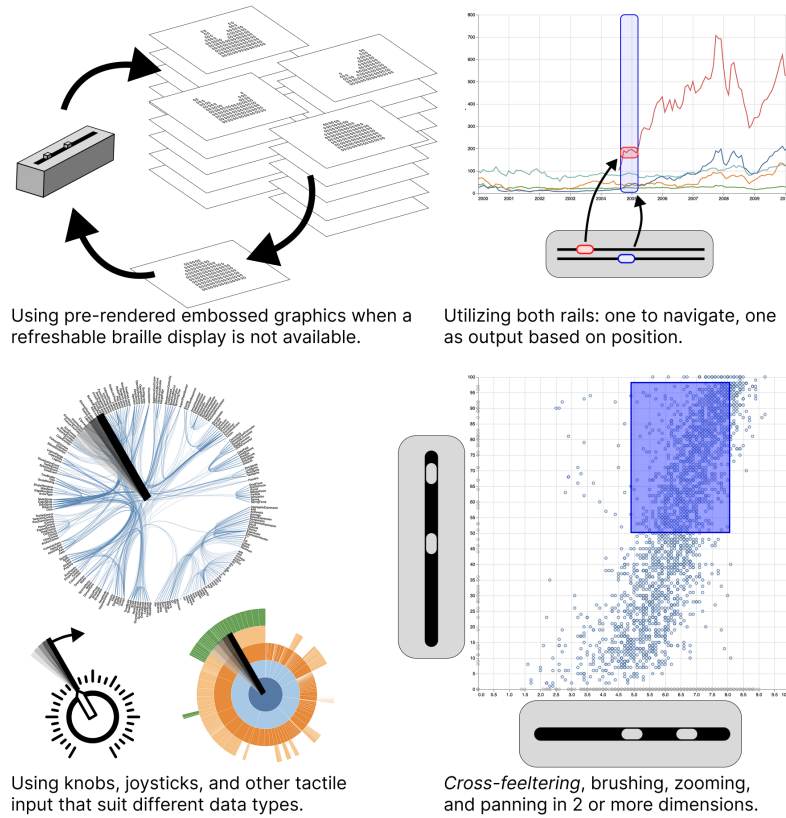


Figure 7.6: Speculative future use cases for our *cross-perception* approach, contributed by our participants.

means the full cross-perception loop is unavailable to users without access to one.

One solution is to pre-render the interaction outputs for a given dataset before a session begins. The cross-feelter’s filter operates over a bounded, discretizable range; it is feasible to pre-compute a set of interaction results at binned filter positions and emboss them onto paper as an “interaction cube” of tactile states, analogous to the data cube that Mosaic pre-computes for fast cross-filtering over large datasets [82]. After each filter manipulation, the user turns to the corresponding page in a booklet of embossed tactile charts, reducing the interaction loop from 3–4 minutes (a full embosser render) to roughly 10 seconds. This approach trades the continuous, real-time update of the powered display for a discrete but physically permanent output, which has its own advantages for users who prefer to annotate or revisit prior states or even compare multiple outputs simultaneously, like in the dual-task comparison vignette in [Section 7.4.1](#).

7.9.2 Sequential Data Navigation with Continuous Positional Feedback: Feelter-Only Chart Interaction

Serial navigation of data arrays using a screen reader is slow because the user must step through observations one at a time without any persistent sense of their position within the full range [45].

A single cross-feelter could replace this with continuous physical scrubbing. Most charts and

datasets have finite, bounded observations in any given dimension; these bounds map naturally onto the full range of a fader rail. One rail would scrub through the ordered observations as the user moves the knob, with the screen reader announcing the current value. A second rail, motorized, would move its knob to a position encoding the corresponding value in a second data dimension, like a rising and falling bar in a bar chart.

More broadly, different input devices could be matched to different data topologies. A rotary dial would suit circular or periodic data (seasonal patterns, directional data); a joystick would suit data with an infinite or continuous range such as a parametric function or model output.

7.9.3 Sonification Navigation via Physical Rail Position

Navigating within a sonification (seeking to a specific region, comparing two positions, returning to a reference point) is currently difficult even on touch-enabled devices, because scrubbing through audio provides no queryable positional state when the hand is not actively in contact with the surface [30, 31, 32]. This is the same object-permanence problem that motivated the cross-feelter’s design for cross-filtering.

A cross-feelter rail could serve as the navigation input for a sonification: physical position on the rail maps to a position within the audio timeline or data domain, and the motorized mechanism maintains that position when the hand is lifted. The user can navigate to a region of interest, remove their hand, let the sonification play, return to the rail, and feel exactly where they left off.

7.9.4 Multi-Dimensional Spatial Interaction: Rails as Input and Output

Two cross-feelter units, one arranged as an x-axis controller and one as a y-axis controller, could support spatial panning, zooming, and brushing over tactile maps or scatterplots, which are interactions that currently lack a satisfactory non-visual implementation because pinch-and-zoom gestures provide no queryable positional state for the zoom extent or pan position. Each feelter unit would maintain the current viewport bounds in its two rails, readable by touch at any time.

More substantially: the motorized rails could serve as *system output* for spatial queries. When a user asks “where is this object?” of a map or image, the system could drive the rails to the corresponding x and y positions and bounds. This would let the user feel the location directly rather than receiving a verbal coordinate and build a spatial mental model of a spatial layout.

7.10 Limitations

Our study’s controlled task design (three-minute timed tasks with multiple-choice answers) prioritized speed and query quantity, which are the right measures for a proof of concept but do not capture the full texture of blind data analysts’ real exploratory workflows. Activities common in open-ended analysis, such as wrangling messy data, identifying outliers, and questioning assumptions, were not represented. An ethnographic or longitudinal study would be better suited to those questions, and we encourage that work.

The Dot Pad’s 500–1200ms render latency falls outside the max 500ms threshold at which Liu and Heer identify measurable degradation in exploratory analysis quality [114]. Our query

count results are therefore a conservative lower bound on what cross-perception could support with faster display hardware.

Novelty effects may have inflated subjective measures. The enjoyment and anxiety improvements may attenuate with extended exposure, though the effect may persist in contexts such as some classroom, work, or onboarding settings, where engagement is itself valuable.

Finally, we compared against a screen reader baseline because screen readers are the dominant interaction tool for blind users working with data and all of our participants were proficient in screen reader use. More granular comparative work against other emerging tactile and voice-based input strategies would further characterize where cross-perception sits in the broader interaction space.

7.11 Conclusion

We introduced cross-perception, a tactile interaction technique that adapts cross-filtering for blind data analysts by preserving the simultaneity of input and output perception that makes the technique analytically valuable. We formalized it through six design goals grounded in principles from visual analytics, spatial cognition, and blind information interaction; instantiated it in the *cross-feelter*, a low-cost motorized fader device; and evaluated it against a screen reader baseline with 15 blind participants.

In our study, cross-perception substantially improves task completion speed, data query generation, and subjective experience, with particularly strong effects for participants without prior data expertise. The result that matters most is not the speed improvement but the query count: participants using the cross-feelter generated nearly three times as many analytical queries during open exploration, reflecting both a quantitative improvement and a qualitative shift away from navigating data toward interrogating it.

We argue that analytical interaction accessibility, the capacity for blind users to filter, conjecture, and test hypotheses through direct manipulation, is an open research problem that deserves sustained attention as a primary site of inquiry, distinct from accessible rendering. The future use cases our participants proposed suggest that extending the design space to include hardware and physicalization opens directions that software-only approaches cannot reach.

7.12 Postface

This project has been framed and reframed several times: should we focus on the prototype? the study? or the interaction design?

Ultimately, the story we want to tell with this research isn't about our *cross-feelter* prototype (despite how much we love its name) or the study (despite how much we love some of our findings). But ultimately, we hope to see more work that takes the core idea here (that *cross-perception* enables new kinds of interaction with information), which is a generative space for new research [9]. We ran speculative sessions with users precisely because we wanted to explore whether the simplicity of our core idea could lead to new directions. And it did.

Often, research papers are defensive about intellectual territory. Papers frame “future work” as a zone of intended control, a domain that the primary author hopes to protect. Some papers are generous in their future work sections, giving new ideas and directions away freely, but most are situated on the style of narrative that the main author intends to explore certain paths forward, perhaps even to the discouragement of others.

Yet research ideas should be generous. If researchers didn't have to worry about competitive grants or the pressures of tenure requirements, the spirit of academia might shine brightest in a paper's future work section. To us, this is where new pathways should be freely given, new directions should be inspired, and unanswered questions are earnestly laid bare. We hope that this is the case with this project, in particular. We want to see more hardware in accessible data interaction. Visualization as a field is situated uniquely to answer this call, too. There are already innovators in hardware and input modalities in our research community. But many haven't turned their gaze towards accessibility. We hope to capture some of their imaginations, and see more work that builds on *cross-perception* in the future.

Part VI

Conclusion

Chapter 9

Findings

The five projects in this thesis were presented as five chapters, each with its own results and its own conclusion. Read separately, they contribute a heuristic resource for evaluation, a structural toolkit for navigation, an authoring environment for non-visual design, an interaction technique and device for blind analysis, and a system and study of end-user personalization. This chapter states what they show *together*. Chapter 2 closed on a lopsided field: a breadth of empirical work and exemplary prototypes on one side, and a near absence of practitioner tools in production environments and practitioner action on the other, with the hardest remaining challenges concentrated in navigation, interaction, and personalization. The findings below are what this thesis learned by working inside that gap.

Each finding is a synthesis across chapters, and each is traceable to results reported in the body of this thesis. We have written them for two audiences at once: practitioners who build, evaluate, and maintain data experiences, and researchers who study how that work happens. Where a promising idea outran the evidence, it is not here; it is in the next chapter, with the rest of the discussion and future work.

9.1 Access Barriers Are Produced for Practitioners, Both With and Without Disabilities

The social model holds that disability is not located in a body but produced in the interaction between a person and a human-made system [124]. This thesis began by applying that lens to end users: visualizations, like curbs, produce barriers. The cross-cutting finding of this thesis is that the same lens applies, with the same force, to the practitioners doing accessibility work, and that the materials of that work are among the systems producing the barriers.

Every project surfaced a version of this. In *Chartability*, the guideline landscape itself was a barrier: standards and research were scattered, often high-level, and difficult to apply to complex interactive data interfaces, and practitioners had little way to judge the quality of what they gathered [102, 165]. Guidelines written to remove barriers for end users limited what practitioners could reasonably understand and act on, and an expert visualization practitioner in our study admitted he did not know where to start. In *Skeleton*'s co-design foundations, *Data Navigator* made better end-user experiences possible, but because its structures were entirely non-visual and code-bound, it produced severe iteration barriers for sighted practitioners: they could not perceive what they were building, and prototypes were misinterpreted or abandoned. And in *Cross-perception*, the screen reader, the most important enabling technology for blind computer use, also limited what blind practitioners could reasonably do analytically: its serial, discrete traversal kept participants in an access-oriented mode that structurally separated interaction from information, with measurable costs in time, completion, and the number of questions they could ask of their data [163].

The operationalized version of this finding: when accessibility work stalls, look for barriers

in the practitioner's tools and materials before attributing the failure to motivation or knowledge. None of the practitioners in these studies lacked care. What they lacked, in each case, was a working relationship with materials that had been shaped without them in mind, and this is true whether the practitioner is sighted, blind, disabled, or not. The same observational stance the field has learned to take toward end users belongs in the study of the workers, too.

9.2 Tools Can Reduce the Work Required to Make Things Accessible

Chapter 2 argued that guidance cannot substitute for means. The constructive half of that argument is that supplying means measurably changes what practitioners can do, and three chapters provide the evidence.

Chartability reduced the perceived difficulty of evaluation and increased the confidence of practitioners new to accessibility, whose audit results were reasonably comparable to the authors' own; beyond the study, it has been applied by policy and governmental organizations and more than 100 companies, across toolchains from Tableau and Excel to D3 and Figma. *Data Navigator* gave previously inaccessible rendering formats a navigable structure: a static raster image gained full multi-input navigation with no automatically detectable conformance failures, and a deterministic ingestion pipeline recreated, and then improved on, an existing toolkit's navigation scheme for rendering modes that previously had none. *Skeleton*'s scaffolding collapsed the spatial authoring of a navigation structure to one or two minutes of refinement, and within single sessions every participant iterated substantively on designs that, in a code-only workflow, our co-design collaborators had struggled to iterate on at all.

None of this is evidence that tools solve access; the boundaries of that claim get their own finding below. It is evidence of something narrower and more useful: the labor of access work is sensitive to tooling, and reductions in that labor are achievable, measurable, and repeatable across very different kinds of tools, from a workbook of heuristics to a rendering-independent toolkit to an authoring interface.

9.3 Making the Invisible Structurally Visible Changes Practice

The strongest through-line in this thesis is what happened when an invisible structure underlying practitioners' work was given a perceivable, inspectable form. Three chapters did this to three different structures. *Chartability* made the evaluative space visible: a scattered landscape of standards and research became a single workbook a practitioner could hold in view, and participants described it as helping them focus, stay consistent between sessions, and aim above compliance. *Skeleton* made navigation structure visible: nodes, edges, and announcements rendered over the practitioner's own chart, and its study found that participants who began by asking compliance questions (does this pass? what does the guideline say?) shifted to asking design questions (is this good? is this too many steps? what is this *for*?). *Software* made the dimensions of prefer-

ence visible: an explicit menu of option categories gave end users something concrete to react to and gave practitioners a representation through which access friction stopped being abstract.

The mechanism is one visualization researchers already trust: reflective practice runs on externalized representations that a designer can perceive and respond to [159]. What this thesis adds is evidence that the loop works on structures the field had left invisible, and that restoring the loop moves practitioners from verification toward judgment. As a guideline, this is directly actionable for toolmakers: when practitioners treat some aspect of accessibility as a checkbox, ask what representation of that aspect they can actually perceive, because a structure no one can see is a structure no one can design.

9.4 Friction From a Tool Can Be Productive

Perhaps in tension with the previous finding that tooling can reduce work, tooling may also *increase* some kinds of work in exchange for quality. The assumption for tooling, in a classic or naïve sense, is that tools exist to make work easier by reducing the labor and friction of certain tasks. But ease of work is only a goal if a baseline exists. If some work isn't being done or new design possibilities open up, then *more*, not less, work might be the goal of an idealized tool. Friction can be useful, even productive, and sometimes necessary. While outside of the scope of this thesis, we would urge fellow researchers to explore the tensions that arise in design when removing friction is the primary, if only, goal of a tool (as it appears to be the case with modern large language models and agentic tooling).

But it *is* a reasonable assumption about practitioner tooling that its job is to make work smoother and easier. The evidence in this thesis complicates that assumption: in several places, the value of a tool came precisely from the resistance it introduced. *Chartability* forced engagement with conflicts between guidelines (such as redundant encodings that help some users while overwhelming others) rather than smoothing those conflicts away; audits remained slow, and participants reported that the slowness helped them focus and prioritize. Evaluating a visualization for accessibility actually *adds* time to the overall time it takes to make a visualization. *Chartability* made it clear that if work isn't being done at all in the first place, introducing new heuristics to a practitioner's workflow ultimately generates friction that wasn't there before. We view this as necessary and good.

Data Navigator arguably introduced so much additional work and friction that it necessitated additional tools to make it easier to work with (in the *Inspector*, *Dimensions API*, and ultimately *Skeleton* of Chapter 6). It became apparent that giving practitioners a nearly-infinite new space of design without clear scoping and bounds for that design space was burdensome and resulted in many additional barriers and design iterations. *Data Navigator* could create *any* discrete navigation experience from relationships in an information space. This is fantastic. But it appears that *scope* in design is, in its own way, a source of friction *and* a relief on the work required of that design: it limits and holds back what can be done. But this scoping friction in turn reduces the labor of exploration required. Infinite, frictionless exploration of a design space appears to have put friction elsewhere in the work: during deliberation, handoffs, testing, and ultimately in producing scope.

Skeleton's sessions opened on an intentionally problematic default structure, and its testing

stage made designs that looked adequate feel inadequate in traversal; participants responded to that resistance not by abandoning the work but by iterating, restructuring dimensions, and in one case restarting from the wizard. The tool introduced *more*, not fewer design iterations. Interestingly, *Skeleton* was intended to reduce friction and make work easier, yet by surfacing and making perceivable and manipulable a space of design elements, practitioners didn't just neatly solve problems in less time or solve more problems, in some cases they problematized their own work and returned to the figurative (and literal) drawing board.

Softerware also produced friction at the level of professional identity: it challenges the traditional separation between the people who design a system and the people who use it, and our practitioners immediately raised the ethical concern that personalization could excuse poor defaults. That concern surfacing early, before any deployment, is the friction doing its work. Practitioners suddenly had to worry about ways that users could self-design into misleading, harmful, or newly inaccessible designs on their own. And yet, practitioners then had to wrestle with the question, "if we give people power over their interfaces, isn't it then their responsibility in the end for what they do with that power?" Our prototype and study didn't neatly resolve these frictions or close out this conversation, but rather these frictions simply served to open up new directions and outline problems presently unsolved.

We would argue that the *cross-feelter* device from *Cross-perception* was a "classic" tool in this sense, unlike the rest: it reduced friction *and* reduced work, which led to more productivity. It is good to hold this project in tension with all of the others in this thesis, which introduced new, productive frictions into accessibility work in some way or another and may or may not have made work easier in some ways in the process. *Cross-perception*, by comparison, primarily served to open up a space of design while we demonstrated one design within that space through our hardware device.

Chapter 2 argued that tools direct the work and the workers who use them. Friction is one of the ways they do it. The finding here is that a tool which removes all resistance from accessibility work can also remove the engagement that the work requires. This is, in some cases, not an ideal outcome. Engagement with particular tensions might be the *best* thing that a particular tool can accomplish. This means that toolmakers can place friction deliberately: at conflicts that deserve a decision, at defaults that deserve suspicion, at moments where compliance would otherwise substitute for judgment, and in places where designerly, considerate work isn't being done but *should* be.

9.5 Tool-use and Tool-making Are Generative

Across the studies in this thesis, tools did not merely execute their users' existing intentions; they expanded what practitioners and users could imagine. In *Skeleton*'s study, five of eight participants reconsidered the design of the visualizations they had brought, not just the navigation built over them: some simplified, some questioned their chart type, and one concluded that the right non-visual design was no navigation at all, a genuine design decision rather than an omission. In *Cross-perception*, formalizing the technique and putting a working device in participants' hands opened an input design space: blind data practitioners proposed pre-rendered tactile interaction cubes, feelter-driven array navigation, sonification scrubbing, and multi-axis spatial interaction,

extensions we present as design starting points. In *Software*, the visible option space provoked interface ideas we had not designed for, including in-context, voice-driven adjustment while navigating, and led practitioners to imagine authoring their own systems through a software interface. Even *Data Navigator*, the lowest-level contribution here, showed this property (and perhaps the strongest): its vocabulary of nodes, edges, and rules gave co-design sessions a shared language in which novel navigable prototypes could be drafted in days.

The generative freedom afforded by tools isn't just *expressiveness*, as defined by existing work on visualization grammars [154, 199], which tends to enable fluency within the bounds of a structured language, but also generation outside of any existing bounds as well [9].

For researchers, this finding doubles as a methodological one, and it is the stance Chapter 2 named: studying practitioners through their tools works because a tool placed in real work is simultaneously an intervention and an instrument. The generative responses it provokes, the reconsiderations, extensions, and refusals, are data about why the work is hard and where it could go next.

And tool-making isn't just generative because tool-users appropriate, re-use, and innovate once they have a new tool, but viewing gaps and problems that practitioners face is also generative for new research directions. This was the cornerstone motivation behind our *action research* approach in Chapter 6: that collaboration with practitioners as an embedded designer, engineer, and researcher in order to join them as they solve problems provides a rich space where novel intellectual gaps can be made both legible and actionable. Tool-making alongside tool-makers and tool-users without having an *a priori* intellectual agenda in mind leaves room for developing new conjectures, evaluating multiple collaborations in aggregate, and ultimately generating research agendas.

9.6 The Hardest Barriers Required Intervention at the Substrate

Chapter 2 characterized navigation, interaction, and personalization as a missing structure, a missing technique, and a missing infrastructure. Having now worked in all three domains, we can state what they share: in each, the barrier sat one level below where practitioners are ordinarily able to act, in the substrate their work is built on, and in each, outcomes changed when the intervention went to that level rather than adding features above it. This finding is held heavily in tension with existing research that tends towards higher levels of software and interaction abstraction, not lower. A massive wave of new research has emerged that essentially asks, “what new layers or wrappers can we create around existing interfaces and user experience problems, using large language models and agents?” While our thesis doesn't engage the effectiveness or intellectual value of those projects, instead we hold this thesis in fundamental contrast to their approach: that some of our most difficult gaps are best engaged below our present levels of software abstraction, not above.

The evidence is consistent across the three parts of this thesis. No amount of feature work on top of canvas-rendered pixels can recover semantics that the rendering substrate never carried, and forcing complex, non-hierarchical data relationships into HTML's hierarchical substrate

limited what was possible to build. This is precisely what a large and passionate resistance to “overlay” technologies engages within both the practical and academic accessibility communities [42, 69, 79, 120]. There may simply be a limit to how much good data, if at all, can be generated from bad or missing data [16, 140].

Data Navigator changed outcomes by supplying a structural substrate of its own, a graph independent of any renderer. This requires underlying data that is supplied with sufficient quality, what Doug Schepers calls an “underlay” rather than an overlay [157].

No software feature on existing consumer hardware approximates persistent physical state and simultaneous perception of input and output, the properties analytical interaction depends on [114]; the *cross-feelter* changed the hardware substrate, and the character of blind analysis changed with it, from accessing data toward interrogating it, at nearly three times the query rate.

And per-chart personalization features fail the effort-to-outcome test that end users themselves applied; what every participant and practitioner in the *Softerware* study named instead were system-class needs [54]: persistence, profile sharing, interoperability, and standards beneath any individual chart.

This is why the substrate matters as a space of intervention: in all three domains, the ceiling on what a motivated, well-informed practitioner could make accessible was set by material beneath their reach, so guidance, effort, and even exemplary prototypes hit that ceiling without moving it. The work in this thesis moved outcomes exactly where it changed the material itself, and the same evidence shows the stages of this work leaning on one another: structure without authoring support went unused, authoring raised evaluative questions practitioners could not answer alone, and personalization without accessible defaults and persistent infrastructure shifts burden onto the people it is meant to serve. Researchers and toolmakers deciding where to spend effort in these three domains have, in this thesis, a consistent answer about where the leverage was found.

9.7 What Tool-making Did Not Do

The findings above would be misread without their boundaries. Three are most apparent in this thesis, which we unpack below.

9.7.1 State of the Art does not Necessitate Adoption

First, a tool’s existence, even if it is the “best” system by some argument or measure, did not cause its adoption. This should seem obvious, and yet technical research appears to operate with an unspoken assumption that *something* practical should happen at scale after a clever or novel prototype is produced. Research ideas are brought into the world and largely abandoned almost immediately [93]. *Data Navigator* was (and perhaps still is) the most flexible and expressive system made for data navigation, and yet translating designs into it remained too complex to simply hand off, which is what motivated our industry collaborations and ultimately *Skeleton*; *Softerware*’s participants made clear that personalization that must be redone per chart is not worth their effort; and the proportion of the web’s most popular pages with critical accessibility failures is essentially unchanged across the six years spanned by this research [195]. Chapter 2

observed that a practitioner cannot simply install a paper; this thesis adds that practitioners will not, and often cannot, adopt even a working tool absent authoring support, incentives, infrastructure, and policy around it. Existence is the beginning of an argument for adoption, not the conclusion or even motivation for one.

9.7.2 Tooling does not *Solve* Access Friction

Second, tools did not solve access friction, and our evidence suggests they may not be able to. Access friction may simply be inevitable, as others already demonstrate and argue [76, 92, 131]. In the *Software* study, no two participants chose the same configuration, participants with the same diagnosed condition needed opposite design treatments, and one participant wanted strong defaults rather than any customization at all. Personalization *relocates* friction and gives people agency within it; but it does not engineer the friction away. Strong accessible defaults, despite being a source of friction in the first place, remain a prerequisite of personalization.

9.7.3 Authorship by People with Disabilities Remains Unresolved

Third, the tools in this thesis did not redistribute authorship. They mostly serve practitioners building *for* disabled users: *Skeleton*'s study evaluated sighted practitioners' engagement, not the experiences their structures produce for blind users, and while *Cross-perception* and *Software* begin to shift who holds the interaction, people with disabilities remain largely excluded as authors, analysts, and people with power over data interfaces. This marks what should be expected from tooling interventions and what should be expected from policy, practice, and the sharing of power [174]. Some existing work has used tooling as a space to explore what de-centering visual authoring might look like [215], but this largely remains an area with significant gaps in practice. The next chapter takes up exactly these questions, of what tools are downstream from, who is responsible for repair, and what this work asks of the field beyond its tools. We view these gaps as generative spaces for future work.

Chapter 10

Discussion & Future Work

10.1 What is a “tool?” A reflection on the social and material identity of tools

In the introduction of this dissertation, we use the example of a hammer: a hammer can destroy and it can construct. So is the *use* of a technology what constitutes it? Do we understand the hammer as the *thing we swing, to destroy and to build*? Should we?

This thesis engages domains of tools and tool-making for accessibility: evaluation, navigation, interaction, and personalization. But these categories for work do not fully characterize the upstream conditions that our software systems and data interfaces inherit.

In our work specifically on accessibility, a larger social reality becomes apparent that shapes the question, “what is a tool?” far more than how an individual might use one, or the domains of work that our tool-making inhabits. Our research has navigated multiple social and political thresholds, from changes to law in the European Union, to the enactment of Title II as part of the update to the Americans with Disabilities Act. These laws have motivated a significant interest in accessibility research, solutions, guidelines, and technologies. In the midst of this, we have seen the rise of overlays and generative AI solutionism [79] and subsequent lawsuits and grass-roots resistance.

For our work, this is mostly good news. Legal change produces motivation, and even with predatory technology attempting to address real problems, pushback is widespread and active. But this paints a picture of the reality that our work inherits: many tools cannot even be used, or cease to be used, if there is not a social, political, and material set of conditions in place motivating those tools, providing resources for their construction, regulating their use, and examining the outcomes of what they accomplish. Tools and technologies are often a response to social, cultural, political, and legal realities that we first negotiate.

I, Frank, recently spoke on this at a keynote in Australia, on how a hammer isn’t *just* a tool and that the idea that “the only thing that matters is how a tool is used” limits how we really understand tools. Instead, I spoke about how a standard, household hammer requires iron and wood. That alone leads to a whole universe of different questions. Western Australia’s conservation efforts were disrupted when a significant amount of natural iron was discovered in a wildlife preserve. So laws were passed and now iron is mined there. That iron is largely exported. And Australia then, whether with Australian iron or not, mostly imports their small tools. Iron is sent out, and through a complex network of trade (likely indirectly related to the iron), hammers are brought in. A “hammer,” to even exist at all, relies on multiple levels of human governance, international relations, and complex infrastructures of trade.

And while my metaphor is largely motivated to encourage younger practitioners to consider the “iron mines” in the technologies they use, such as modern generative AI, it is also an area that is not adequately explored and addressed in terms of accessibility research.

Research on accessibility is dependent on funding, which is often dependent on political

priorities and action. Depending on the current social and political state of the world at large, accessibility research itself may never gain the opportunities required in order to innovate and produce new tools at all. And as the US's 2025 federal cuts to research demonstrated, millions of dollars devoted to accessibility research can be lost to political agendas. It is for this reason that engagement with policy recommendation and guidance is essential. Personal political activity and involvement is also essential. Researchers who genuinely believe in accessibility as a human right or as a dignity that all people deserve should work with policymakers to ensure that there are material and structural resources in place for this work to continue. We cannot naively believe that technology, divorced from the realm of social and political forces, is capable of solving accessibility barriers [174]. Without enforcement and threat of litigation, very little accessibility work has been accomplished in the past by technology companies alone.

Featured in these chapters (in their pre and postfaces) is the policy and outreach work involved in seeing that work like *Chartability* and *Data Navigator* were used in real contexts, including by organizations that govern and influence the lives of many people. Immediate incentives to produce novelty may not be enough to sustain the larger socio-cultural and political ecosystems that our work participates in and is downstream from. We must also get involved.

10.2 “Applied” accessibility work and *low research*

I came into an academic research environment already as an engineer. I had questions to big problems, especially about how tools might shape human behavior and how we can use that towards ethically good outcomes. But *Chartability* was a project I made first for myself, to engage the problems I, personally, was facing.

And *Data Navigator* was motivated by my existing experience making visualizations navigable. I knew we had to build better tools, because continually trying to make a hierarchical substrate like HTML work for complex, non-hierarchical relationships simply limited what was possible and easy to do.

These two projects then led to many opportunities to test them in applied contexts. Our CZI grant was motivated towards the application of both, Adobe was interested in *Data Navigator*, and the University of Wisconsin's GIS folks wanted to figure out how to make a complex map navigable, *Data Navigator* or not. Yet, as a researcher I wasn't incentivized to pursue these projects. In fact, they all proved to have no immediately apparent novelty. It was a risk to spend my time in these spaces.

Applied work is structurally and socially devalued in academic research environments, seen often as a lower form of research and design compared to more “pure” forms of knowledge production. (We conjecture that in part this is due to the scope of knowledge that practitioners produce: they work to take broad knowledge and apply it to specific, small problem spaces. Foundational research tends to assume that broad knowledge does not then bubble up from this work, that it is a one-way flow.)

In my first year as a PhD student I was given the advice by a senior researcher not to hold my experience as a practitioner too highly because, “practitioners often don't know what they are doing.” But I did, of course, to some degree. Yet, this attitude is pervasive in some spaces and is a problem because it limits what kinds of knowledge we care to attend to.

Initially, I felt that with *Chartability* I had lucked out. But after four more collaborations which resulted in funding and two research publications, we can now argue that this method has been reliable. In *Softerware* and *Skeleton*, pursuing practical problems in applied environments *did* serve as a fruitful source of generalizable knowledge and advancements in the state of the art. But these projects, in addition to *Chartability*, were all motivated by immediate, non-novel problems: people knew their charts were inaccessible and wanted some design guidance or engineering solutions.

So perhaps “low research” works.

Yet, some research communities, especially *ASSETS*, have yet to reckon with “applied” work like this. If you trace the lineage of *action research* from Hayes [80] (which is how we frame our applied work in *Skeleton*), it leads into fantastic areas of participatory and community-based work, often with or by marginalized people and people with disabilities [11, 92, 117, 122, 174, 177, 178, 187, 202]. We would even argue that this lineage of work, which prioritizes and centers the lived experience of a person, is also why we have seen so many successful auto-biographical and auto-innovation styles of research as well recently [15, 59, 61, 87, 160, 200, 201].

And yet, action research largely does not touch on how practitioners (particularly those without disabilities) do applied work in accessibility [102, 165]. We believe that significant gaps exist in the space between traditional “high” knowledge production and applied, “lowly” spaces. Researchers tend to pursue novelty and the lifecycle of research production is highly risk averse. And yet, relatively little work has been done within the accessibility research community to intellectually address the actual social and cultural risks involved in producing decades of useless, lost, and forgotten prototypes and design guidelines [93, 97]. We would conjecture that research-producing academics are overly concerned with academic readership relative to the myriad of ways that their knowledge production actually is used by the rest of the world [132]. We would challenge *ASSETS* and the AT/accessibility communities of *CHI* in particular to take seriously the lack of applied (industry, government, and non-academic) work it produces as a community.

10.3 Who is responsible for repair?

Lastly, we want to revisit one of this thesis’s opening points, where we argue that the *tool-makers* are first responsible for repair. This is true. However, the most pressing issue we have faced in recent years is mostly unmentioned across these research projects: tool-makers might be responsible, but this is because they are the only ones who have the *power* to make things accessible. Does this always need to be the case? Can we imagine an artifact’s authority over the user’s ability to access being designed towards self-subversion [65] or de-centralized agency [28, 106, 133], instead? What might that look like, concretely?

In *Softerware*, we begin to engage this larger problem in terms of an idealized state where a user can repair or re-design their own experiences. But to me, this self-repair is like laying down train tracks for yourself as you move a locomotive, but then lifting up your own tracks behind you as you go. You’re the only one helping yourself. This is not ideal, for you or others.

What we need are broad, lasting, infrastructural changes. On the web, this problem becomes quite difficult to solve. A personal computer or device? Again, someone can auto-design their interfaces into a better state. But back when we started *Chartability*, the WebAIM Million’s report

showed more than 95% of the top one million website home pages contain at least one critical accessibility error. And now, more than six years later, that proportion is unchanged [195].

Some had imagined that generative AI would solve the massive infrastructural repair problems we face. But unfortunately, the latest WebAIM Million report shows that since 2020, ARIA usage has increased and correlates to more errors, while use of `tabindex` on a page has increased nearly 300% and also correlates to more errors on a page. If anything, during the age of generative AI, we have seen existing bad patterns worsen in prevalence and complexity.

We firmly believe that a tools-based approach is not enough on its own. Tool-making cannot be the *only* intervention on inaccessibility. Tools and tool-making, as our thesis argues, have a powerful role to play. But we simply can't tool our way out of failed infrastructure and inadequate policy when someone else *owns* the tools and tool-making. Visiting a website is like going into someone else's home: arranged according to their effort, tastes, and so on. If you can't access their home, you essentially need to request that they let you in personally. Website repair always falls to the owner and maintainer of a website, and they largely don't take any meaningful action.

Sidewalks outside of homes are a good parallel to this problem. Sidewalk accessibility is a massive infrastructural problem [150], and yet localities treat sidewalk maintenance in different ways: some, like where I, Frank, presently live in the south hills of Pittsburgh, put the onus on the homeowner whose house and property the sidewalk touches. In other places, sidewalks are considered a public path, similar to a roadway, and are maintained through public tax and resource management. To no surprise, privately-maintained, public-access sidewalks are worse for people in pretty much every way than publicly-maintained ones [207]. This is because private homeowners don't care about sidewalk maintenance unless the city manages to fine them or they get sued.

And the web is a collection of private spaces that you visit privately. There is no truly shared, universally democratic, public space on the web. Centralization is partly to blame: sharing space while scaling leads to consolidation.

So our future work will continue to wrestle with the same tensions of scale, repair, and anti-consolidation of power, motivated by the same WebAIM Million report. But now we look to questions of *democratic* and *radical* access to accessibility repair. The barriers we hope to tackle in the future are political and infrastructural. Perhaps tool-making will participate in this work, but it seems clear now from our work that the upstream technical problems and socio-political conditions that tools inherit, will likely not be addressed by tools alone.

What does “democratic” and “radical” infrastructure work look like? It will probably be an extension of *Softerware*, to some degree. We imagine future research that explores public-first spaces, ones where access is socially negotiated and repair belongs to all of us. Is this an autonomous space, like an autonomous zone [14] separate from the web? Above it, looking down into it, like shared annotation tools but capable of sharing the manipulation of websites [149]? A space with ambient co-repair, modeled after projects that bring people together [161] or that allow community “fixing” of misinformation [98]? Perhaps feminist thought on the ethics of care can help us [83, 119]? Or maybe it will be something else entirely; I'm not yet sure. But what made the web fantastic years ago is long gone; most of it has been hedged into corporate spaces that are controlled, maintained, and repaired by corporate power. And these entities are notoriously bad at repair. What we imagine in the future involves reclaiming a sense that the web is *ours*, belongs to *us*, and that ultimately *we* are responsible for making it accessible.

References

- [1] Christopher Ahlberg and Ben Shneiderman. Visual information seeking: tight coupling of dynamic query filters with starfield displays. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI94, page 313–317, New York, NY, April 1994. ACM. doi: 10.1145/191666.191775. URL <https://doi.org/10.1145/191666.191775>. 7.2, 7.3.1, G4:, G6:
- [2] Sara Ahmed. *Queer Phenomenology: Orientations, Objects, Others*. Duke University Press, Durham, NC, 2006. ISBN 9780822339144. ISBN: 978-0-8223-3914-4. 2.1.2
- [3] Sara Ahmed. *What’s the Use? On the Uses of Use*. Duke University Press, Durham, NC, 2019. ISBN 9781478006503. ISBN: 978-1478006503. 2.1.2
- [4] Obead Alhadreti and Pam Mayhew. Rethinking thinking aloud: A comparison of three think-aloud protocols. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI ’18, page 1–12. ACM, April 2018. doi: 10.1145/3173574.3173618. URL <https://doi.org/10.1145/3173574.3173618>. 6.6.2, 7.6.6.2
- [5] R. Amar, J. Eagan, and J. Stasko. Low-level components of analytic activity in information visualization. In *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.*, page 111–117. IEEE, November 2005. doi: 10.1109/infvis.2005.1532136. URL <https://doi.org/10.1109/infvis.2005.1532136>. 3, 7.2
- [6] Apple Inc. Audio graphs. <https://developer.apple.com/documentation/accessibility/audio-graphs>, 2021. Apple Developer Documentation. Feature introduced at WWDC 2021. Accessed: 2026-06-12. 2.2.1
- [7] Catherine M. Baker, Lauren R. Milne, Ryan Drapeau, Jeffrey Scofield, Cynthia L. Bennett, and Richard E. Ladner. Tactile graphics with a voice. *ACM Transactions on Accessible Computing*, 8(1):1–22, January 2016. ISSN 1936-7236. doi: 10.1145/2854005. URL <https://doi.org/10.1145/2854005>. 2.2.1
- [8] BANA. Guidelines and Standards for Tactile Graphics. Braille standard, Braille Authority of North America, 2010. Accessed: 2021-09-03. 2.2.1
- [9] Michel Beaudouin-Lafon, Susanne Bødker, and Wendy E. Mackay. Generative theories of interaction. *ACM Transactions on Computer-Human Interaction*, 28(6):1–54, November 2021. ISSN 1557-7325. doi: 10.1145/3468505. URL <https://doi.org/10.1145/3468505>. 2.1.2, 7.12, 9.5
- [10] Richard A. Becker and William S. Cleveland. Brushing scatterplots. *Technometrics*, 29(2):127–142, May 1987. ISSN 1537-2723. doi: 10.1080/00401706.1987.10488204. URL

<https://doi.org/10.1080/00401706.1987.10488204>. 7.2, 7.3.1

- [11] Cynthia L. Bennett, Erin Brady, and Stacy M. Branham. Interdependence as a Frame for Assistive Technology Research and Design. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '18, page 161–173, New York, NY, USA, October 2018. Association for Computing Machinery, ACM. doi: 10.1145/3234695.3236348. URL <https://doi.org/10.1145/3234695.3236348>. Accessed: 2021-09-03. 1.1, 2.3, 10.2
- [12] Cynthia L. Bennett, Burren Peil, and Daniela K. Rosner. Biographical prototypes. In *Proceedings of the 2019 on Designing Interactive Systems Conference*, pages 35–47. ACM, 2019. doi: 10.1145/3322276.3322376. 2.3
- [13] Dan Bennett, Alan Dix, Parisa Eslambolchilar, Feng Feng, Tom Froese, Vassilis Kostakos, Sebastien Lericque, and Niels van Berkel. Emergent interaction: Complexity, dynamics, and enaction in hci. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, page 1–7. ACM, May 2021. doi: 10.1145/3411763.3441321. URL <https://doi.org/10.1145/3411763.3441321>. 2.1.2
- [14] Hakim Bey. *TAZ: The temporary autonomous zone, ontological anarchy, poetic terrorism*. Autonomedia, 1991. 10.3
- [15] Brandon Biggs, Christopher Toth, Tony Stockman, James M. Coughlan, and Bruce N. Walker. Evaluation of a non-visual auditory choropleth and travel map viewer. In *Proceedings of the 27th International Conference on Auditory Display (ICAD 2022)*, pages 82–90. International Community for Auditory Display, June 2022. doi: 10.21785/icad2022.027. URL <https://doi.org/10.21785/icad2022.027>. PMCID: PMC10010675. 2.2.1, 6.4.1, 10.2
- [16] Abeba Birhane, Vishnu Boddeti, Sanghyun Han, Sasha Luccioni, and Vinay Prabhu. Into the laion’s den: Investigating hate in multimodal datasets. In *Advances in Neural Information Processing Systems 36*, NeurIPS 2023, page 21268–21284. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2023. doi: 10.52202/075280-0930. URL <http://dx.doi.org/10.52202/075280-0930>. 9.6
- [17] Matt Blanco, Jonathan Zong, and Arvind Satyanarayan. Olli: An Extensible Visualization Library for Screen Reader Accessibility. In *IEEE VIS Posters*, 2022. URL <https://vis.csail.mit.edu/pubs/olli>. Accessed: 2026-03-30, <https://vis.csail.mit.edu/pubs/olli>. 2.2.1, 2.4.1, 6.3.3, 6.4.4.2
- [18] Jens Bornschein, Denise Prescher, and Gerhard Weber. Collaborative Creation of Digital Tactile Graphics. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility - ASSETS '15*, ASSETS '15, page 117–126, New York, NY, USA, October 2015. Association for Computing Machinery, ACM Press. doi: 10.1145/2700648.2809869. URL <https://doi.org/10.1145/2700648.2809869>. 6.3.3
- [19] M. Bostock, V. Ogievetsky, and J. Heer. D³ data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, 17(12):2301–2309, December 2011. ISSN 1077-2626. doi: 10.1109/tvcg.2011.185. URL <https://doi.org/10.1109/tvcg.2011.185>. 1.2, 2.2.3, 6.3.1

- [20] Geoffrey C. Bowker and Susan Leigh Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999. ISBN 9780262522953. ISBN: 978-0-262-52295-3. [2.1.2](#)
- [21] Virginia Braun and Victoria Clarke. Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2):77–101, January 2006. ISSN 1478-0895. doi: 10.1191/1478088706qp063oa. URL <https://doi.org/10.1191/1478088706qp063oa>. [6.6.3](#)
- [22] Albert S. Bregman. *Auditory Scene Analysis: The Perceptual Organization of Sound*. The MIT Press, Cambridge, MA, May 1990. ISBN 9780262269209. doi: 10.7551/mitpress/1486.001.0001. URL <https://doi.org/10.7551/mitpress/1486.001.0001>. [7.3.2](#)
- [23] S. Brewster. Visualization tools for blind people using multiple modalities. *Disability and Rehabilitation*, 24(11-12):613–621, January 2002. ISSN 1464-5165. doi: 10.1080/09638280110111388. URL <https://doi.org/10.1080/09638280110111388>. [2.2.1](#), [6.3.2](#)
- [24] Nicola J. Broadbent, Larry R. Squire, and Robert E. Clark. Spatial memory, recognition memory, and the hippocampus. *Proceedings of the National Academy of Sciences*, 101(40):14515–14520, October 2004. ISSN 1091-6490. doi: 10.1073/pnas.0406344101. URL <https://doi.org/10.1073/pnas.0406344101>. [7.4.1](#)
- [25] A. Buja, J.A. McDonald, J. Michalak, and W. Stuetzle. Interactive data visualization using focusing and linking. In *Proceeding Visualization '91*, pages 156–163,, Los Alamitos, CA, December 2002. IEEE Comput. Soc. Press. doi: 10.1109/visual.1991.175794. URL <https://doi.org/10.1109/visual.1991.175794>. [7.2](#), [7.3.1](#), [G4](#):
- [26] Jennifer L. Burke, Matthew S. Prewett, Ashley A. Gray, Liuquin Yang, Frederick R. B. Stilson, Michael D. Coover, Linda R. Elliot, and Elizabeth Redden. Comparing the effects of visual-auditory and visual-tactile feedback on user performance: a meta-analysis. In *Proceedings of the 8th international conference on Multimodal interfaces, ICMi06*, page 108–117. ACM, November 2006. doi: 10.1145/1180995.1181017. URL <https://doi.org/10.1145/1180995.1181017>. [1.1](#)
- [27] Matthew Butler, Leona M Holloway, Samuel Reinders, Cagatay Goncu, and Kim Marriott. Technology developments in touch-based accessible graphics: A systematic review of research 2010-2020. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, CHI '21*, page 1–15. ACM, May 2021. doi: 10.1145/3411764.3445207. URL <https://doi.org/10.1145/3411764.3445207>. [6.3.2](#), [7.2](#), [7.3.2](#)
- [28] Katherine Casey. Radical decentralization: Does community-driven development work? *Annual Review of Economics*, 10(1):139–163, August 2018. ISSN 1941-1391. doi: 10.1146/annurev-economics-080217-053339. URL <http://dx.doi.org/10.1146/annurev-economics-080217-053339>. [10.3](#)
- [29] James I. Charlton. *Nothing About Us Without Us: Disability Oppression and Empowerment*. University of California Press, Berkeley, CA, March 1998. ISBN 9780520925441. doi: 10.1525/9780520925441. URL <https://doi.org/10.1525/9780520925441>

- [30] Pramod Chundury, Biswaksen Patnaik, Yasmin Reyazuddin, Christine Tang, Jonathan Lazar, and Niklas Elmqvist. Towards understanding sensory substitution for accessible visualization: An interview study. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):1084–1094, January 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2021.3114829. URL <https://doi.org/10.1109/tvcg.2021.3114829>. 2.2.1, 6.3.2, 7.9.3
- [31] Pramod Chundury, Urja Thakkar, Yasmin Reyazuddin, J. Bern Jordan, Niklas Elmqvist, and Jonathan Lazar. Understanding the visualization and analytics needs of blind and low-vision professionals. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '24, page 1–5. ACM, October 2024. doi: 10.1145/3663548.3688496. URL <https://doi.org/10.1145/3663548.3688496>. 2.1.1, 7.9.3
- [32] Pramod Chundury, J. Bern Jordan, Yasmin Reyazuddin, Niklas Elmqvist, and Jonathan Lazar. Sound, touch, or the full monty? a comparative study of accessible data exploration systems for blind users. *ACM Transactions on Accessible Computing*, 19(1):1–41, March 2026. ISSN 1936-7236. doi: 10.1145/3798100. URL <https://doi.org/10.1145/3798100>. 7.3.2, 7.9.3
- [33] Andy Cockburn and Bruce McKenzie. Evaluating the effectiveness of spatial memory in 2d and 3d physical and virtual environments. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI02, page 203–210. ACM, April 2002. doi: 10.1145/503376.503413. URL <https://doi.org/10.1145/503376.503413>. 7.4.1
- [34] Maxime Cordeil, Benjamin Bach, Andrew Cunningham, Bastian Montoya, Ross T. Smith, Bruce H. Thomas, and Tim Dwyer. Embodied axes: Tangible, actuated interaction for 3d augmented reality data spaces. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, page 1–12. ACM, April 2020. doi: 10.1145/3313831.3376613. URL <https://doi.org/10.1145/3313831.3376613>. 7.3.4, 7.5.2
- [35] Clare Cullen and Oussama Metatla. Co-designing inclusive multisensory story mapping with children with mixed visual abilities. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children*, IDC '19, pages 361–373. ACM, June 2019. doi: 10.1145/3311927.3323146. URL <https://doi.org/10.1145/3311927.3323146>. 2.2.1, 6.4
- [36] Lilian de Greef, Dominik Moritz, and Cynthia Bennett. Interdependent variables: Remotely designing tactile graphics for an accessible workflow. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '21, page 1–6. ACM, October 2021. doi: 10.1145/3441852.3476468. URL <https://doi.org/10.1145/3441852.3476468>. 2.2.1, 6.4
- [37] Sergio Della Sala, Colin Gray, Alan Baddeley, Nadia Allamano, and Lindsey Wilson. Pattern span: a tool for unwelding visuo-spatial memory. *Neuropsychologia*, 37(10):

- 1189–1199, September 1999. ISSN 0028-3932. doi: 10.1016/s0028-3932(98)00159-6. URL [https://doi.org/10.1016/s0028-3932\(98\)00159-6](https://doi.org/10.1016/s0028-3932(98)00159-6). 7.4.1
- [38] Deque. Automated Testing Identifies 57 Percent of Digital Accessibility Issues. Technical report, Deque, March 2021. Accessed: 2021-09-06. 3
- [39] Alan Dix. Designing for appropriation. In *Electronic Workshops in Computing*. BCS Learning & Development, September 2007. doi: 10.14236/ewic/hci2007.53. URL <https://doi.org/10.14236/ewic/hci2007.53>. 2.1.2
- [40] Sebastian Draxler, Gunnar Stevens, Martin Stein, Alexander Boden, and David Randall. Supporting the social context of technology appropriation: on a synthesis of sharing tools and tool knowledge. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI ’12, page 2835–2844. ACM, May 2012. doi: 10.1145/2207676.2208687. URL <https://doi.org/10.1145/2207676.2208687>. 2.1.2
- [41] Anthony Dunne and Fiona Raby. *Speculative everything*. The MIT Press. MIT Press, London, England, December 2013. ISBN 9780262019842. 7.8.4, 7.9
- [42] Niklas Egger, Gottfried Zimmermann, and Christophe Strobbe. *Overlay Tools as a Support for Accessible Websites – Possibilities and Limitations*, page 6–17. Springer International Publishing, 2022. ISBN 9783031086458. doi: 10.1007/978-3-031-08645-8_2. URL http://dx.doi.org/10.1007/978-3-031-08645-8_2. 9.6
- [43] Frank Elavsky and Cindy Xiong Bearfield. Playing telephone with generative models: “verification disability,” “compelled reliance,” and accessibility in data visualization. In *2025 IEEE Workshop on Accessible Data Visualization (AccessViz)*, page 14–24. IEEE, November 2025. doi: 10.1109/accessviz68666.2025.00008. URL <https://doi.org/10.1109/accessviz68666.2025.00008>. 6.3.2, 7.1.1
- [44] Frank Elavsky, Cynthia Bennett, and Dominik Moritz. How accessible is my visualization? evaluating visualization accessibility with chartability. *Computer Graphics Forum*, 41(3):57–70, June 2022. doi: 10.1111/cgf.14522. URL <https://doi.org/10.1111/cgf.14522>. 6.4.3, 6.8.2, **G3**:
- [45] Frank Elavsky, Lucas Nadolskis, and Dominik Moritz. Data navigator: An accessibility-centered data navigation toolkit. *IEEE Transactions on Visualization and Computer Graphics*, page 1–11, 2023. ISSN 2160-9306. doi: 10.1109/tvcg.2023.3327393. URL <https://doi.org/10.1109/tvcg.2023.3327393>. 6.2, 6.3.2, 6.3.3, 6.4, 7.2, 7.3.3, 7.9.2
- [46] A. Endert, L. Bradel, and C. North. Beyond control panels: Direct manipulation for visual analytics. *IEEE Computer Graphics and Applications*, 33(4):6–13, July 2013. ISSN 0272-1716. doi: 10.1109/mcg.2013.53. URL <https://doi.org/10.1109/mcg.2013.53>. 6.2
- [47] Christin Engel and Gerhard Weber. Analysis of tactile chart design. In *Proceedings of the 10th International Conference on Pervasive Technologies Related to Assistive Environments*, PETRA ’17, pages 197–200. ACM, June 2017. doi: 10.1145/3056540.3064955. URL <https://doi.org/10.1145/3056540.3064955>. 7.3.2

- [48] Pavithra Eswaramoorthy, Frank Elavsky, Tania Allard, and gabalafou. Quansight-Labs/bokeh-a11y-audit: v1.0.0, February 2025. URL <https://doi.org/10.5281/zenodo.14923642>. 6.4.3
- [49] Danyang Fan, Alexa Fay Siu, Wing-Sum Adrienne Law, Raymond Ruihong Zhen, Sile O’Modhrain, and Sean Follmer. Slide-tone and tilt-tone: 1-dof haptic techniques for conveying shape characteristics of graphs to blind users. In *CHI Conference on Human Factors in Computing Systems*, CHI ’22, page 1–19. ACM, April 2022. doi: 10.1145/3491102.3517790. URL <https://doi.org/10.1145/3491102.3517790>. 7.3.2
- [50] Danyang Fan, Alexa Fay Siu, Hrishikesh Rao, Gene Sung-Ho Kim, Xavier Vazquez, Lucy Greco, Sile O’Modhrain, and Sean Follmer. The accessibility of data visualizations on the web for screen reader users: Practices and experiences during covid-19. *ACM Transactions on Accessible Computing*, 16(1):1–29, March 2023. ISSN 1936-7236. doi: 10.1145/3557899. URL <https://doi.org/10.1145/3557899>. 1.1, 6.3.3
- [51] Ali Mazraeh Farahani, Peyman Adibi, Mohammad Saeed Ehsani, Hans-Peter Hutter, and Alireza Darvishy. Automatic chart understanding: A review. *IEEE Access*, 11: 76202–76221, 2023. ISSN 2169-3536. doi: 10.1109/access.2023.3298050. URL <https://doi.org/10.1109/access.2023.3298050>. 6.3.2
- [52] Stephen Few. Visual Business Intelligence - Data Visualization and the Blind. <https://www.perceptualedge.com/blog/?p=1756>, 2013. [Accessed 05-04-2025]. 1.1
- [53] Danyel Fisher, Igor Popov, Steven Drucker, and m.c. schraefel. Trust me, i’m partially right: incremental visualization lets analysts explore large datasets faster. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI ’12, page 1673–1682. ACM, May 2012. doi: 10.1145/2207676.2208294. URL <https://doi.org/10.1145/2207676.2208294>. 6.2, 7.2, 7.3.1, **G5**
- [54] Chris Fleizach and Jeffrey P. Bigham. System-class accessibility: The architectural support for making a whole system usable by people with disabilities. *Queue*, 22(5): 28–39, October 2024. ISSN 1542-7749. doi: 10.1145/3704627. URL <http://dx.doi.org/10.1145/3704627>. 2.2.2, 9.6
- [55] John H. Flowers, Dion C. Buhman, and Kimberly D. Turnage. Cross-Modal Equivalence of Visual and Auditory Scatterplots for Exploring Bivariate Data Samples. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 39(3):341–351, September 1997. doi: 10.1518/001872097778827151. Accessed: 2021-09-03. 2.2.1, 3
- [56] Steven L. Franconeri, Lace M. Padilla, Priti Shah, Jeffrey M. Zacks, and Jessica Hullman. The science of visual data communication: What works. *Psychological Science in the Public Interest*, 22(3):110–161, December 2021. ISSN 1539-6053. doi: 10.1177/15291006211051956. URL <https://doi.org/10.1177/15291006211051956>. 1.2
- [57] Krzysztof Z. Gajos, Daniel S. Weld, and Jacob O. Wobbrock. Automatically generating personalized user interfaces with SUPPLE. *Artificial Intelligence*, 174(12–13):910–950, August 2010. ISSN 0004-3702. doi: 10.1016/j.artint.2010.05.005. URL <https://doi.org/10.1016/j.artint.2010.05.005>. 2.4.3

- [58] Sara Garcia-Navarra, Lucia Solares, and Marta Mendez. Virtual reality and mixed reality in the assessment of spatial memory. *Exploration of Digital Health Technologies*, 3, June 2025. ISSN 2996-9409. doi: 10.37349/edht.2025.101151. URL <https://doi.org/10.37349/edht.2025.101151>. 7.4.1, **G1**:
- [59] Paula Gardner and Jess Rauchberg. *Feminist, Postcolonial, and Crip Approaches to Human-machine Communication Methodology*. Sage Publishing, 2023. 10.2
- [60] Elena Gherri and Martin Eimer. Active listening impairs visual perception and selectivity: An erp study of auditory dual-task costs on visual attention. *Journal of Cognitive Neuroscience*, 23(4):832–844, April 2011. ISSN 1530-8898. doi: 10.1162/jocn.2010.21468. URL <https://doi.org/10.1162/jocn.2010.21468>. 7.3.2, **G2**:
- [61] Kate S Glazko, Momona Yamagami, Aashaka Desai, Kelly Avery Mack, Venkatesh Potluri, Xuhai Xu, and Jennifer Mankoff. An autoethnographic case study of generative artificial intelligence’s utility for accessibility. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’23, page 1–8. ACM, October 2023. doi: 10.1145/3597638.3614548. URL <http://dx.doi.org/10.1145/3597638.3614548>. 10.2
- [62] Michele E. Gloede and Melissa K. Gregg. The fidelity of visual and auditory memory. *Psychonomic Bulletin & Review*, 26(4):1325–1332, April 2019. ISSN 1531-5320. doi: 10.3758/s13423-019-01597-7. URL <https://doi.org/10.3758/s13423-019-01597-7>. 1.1
- [63] A. Jonathan R. Godfrey, Paul Murrell, and Volker Sorge. An accessible interaction model for data visualisation in statistics. In *Lecture Notes in Computer Science*, page 590–597. Springer International Publishing, 2018. ISBN 9783319942773. doi: 10.1007/978-3-319-94277-3_92. URL https://doi.org/10.1007/978-3-319-94277-3_92. 2.2.1
- [64] Jenna L. Gorlewicz, Jennifer L. Tennison, P. Merlin Uesbeck, Margaret E. Richard, Hari P. Palani, Andreas Stefik, Derrick W. Smith, and Nicholas A. Giudice. Design guidelines and recommendations for multimodal, touchscreen-based graphics. *ACM Transactions on Accessible Computing*, 13(3):1–30, August 2020. ISSN 1936-7236. doi: 10.1145/3403933. URL <https://doi.org/10.1145/3403933>. 7.3.2, 7.4.1, 7.5.2
- [65] David Graeber and Charlie Rose. Never cower to authority, 2006. URL <https://www.youtube.com/watch?v=qt5op6wft-8>. Accessed [2026]. 10.3
- [66] R.L. Graham. An efficient algorithm for determining the convex hull of a finite planar set. *Information Processing Letters*, 1(4):132–133, 1972. ISSN 0020-0190. doi: 10.1016/0020-0190(72)90045-2. URL [http://dx.doi.org/10.1016/0020-0190\(72\)90045-2](http://dx.doi.org/10.1016/0020-0190(72)90045-2). 6.5.4.2
- [67] David Gray Widder, Sarah West, and Meredith Whittaker. Open (for business): Big tech, concentrated power, and the political economy of open ai. *SSRN Electronic Journal*, 2023. ISSN 1556-5068. doi: 10.2139/ssrn.4543807. URL <https://doi.org/10.2139/ssrn.4543807>. 1.2
- [68] Catherine Grevet and Eric Gilbert. Piggyback prototyping: Using existing, large-scale

social computing systems to prototype new ones. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, CHI '15, page 4047–4056. ACM, April 2015. doi: 10.1145/2702123.2702395. URL <https://doi.org/10.1145/2702123.2702395>. 2.1.2

- [69] Karl Groves. Overlay fact sheet: What it is and how to make it work for you - karl groves, March 2023. URL <https://karlgroves.com/overlay-fact-sheet-what-it-is-and-how-to-make-it-work-for-you/>. 9.6
- [70] Nathan Hahn, Joseph Chang, Ji Eun Kim, and Aniket Kittur. The knowledge accelerator: Big picture thinking in small pieces. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, CHI'16, page 2258–2270. ACM, May 2016. doi: 10.1145/2858036.2858364. URL <https://doi.org/10.1145/2858036.2858364>. 2.1.1
- [71] Jen Hale. Extensive digitization of tactile map collection. <https://www.perkins.org/extensive-digitization-of-tactile-map-collection/>, 2016. Perkins Archives Blog, Perkins School for the Blind. July 2016. Accessed: 2022-02-23. 2.2.1
- [72] Foad Hamidi, Patrick Mbullo, Deurence Onyango, Michaela Hynie, Susan McGrath, and Melanie Baljko. Participatory design of diy digital assistive technology in western kenya. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities*, AfriCHI '18, page 1–11. ACM, December 2018. doi: 10.1145/3283458.3283478. URL <https://doi.org/10.1145/3283458.3283478>. 2.1.1
- [73] Noor Hammad, Frank Elavsky, Sanika Moharana, Jessie Chen, Seyoung Lee, Patrick Carrington, Dominik Moritz, Jessica Hammer, and Erik Harpstead. Exploring the affordances of game-aware streaming to support blind and low vision viewers: A design probe study. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '24, page 1–13. ACM, October 2024. doi: 10.1145/3663548.3675665. URL <https://doi.org/10.1145/3663548.3675665>. 6.1.1, 6.2
- [74] A. Hamraie. Making access critical: Disability, race, and gender in environmental design. <https://belonging.berkeley.edu/aimi-hamraie-making-access-critical-disability-race-and-gender-environmental-design/>, 2019. [Online]. 2.3, 3
- [75] Aimi Hamraie. *Building Access: Universal Design and the Politics of Disability*. University of Minnesota Press, Minneapolis, MN, 2017. ISBN 9781517901639. ISBN: 978-1-5179-0163-9. 2.3, 2.4.3
- [76] Aimi Hamraie. Designing collective access: A feminist disability theory of universal design. In *Disability, space, architecture*, pages 78–297. Routledge, 2017. 2.3, 9.7.2
- [77] Aimi Hamraie and Kelly Fritsch. Crip technoscience manifesto. *Catalyst: Feminism, Theory, Technoscience*, 5(1):1–33, April 2019. ISSN 2380-3312. doi: 10.28968/cftt.v5i1.29607. URL <https://doi.org/10.28968/cftt.v5i1.29607>. 2.3
- [78] Gunnar Harboe and Elaine M. Huang. Real-world affinity diagramming practices: Bridging the paper-digital gap. In *Proceedings of the 33rd Annual ACM Conference on Human*

Factors in Computing Systems, CHI '15, page 95–104. ACM, April 2015. doi: 10.1145/2702123.2702561. URL <https://doi.org/10.1145/2702123.2702561>. 6.6.3, 7.6.7

- [79] Parker Hartman and Tim Gorichanaz. Evaluating ai-powered website accessibility overlays. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '25, page 1–4. ACM, October 2025. doi: 10.1145/3663547.3759761. URL <http://dx.doi.org/10.1145/3663547.3759761>. 9.6, 10.1
- [80] Gillian R. Hayes. The relationship of action research to human-computer interaction. *ACM Transactions on Computer-Human Interaction*, 18(3):1–20, July 2011. ISSN 1073-0516. doi: 10.1145/1993060.1993065. URL <https://doi.org/10.1145/1993060.1993065>. 6.2, 6.4, 10.2
- [81] Jeffrey Heer and Maneesh Agrawala. Design considerations for collaborative visual analytics. In *2007 IEEE Symposium on Visual Analytics Science and Technology*, page 171–178. IEEE, October 2007. doi: 10.1109/vast.2007.4389011. URL <https://doi.org/10.1109/vast.2007.4389011>. 6.2, 7.3.1
- [82] Jeffrey Heer and Dominik Moritz. Mosaic: An architecture for scalable & interoperable data views. *IEEE Transactions on Visualization and Computer Graphics*, page 1–11, 2023. ISSN 2160-9306. doi: 10.1109/tvcg.2023.3327189. URL <https://doi.org/10.1109/tvcg.2023.3327189>. 1.2, 2.2.3, 2.4.2, 3, 7.3.1, G5:, 7.5.3, 7.9.1
- [83] Ana O Henriques, Anna R. L. Carter, Beatriz Severes, Reem Talhouk, Angelika Strohmayer, Ana Cristina Pires, Colin M. Gray, Kyle Montague, and Hugo Nicolau. A feminist care ethics toolkit for community-based design: Bridging theory and practice. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, page 1–26. ACM, April 2025. doi: 10.1145/3706598.3713950. URL <http://dx.doi.org/10.1145/3706598.3713950>. 10.3
- [84] Thomas Hermann, Andy Hunt, and John G Neuhoff, editors. *The Sonification Handbook*. Logos Verlag Berlin, Berlin, Germany, December 2011. ISBN 9783832528195. ISBN: 978-3-8325-2819-5. 2.2.1, 6.3.2, 7.3.2
- [85] Jaylin Herskovitz, Andi Xu, Rahaf Alharbi, and Anhong Guo. Hacking, switching, combining: Understanding and supporting diy assistive technology design by blind people. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, page 1–17. ACM, April 2023. doi: 10.1145/3544548.3581249. URL <https://doi.org/10.1145/3544548.3581249>. 2.1.2
- [86] Highsoft. Highcharts - interactive charting library for developers. <https://www.highcharts.com/>, 2009. [Online]. 2.2.1
- [87] Megan Hofmann, Devva Kasnitz, Jennifer Mankoff, and Cynthia L Bennett. Living disability theory: Reflections on access, research, and design. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '20, page 1–13. ACM, October 2020. doi: 10.1145/3373625.3416996. URL <http://dx.doi.org/10.1145/3373625.3416996>. 2.3, 10.2
- [88] Fred Hohman, Mary Beth Kery, Donghao Ren, and Dominik Moritz. Model compres-

sion in practice: Lessons learned from practitioners creating on-device machine learning experiences. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, page 1–18. ACM, May 2024. doi: 10.1145/3613904.3642109. URL <http://dx.doi.org/10.1145/3613904.3642109>. 2.1.2

- [89] Leona Holloway, Peter Cracknell, Kate Stephens, Melissa Fanshawe, Samuel Reinders, Kim Marriott, and Matthew Butler. Refreshable tactile displays for accessible data visualisation, 2024. URL <https://doi.org/10.48550/arxiv.2401.15836>. 6.3.2, 7.2
- [90] Leona M Holloway, Cagatay Goncu, Alon Ilisar, Matthew Butler, and Kim Marriott. Infosonics: Accessible infographics for people who are blind using sonification and voice. In *CHI Conference on Human Factors in Computing Systems*, CHI '22, page 1–13. ACM, April 2022. doi: 10.1145/3491102.3517465. URL <https://doi.org/10.1145/3491102.3517465>. 6.3.2, 7.2
- [91] Jonathan Hook, Sanne Verbaan, Abigail Durrant, Patrick Olivier, and Peter Wright. A study of the challenges related to diy assistive technology in the context of children with disabilities. In *Proceedings of the 2014 conference on Designing interactive systems*, DIS '14, pages 597–606. ACM, June 2014. doi: 10.1145/2598510.2598530. URL <https://doi.org/10.1145/2598510.2598530>. 2.1.1
- [92] Stacy Hsueh, Beatrice Vincenzi, Akshata Murdeshwar, and Marianela Ciolfi Felice. Crip-ping data visualizations: Crip technoscience as a critical lens for designing digital access. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '23, page 1–16. ACM, October 2023. doi: 10.1145/3597638.3608427. URL <http://dx.doi.org/10.1145/3597638.3608427>. 1.1, 2.3, 3, 9.7.2, 10.2
- [93] Amy Hurst and Shaun Kane. Making "making" accessible. In *Proceedings of the 12th International Conference on Interaction Design and Children*, IDC '13, pages 635–638, New York, NY, USA, June 2013. Association for Computing Machinery, ACM. doi: 10.1145/2485760.2485883. Accessed: 2021-09-03. 2.1.1, 9.7.1, 10.2
- [94] Amy Hurst and Jasmine Tobias. Empowering individuals with do-it-yourself assistive technology. In *The proceedings of the 13th international ACM SIGACCESS conference on Computers and accessibility*, ASSETS '11, page 11–18. ACM, October 2011. doi: 10.1145/2049536.2049541. URL <https://doi.org/10.1145/2049536.2049541>. 2.1.1, 7.5
- [95] Edwin L. Hutchins, James D. Hollan, and Donald A. Norman. Direct manipulation interfaces. *Human–Computer Interaction*, 1(4):311–338, December 1985. ISSN 1532-7051. doi: 10.1207/s15327051hci0104.2. URL https://doi.org/10.1207/s15327051hci0104_2. 6.2, 6.3.1
- [96] Hilary Hutchinson, Wendy Mackay, Bo Westerlund, Benjamin B. Bederson, Allison Druin, Catherine Plaisant, Michel Beaudouin-Lafon, Stéphane Conversy, Helen Evans, Heiko Hansen, Nicolas Roussel, and Björn Eiderbäck. Technology probes: inspiring design for and with families. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI03, page 17–24. ACM, April 2003. doi:

10.1145/642611.642616. URL <https://doi.org/10.1145/642611.642616>. 6.1.1, 6.2

- [97] Liz Jackson, Alex Haagaard, and Rua Williams. Disability Dongle. <https://blog.castac.org/2022/04/disability-dongle/>, 2022. 10.2
- [98] Farnaz Jahanbakhsh, Amy X. Zhang, Karrie Karahalios, and David R. Karger. Our browser extension lets readers change the headlines on news articles, and you won't believe what they did! *Proceedings of the ACM on Human-Computer Interaction*, 6 (CSCW2):1–33, November 2022. ISSN 2573-0142. doi: 10.1145/3555643. URL <http://dx.doi.org/10.1145/3555643>. 10.3
- [99] Bin Jiang, Ferjan Ormeling, and Wolfgang Kainz. Visualization support for fuzzy spatial analysis. In *1995 ACSM/ASPRS Annual Convention & Exposition: Technical Papers*, volume 4: Auto Carto 12, pages 291–310, United States, 1994. American Congress on Surveying and Mapping. ISBN 1-57083-019-3. 12th International Symposium on Computer-Assisted Cartography, Auto-Carto 1995, Auto-Carto XII ; Conference date: 27-02-1995 Through 01-03-1995. 7.8.2
- [100] Chutian Jiang, Wentao Lei, Emily Kuang, Teng Han, and Mingming Fan. Understanding strategies and challenges of conducting daily data analysis (dda) among blind and low-vision people. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '23, page 1–15. ACM, October 2023. doi: 10.1145/3597638.3608423. URL <https://doi.org/10.1145/3597638.3608423>. 2.1.1
- [101] Shuli Jones, Isabella Pedraza Pineros, Daniel Hajas, Jonathan Zong, and Arvind Satyanarayan. “customization is key”: Reconfigurable textual tokens for accessible data visualizations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, page 1–14. ACM, May 2024. doi: 10.1145/3613904.3641970. URL <https://doi.org/10.1145/3613904.3641970>. 2.4.3, 3, 6.3.2, 6.5.2
- [102] Shakila Cherise S Joyner, Amalia Riegelhuth, Kathleen Garrity, Yea-Seul Kim, and Nam Wook Kim. Visualization accessibility in the wild: Challenges faced by visualization designers. In *CHI Conference on Human Factors in Computing Systems*, CHI '22, page 1–19. ACM, April 2022. doi: 10.1145/3491102.3517630. URL <https://doi.org/10.1145/3491102.3517630>. 2.2.3, 3, 6.2, 6.3.3, 6.8.2, 9.1, 10.2
- [103] Crescentia Jung, Shubham Mehta, Atharva Kulkarni, Yuhang Zhao, and Yea-Seul Kim. Communicating visualizations without visuals: Investigation of visualization alternative text for people with visual impairments. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):1095–1105, January 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2021.3114846. URL <https://doi.org/10.1109/tvcg.2021.3114846>. 2.2.1, 2.4, 6.3.2, 7.2, 7.3.2
- [104] Shaun K. Kane, Jeffrey P. Bigham, and Jacob O. Wobbrock. Slide rule: Making mobile touch screens accessible to blind people using multi-touch interaction techniques. In *Proceedings of the 10th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS08, page 73–80. ACM, October 2008. doi: 10.1145/1414471.1414

487. URL <https://doi.org/10.1145/1414471.1414487>. 2.4.2

- [105] Hyeonsu B. Kang, Xin Qian, Tom Hope, Dafna Shahaf, Joel Chan, and Aniket Kittur. Augmenting scientific creativity with an analogical search engine. *ACM Transactions on Computer-Human Interaction*, 29(6):1–36, November 2022. ISSN 1557-7325. doi: 10.1145/3530013. URL <https://doi.org/10.1145/3530013>. 2.1.1
- [106] Os Keyes, Josephine Hoy, and Margaret Drouhard. Human-computer insurrection: Notes on an anarchist hci. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI ’19, page 1–13. ACM, May 2019. doi: 10.1145/3290605.3300569. URL <http://dx.doi.org/10.1145/3290605.3300569>. 10.3
- [107] Gyeongri Kim, Jiho Kim, and Yea-Seul Kim. “explain what a treemap is”: Exploratory investigation of strategies for explaining unfamiliar chart to blind and low vision users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI ’23, page 1–13. ACM, April 2023. doi: 10.1145/3544548.3581139. URL <https://doi.org/10.1145/3544548.3581139>. 6.3.2
- [108] Hyeok Kim, Yea-Seul Kim, and Jessica Hullman. Erie: A declarative grammar for data sonification. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI ’24, page 1–19. ACM, May 2024. doi: 10.1145/3613904.3642442. URL <https://doi.org/10.1145/3613904.3642442>. 6.3.2, 6.3.3, 7.2, 7.3.2
- [109] Jiho Kim, Arjun Srinivasan, Nam Wook Kim, and Yea-Seul Kim. Exploring chart question answering for blind and low vision users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI ’23, page 1–15. ACM, April 2023. doi: 10.1145/3544548.3581532. URL <https://doi.org/10.1145/3544548.3581532>. 2.2.1, 6.3.2, 7.3.3
- [110] N. W. Kim, S. C. Joyner, A. Riegelhuth, and Y. Kim. Accessible visualization: Design space, opportunities, and challenges. *Computer Graphics Forum*, 40(3):173–188, June 2021. doi: 10.1111/cgf.14298. URL <https://doi.org/10.1111/cgf.14298>. 2.2.3, 6.3.2
- [111] N. W. Kim, G. Ataguba, S. C. Joyner, Chuangdian Zhao, and Hyejin Im. Beyond alternative text and tables: Comparative analysis of visualization tools and accessibility methods. *Computer Graphics Forum*, 42(3):323–335, June 2023. ISSN 1467-8659. doi: 10.1111/cgf.14833. URL <https://doi.org/10.1111/cgf.14833>. 2.4.3, 6.3.2, 6.3.3
- [112] David Ledo, Steven Houben, Jo Vermeulen, Nicolai Marquardt, Lora Oehlberg, and Saul Greenberg. Evaluation strategies for HCI toolkit research. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI ’18, page 1–17. ACM, April 2018. doi: 10.1145/3173574.3173610. URL <https://doi.org/10.1145/3173574.3173610>. 2.1.2, 2.2.3
- [113] Xiaou Li, Laura A. Carlson, Weimin Mou, Mark R. Williams, and Jared E. Miller. Describing spatial locations from perception and memory: The influence of intrinsic axes on reference object selection. *Journal of Memory and Language*, 65(2):222–236, August 2011. ISSN 0749-596X. doi: 10.1016/j.jml.2011.04.001. URL <https://doi.org/10.1016/j.jml.2011.04.001>

[//doi.org/10.1016/j.jml.2011.04.001](https://doi.org/10.1016/j.jml.2011.04.001). 7.4.1

- [114] Zhicheng Liu and Jeffrey Heer. The effects of interactive latency on exploratory visual analysis. *IEEE Transactions on Visualization and Computer Graphics*, 20(12):2122–2131, December 2014. ISSN 1077-2626. doi: 10.1109/tvcg.2014.2346452. URL <https://doi.org/10.1109/tvcg.2014.2346452>. 2.4.2, 3, 6.2, 7.2, 7.3.1, **G5**, 7.10, 9.6
- [115] Tania Llana, Marta Mendez, M.-Carmen Juan, and Magdalena Mendez-Lopez. Navigational object-location memory assessment in real and virtual environments: A systematic review. *Behavioural Brain Research*, 480:115388, March 2025. ISSN 0166-4328. doi: 10.1016/j.bbr.2024.115388. URL <https://doi.org/10.1016/j.bbr.2024.115388>. 7.4.1, **G1**:
- [116] Alan Lundgard and Arvind Satyanarayan. Accessible visualization via natural language descriptions: A four-level model of semantic content. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):1073–1083, January 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2021.3114770. URL <https://doi.org/10.1109/tvcg.2021.3114770>. 2.2.1, 2.4, 6.3.2, 7.2, 7.3.2
- [117] Alan Lundgard, Crystal Lee, and Arvind Satyanarayan. Sociotechnical considerations for accessible visualization design. In *2019 IEEE Visualization Conference (VIS)*, page 16–20. IEEE, October 2019. doi: 10.1109/visual.2019.8933762. URL <https://doi.org/10.1109/visual.2019.8933762>. 6.3.3, 10.2
- [118] Kelly Mack, Emma McDonnell, Dhruv Jain, Lucy Lu Wang, Jon E. Froehlich, and Leah Findlater. What do we mean by “accessibility research”? A literature survey of accessibility papers in chi and assets from 1994 to 2019. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI ’21, page 1–18, New York, NY, USA, May 2021. ACM. ISBN 978-1-4503-8096-6. doi: 10.1145/3411764.3445412. URL <https://doi.org/10.1145/3411764.3445412>. Accessed: 2022-02-22. 2.2.3
- [119] Els Maeckelberghe. Feminist ethic of care: A third alternative approach. *Health Care Analysis*, 12(4):317–327, December 2004. ISSN 1573-3394. doi: 10.1007/s10728-004-6639-6. URL <http://dx.doi.org/10.1007/s10728-004-6639-6>. 10.3
- [120] Tlameo Makati, Garreth W. Tigwell, and Kristen Shinohara. The promise and pitfalls of web accessibility overlays for blind and low vision users. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’24, page 1–12. ACM, October 2024. doi: 10.1145/3663548.3675650. URL <http://dx.doi.org/10.1145/3663548.3675650>. 9.6
- [121] R G Maling and D C Clarkson. Electronic controls for the tetraplegic (possum) (patient operated selector mechanisms—p.o. s.m.). *Spinal Cord*, 1(3):161–174, November 1963. ISSN 1476-5624. doi: 10.1038/sc.1963.15. URL <http://dx.doi.org/10.1038/sc.1963.15>. 3
- [122] Jennifer Mankoff, Gillian R. Hayes, and Devva Kasnitz. Disability studies as a source of critical inquiry for the field of assistive technology. In *Proceedings of the 12th interna-*

- tional ACM SIGACCESS conference on Computers and accessibility, ASSETS '10, page 3–10, New York, NY, USA, October 2010. Association for Computing Machinery, ACM. doi: 10.1145/1878803.1878807. URL <https://doi.org/10.1145/1878803.1878807>. Accessed: 2021-09-06. 10.2
- [123] Douglass L. Mansur, Merra M. Blattner, and Kenneth I. Joy. Sound graphs: A numerical data analysis method for the blind. *Journal of Medical Systems*, 9(3):163–174, June 1985. ISSN 1573-689X. doi: 10.1007/bf00996201. URL <https://doi.org/10.1007/bf00996201>. 2.2.1, 6.3.2
 - [124] Deborah Marks. Models of disability. *Disability and Rehabilitation*, 19(3):85–91, January 1997. ISSN 1464-5165. doi: 10.3109/09638289709166831. URL <https://doi.org/10.3109/09638289709166831>. 1.1, 9.1
 - [125] Kim Marriott, Bongshin Lee, Matthew Butler, Ed Cutrell, Kirsten Ellis, Cagatay Goncu, Marti Hearst, Kathleen McCoy, and Danielle Albers Szafr. Inclusive data visualization for people with disabilities: a call to action. *Interactions*, 28(3):47–51, April 2021. ISSN 1558-3449. doi: 10.1145/3457875. URL <https://doi.org/10.1145/3457875>. 2.2.3, 6.3.2, 7.2
 - [126] Kim Marriott, Matthew Butler, Leona Holloway, William Jolley, Bongshin Lee, Bruce Maguire, and Danielle Albers Szafr. From vision to touch: Bridging visual and tactile principles for accessible data representation. *IEEE Transactions on Visualization and Computer Graphics*, 32(1):659–669, January 2026. ISSN 2160-9306. doi: 10.1109/tvcg.2025.3634254. URL <https://doi.org/10.1109/tvcg.2025.3634254>. 6.3.2, 7.2
 - [127] David K. McGookin and Stephen A. Brewster. Soundbar: Exploiting multiple views in multimodal graph browsing. In *Proceedings of the 4th Nordic conference on Human-computer interaction: Changing roles*, NORDICHI06, page 145–154, New York, NY, USA, October 2006. Association for Computing Machinery, ACM. doi: 10.1145/1182475.1182491. URL <https://doi.org/10.1145/1182475.1182491>. Accessed: 2021-09-03. 2.2.1
 - [128] Andrew M. McNutt. No grammar to rule them all: A survey of json-style dsls for visualization. *IEEE Transactions on Visualization and Computer Graphics*, page 1–11, 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2022.3209460. URL <https://doi.org/10.1109/tvcg.2022.3209460>. 2.1.2, 6.2
 - [129] Andrew Michael McNutt. *Understanding and Enhancing JSON-based DSL Interfaces for Visualization*. PhD thesis, University of Chicago, 2023. URL <https://doi.org/10.31219/osf.io/fy246>. 2.1.2
 - [130] Catherine Mei*, Josh Pollock*, Daniel Hajas, Jonathan Zong, and Arvind Satyanarayan. Benthic: Perceptually congruent structures for accessible charts and diagrams. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '25, page 1–17. ACM, October 2025. doi: 10.1145/3663547.3746342. URL <https://doi.org/10.1145/3663547.3746342>. 2.2.1, 6.3.2, 6.3.3, 7.2
 - [131] Mia Mingus. Access intimacy: The missing link. <https://leavingevidence>.

wordpress.com/2011/05/05/access-intimacy-the-missing-link/, May 2011. Blog post, *Leaving Evidence*. 1.1, 2.3, 9.7.2

- [132] Ehsan Mohammadi, Mike Thelwall, Stefanie Haustein, and Vincent Larivière. Who reads research articles? an altmetrics analysis of `scpm/scpendeley` user categories. *Journal of the Association for Information Science and Technology*, 66(9):1832–1846, April 2015. ISSN 2330-1643. doi: 10.1002/asi.23286. URL <http://dx.doi.org/10.1002/asi.23286>. 10.2
- [133] Giles Mohan and Kristian Stokke. Participatory development and empowerment: The dangers of localism. *Third World Quarterly*, 21(2):247–268, April 2000. ISSN 1360-2241. doi: 10.1080/01436590050004346. URL <http://dx.doi.org/10.1080/01436590050004346>. 10.3
- [134] Dominik Moritz, Bill Howe, and Jeffrey Heer. Falcon: Balancing interactive latency and resolution sensitivity for scalable linked visualizations. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI ’19, page 1–11. ACM, May 2019. doi: 10.1145/3290605.3300924. URL <https://doi.org/10.1145/3290605.3300924>. 3, 7.3.1
- [135] Peya Mowar, Aaron Steinfeld, and Jeffrey P. Bigham. iTagPDF: Towards finally automating PDF accessibility. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI ’26, page 1–17. ACM, April 2026. doi: 10.1145/3772318.3790289. URL <https://doi.org/10.1145/3772318.3790289>. Best Paper Award. 6.3.3
- [136] Brad A. Myers. Taxonomies of visual programming and program visualization. *Journal of Visual Languages & Computing*, 1(1):97–123, March 1990. ISSN 1045-926X. doi: 10.1016/S1045-926X(05)80036-9. URL [https://doi.org/10.1016/S1045-926X\(05\)80036-9](https://doi.org/10.1016/S1045-926X(05)80036-9). 6.3.1
- [137] I. Nassi and B. Shneiderman. Flowchart techniques for structured programming. *ACM SIGPLAN Notices*, 8(8):12–26, August 1973. ISSN 1558-1160. doi: 10.1145/953349.953350. URL <https://doi.org/10.1145/953349.953350>. 6.3.1
- [138] Catherine A. Okoro, NaTasha D. Hollis, Alissa C. Cyrus, and Shannon Griffin-Blake. Prevalence of disabilities and health care access by disability status and type among adults — united states, 2016. *MMWR. Morbidity and Mortality Weekly Report*, 67(32):882–887, August 2018. ISSN 1545-861X. doi: 10.15585/mmwr.mm6732a3. URL <https://doi.org/10.15585/mmwr.mm6732a3>. 1.1
- [139] Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, January 1979. ISSN 2168-2909. doi: 10.1109/tsmc.1979.4310076. URL <http://dx.doi.org/10.1109/TSMC.1979.4310076>. 6.5.4.3
- [140] Amandalynne Paullada, Inioluwa Deborah Raji, Emily M. Bender, Emily Denton, and Alex Hanna. Data and its (dis)contents: A survey of dataset development and use in machine learning research. *Patterns*, 2(11):100336, November 2021. ISSN 2666-3899. doi: 10.1016/j.patter.2021.100336. URL <http://dx.doi.org/10.1016/j.pat>

ter.2021.100336. 9.6

- [141] Jorge Poco and Jeffrey Heer. Reverse-engineering visualizations: Recovering visual encodings from chart images. *Computer Graphics Forum*, 36(3):353–363, 2017. ISSN 1467-8659. doi: 10.1111/cgf.13193. URL <http://dx.doi.org/10.1111/cgf.13193>. 6.5.4.3
- [142] Christopher Power, André Freire, Helen Petrie, and David Swallow. Guidelines are only half of the story: Accessibility problems encountered by blind users on the web. In *Proceedings of the SIGCHI conference on human factors in computing systems*, CHI ’12, page 433–442, New York, NY, USA, May 2012. Association for Computing Machinery, ACM. doi: 10.1145/2207676.2207736. URL <https://doi.org/10.1145/2207676.2207736>. 2.2.2, 2.3, 3
- [143] Lauren Race, Chancey Fleet, Danielle Montour, Lindsay Yazzolino, Marco Salsiccia, Claire Ferrari, Sheri Wells-Jensen, and Amy Hurst. Designing while blind: Nonvisual tools and inclusive workflows for tactile graphic creation. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’23, page 1–8. ACM, October 2023. doi: 10.1145/3597638.3614546. URL <https://doi.org/10.1145/3597638.3614546>. 6.4, 7.3.2
- [144] J.E. Rawling, E.C. Carson, J.W. Attig, D.M. Mickelson, W.N. Mode, M.D. Johnson, and K.M. Syverson. Quaternary geology of wisconsin. Technical report, Wisconsin Geological and Natural History Survey, 2025. URL <https://doi.org/10.54915/xqpw9883>. 6.4.1
- [145] Blake Ellis Reid. The curb-cut effect and the perils of accessibility without disability. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4262991. URL <https://doi.org/10.2139/ssrn.4262991>. 1.1
- [146] Samuel Reinders, Matthew Butler, Ingrid Zukerman, Bongshin Lee, Lizhen Qu, and Kim Marriott. When refreshable tactile displays meet conversational agents: Investigating accessible data presentation and analysis with touch and speech. *IEEE Transactions on Visualization and Computer Graphics*, 31(1):864–874, January 2025. ISSN 2160-9306. doi: 10.1109/tvcg.2024.3456358. URL <https://doi.org/10.1109/tvcg.2024.3456358>. 6.3.2, 7.2, 7.3.3
- [147] Rod D Roscoe, Erin K Chiou, and Abigail R Wooldridge, editors. *Advancing diversity, inclusion, and social justice through human systems engineering*. CRC Press, London, England, December 2019. 1.1
- [148] Azriel Rosenfeld and John L. Pfaltz. Sequential operations in digital picture processing. *Journal of the ACM*, 13(4):471–494, October 1966. ISSN 1557-735X. doi: 10.1145/321356.321357. URL <http://dx.doi.org/10.1145/321356.321357>. 6.5.4.3
- [149] Heather Ruland Staines. Digital open annotation with hypothesis: Supplying the missing capability of the web. *Journal of Scholarly Publishing*, 49(3):345–365, April 2018. ISSN 1710-1166. doi: 10.3138/jsp.49.3.04. URL <http://dx.doi.org/10.3138/jsp.49.3.04>. 10.3
- [150] Manaswi Saha, Michael Saugstad, Hanuma Teja Maddali, Aileen Zeng, Ryan Holland,

- Steven Bower, Aditya Dash, Sage Chen, Anthony Li, Kotaro Hara, and Jon Froehlich. Project sidewalk: A web-based crowdsourcing tool for collecting sidewalk accessibility data at scale. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, page 1–14. ACM, May 2019. doi: 10.1145/3290605.3300292. URL <http://dx.doi.org/10.1145/3290605.3300292>. 10.3
- [151] Antti Salovaara. Inventing new uses for tools: A cognitive foundation for studies on appropriation. *Human Technology: An Interdisciplinary Journal on Humans in ICT Environments*, 4(2):209–228, November 2008. ISSN 1795-6889. doi: 10.17011/ht/urn.200811065856. URL <https://doi.org/10.17011/ht/urn.200811065856>. 2.1.2
- [152] Elizabeth B.-N. Sanders and Pieter Jan Stappers. Probes, toolkits and prototypes: three approaches to making in codesigning. *CoDesign*, 10(1):5–14, January 2014. ISSN 1745-3755. doi: 10.1080/15710882.2014.888183. URL <https://doi.org/10.1080/15710882.2014.888183>. 2.1.1
- [153] SAS. Sas Graphics Accelerator. <https://support.sas.com/software/products/graphics-accelerator/>, 2022. Accessed: 2021-09-08. 2.2.1
- [154] Arvind Satyanarayan, Dominik Moritz, Kanit Wongsuphasawat, and Jeffrey Heer. Vegalite: A grammar of interactive graphics. *IEEE Transactions on Visualization and Computer Graphics*, 23(1):341–350, January 2017. ISSN 1077-2626. doi: 10.1109/tvcg.2016.2599030. URL <https://doi.org/10.1109/tvcg.2016.2599030>. 1.2, 2.2.3, 6.2, 6.3.1, 6.5.2, 6.5.4.1, 9.5
- [155] Manolis Savva, Nicholas Kong, Arti Chhajta, Li Fei-Fei, Maneesh Agrawala, and Jeffrey Heer. Revision: automated classification, analysis and redesign of chart images. In *Proceedings of the 24th annual ACM symposium on User interface software and technology*, UIST '11, page 393–402. ACM, October 2011. doi: 10.1145/2047196.2047247. URL <http://dx.doi.org/10.1145/2047196.2047247>. 6.5.4.3
- [156] Anastasia Schaadhardt, Alexis Hiniker, and Jacob O. Wobbrock. Understanding Blind Screen-Reader Users' Experiences of Digital Artboards. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, New York, NY, USA, May 2021. Association for Computing Machinery. Accessed: 2021-09-06. 2.2.1
- [157] Doug Schepers. On underlays. Internet Archive, 2026. URL https://web.archive.org/web/20260616140431/https://www.linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL?utm_source=share&utm_medium=member_desktop&rcm=ACoAADDaWbKBOdoW1lI9B5DHy57VfR5jIs33Kq0. Archived version available at https://web.archive.org/web/20260616140431/https://www.linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL?utm_source=share&utm_medium=member_desktop&rcm=ACoAADDaWbKBOdoW1lI9B5DHy57VfR5jIs33Kq0; original URL: [linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL](https://www.linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL). Accessed: 2026-06-16. 9.6

- [158] William Schiff and Emerson Foulke, editors. *Tactual Perception*. Cambridge University Press, Cambridge, England, March 1982. ISBN 0521240956. 2.2.1, 7.3.2
- [159] Donald Schön and John Bennett. Reflective conversation with materials, April 1996. URL <https://doi.org/10.1145/229868.230044>. 6.8.1, 9.3
- [160] JooYoung Seo, Yilin Xia, Bongshin Lee, Sean Mccurry, and Yu Jun Yam. Maidr: Making statistical visualizations accessible with multimodal data representation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, page 1–22. ACM, May 2024. doi: 10.1145/3613904.3642730. URL <https://doi.org/10.1145/3613904.3642730>. 6.3.2, 10.2
- [161] Hanieh Shakeri, Ye Yuan, Benett Axtell, Denise Y. Geiskkovitch, and Carman Neustaedter. Designing smart home technology for passive co-presence over distance. In *Designing Interactive Systems Conference*, DIS '24, pages 3389–3406. ACM, July 2024. doi: 10.1145/3643834.3661508. URL <http://dx.doi.org/10.1145/3643834.3661508>. 10.3
- [162] Ather Sharif and Babak Forouraghi. evographs — a jquery plugin to create web accessible graphs. In *2018 15th IEEE Annual Consumer Communications & Networking Conference (CCNC)*, page 1–4. IEEE, January 2018. doi: 10.1109/ccnc.2018.8319239. URL <https://doi.org/10.1109/ccnc.2018.8319239>. 6.3.3
- [163] Ather Sharif, Sanjana Shivani Chintalapati, Jacob O. Wobbrock, and Katharina Reinecke. Understanding screen-reader users' experiences with online data visualizations. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '21, page 1–16. ACM, October 2021. doi: 10.1145/3441852.3471202. URL <https://doi.org/10.1145/3441852.3471202>. 2.2.1, 6.3.2, 6.3.3, 7.3.3, 9.1
- [164] Ather Sharif, Olivia H. Wang, Alida T. Muongchan, Katharina Reinecke, and Jacob O. Wobbrock. Voxlens: Making online data visualizations accessible with an interactive javascript plug-in. In *CHI Conference on Human Factors in Computing Systems*, CHI '22, page 1–19. ACM, April 2022. doi: 10.1145/3491102.3517431. URL <https://doi.org/10.1145/3491102.3517431>. 2.2.1, 6.3.3
- [165] Ather Sharif, Joo Gyeong Kim, Jessie Zijia Xu, and Jacob O. Wobbrock. Understanding and reducing the challenges faced by creators of accessible online data visualizations. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '24, page 1–20. ACM, October 2024. doi: 10.1145/3663548.3675625. URL <https://doi.org/10.1145/3663548.3675625>. 2.2.3, 3, 6.2, 6.3.3, 6.8.2, 9.1, 10.2
- [166] Ather Sharif, Venkatesh Potluri, Jazz Rui Xia Ang, Jacob O. Wobbrock, and Jennifer Mankoff. Touchpad mapper: Examining information consumption from 2d digital content using touchpads by screen-reader users. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '24, page 1–4. ACM, October 2024. doi: 10.1145/3663548.3688505. URL <https://doi.org/10.1145/3663548.3688505>. 7.3.4, 7.5.2

- [167] Ashley Shew. *Against Technoableism: Rethinking Who Needs Improvement*. W. W. Norton, New York, NY, 2023. ISBN 9781324036661. ISBN: 978-1-324-03666-1. [2.3](#)
- [168] Shneiderman. Direct manipulation: A step beyond programming languages. *Computer*, 16(8):57–69, August 1983. ISSN 0018-9162. doi: 10.1109/mc.1983.1654471. URL <https://doi.org/10.1109/mc.1983.1654471>. [6.3.1](#)
- [169] Ben Shneiderman, Christopher Williamson, and Christopher Ahlberg. Dynamic queries: database searching by direct manipulation. In *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '92*, CHI '92, page 669–670. ACM Press, 1992. doi: 10.1145/142750.143082. URL <https://doi.org/10.1145/142750.143082>. [6.2](#), [7.3.1](#), [G6](#):
- [170] Jim Smiley, Benjamin Lee, Siddhant Tandon, Maxime Cordeil, Lonni Besançon, Jarrod Knibbe, Bernhard Jenny, and Tim Dwyer. The made-axis: A modular actuated device to embody the axis of a data dimension. *Proceedings of the ACM on Human-Computer Interaction*, 5(ISS):1–23, November 2021. ISSN 2573-0142. doi: 10.1145/3488546. URL <https://doi.org/10.1145/3488546>. [7.3.4](#), [7.5.2](#)
- [171] Jim Smiley, Swamy Ananthanarayan, Leona M Holloway, Matthew Butler, and Tim Dwyer. Magnepins: A modular, affordable, and diy refreshable braille and tactile display. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, UIST '25, pages 1–12. ACM, September 2025. doi: 10.1145/3746059.3747692. URL <https://doi.org/10.1145/3746059.3747692>. [7.2](#), [7.5](#)
- [172] David Canfield Smith. *Pygmalion: A Computer Program to Model and Stimulate Creative Thought*. Birkhäuser Basel, 1977. ISBN 9783034857444. doi: 10.1007/978-3-0348-5744-4-4. URL <https://doi.org/10.1007/978-3-0348-5744-4>. Reissue of the author’s 1975 Stanford University doctoral dissertation. [6.3.1](#)
- [173] Volker Sorge. Polyfilling accessible chemistry diagrams. In *Lecture Notes in Computer Science*, page 43–50. Springer International Publishing, 2016. ISBN 9783319412641. doi: 10.1007/978-3-319-41264-1_6. URL https://doi.org/10.1007/978-3-319-41264-1_6. [2.2.1](#), [6.3.2](#)
- [174] Katta Spiel, Kathrin Gerling, Cynthia L. Bennett, Emeline Brulé, Rua M. Williams, Jennifer Rode, and Jennifer Mankoff. Nothing about us without us: Investigating the role of critical disability studies in hci. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, page 1–8. ACM, April 2020. doi: 10.1145/3334480.3375150. URL <https://doi.org/10.1145/3334480.3375150>. [1.1](#), [2.3](#), [9.7.3](#), [10.1](#), [10.2](#)
- [175] Clay Spinuzzi. The methodology of participatory design. *Technical communication*, 52(2):163–174, 2005. [2.1.1](#)
- [176] Sebastian Paul Suggate. Beyond self-report: Measuring visual, auditory, and tactile mental imagery using a mental comparison task. *Behavior Research Methods*, 56(8):8658–8676, September 2024. ISSN 1554-3528. doi: 10.3758/s13428-024-02496-z. URL <https://doi.org/10.3758/s13428-024-02496-z>. [1.1](#)
- [177] Cella M Sum, Rahaf Alharbi, Franchesca Spektor, Cynthia L Bennett, Christina N Har-

- rington, Katta Spiel, and Rua Mae Williams. Dreaming disability justice in hci. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, CHI '22, page 1–5. ACM, April 2022. doi: 10.1145/3491101.3503731. URL <http://dx.doi.org/10.1145/3491101.3503731>. 2.3, 10.2
- [178] Cella M. Sum, Franchesca Spektor, Rahaf Alharbi, Leya Breanna Baltaxe-Admony, Erika Devine, Hazel Anneke Dixon, Jared Duval, Tessa Eagle, Frank Elavsky, Kim Fernandes, Leandro S. Guedes, Serena Hillman, Vaishnav Kameswaran, Lynn Kirabo, Tamanna Motahar, Kathryn E. Ringland, Anastasia Schaadhardt, Laura Scheepmaker, and Alicia Williamson. Challenging ableism: A critical turn toward disability justice in hci. *XRDS: Crossroads, The ACM Magazine for Students*, 30(4):50–55, June 2024. ISSN 1528-4980. doi: 10.1145/3665602. URL <http://dx.doi.org/10.1145/3665602>. 2.3, 10.2
- [179] Ruoting Sun, Rong Luo, Xiwen Yao, Xinran She, Kotaro Hara, and Yang Jiao. Tactile data comics: A step-by-step multimodal presentation method on a refreshable tactile display for blind and visually impaired individuals. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, CHI EA '25, page 1–8, New York, NY, April 2025. ACM. doi: 10.1145/3706599.3720192. URL <https://doi.org/10.1145/3706599.3720192>. 7.2, 7.3.2
- [180] Ivan E. Sutherland. Sketchpad: A man-machine graphical communication system. In *Proceedings of the May 21-23, 1963, spring joint computer conference on - AFIPS '63 (Spring)*, AFIPS '63 (Spring), page 329, New York, NY, USA, 1963. ACM Press. doi: 10.1145/1461551.1461591. URL <https://doi.org/10.1145/1461551.1461591>. 6.3.1
- [181] Sarit Felicia Anais Szpiro, Shafeka Hashash, Yuhang Zhao, and Shiri Azenkot. How people with low vision access computing devices: Understanding challenges and opportunities. In *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '16, page 171–180, New York, NY, USA, October 2016. ACM. ISBN 9781450341240. doi: 10.1145/2982142.2982168. URL <https://doi.org/10.1145/2982142.2982168>. 2.2.3
- [182] Pierre Tchounikine. Designing for appropriation: A theoretical account. *Human-Computer Interaction*, 32(4):155–195, July 2016. ISSN 1532-7051. doi: 10.1080/07370024.2016.1203263. URL <https://doi.org/10.1080/07370024.2016.1203263>. 2.1.2
- [183] Tim Teitelbaum and Thomas Reps. The cornell program synthesizer. *Communications of the ACM*, 24(9):563–573, 1981. doi: 10.1145/358746.358755. URL <https://doi.org/10.1145/358746.358755>. 6.3.1
- [184] Yuanyang (YY) Teng and Darren Gergle. Access is not enough: Toward developmental flourishing. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI EA '26, page 1–6. ACM, April 2026. doi: 10.1145/3772363.3798552. URL <http://dx.doi.org/10.1145/3772363.3798552>. 1.1
- [185] J. Thatcher. Screen reader/2: access to os/2 and the graphical user interface. In *Pro-*

- ceedings of the first annual ACM conference on Assistive technologies - Assets '94*, Assets '94, page 39–46. ACM Press, 1994. doi: 10.1145/191028.191039. URL <http://dx.doi.org/10.1145/191028.191039>. 6.3.1
- [186] John R Thompson, Jesse J Martinez, Alper Sarikaya, Edward Cutrell, and Bongshin Lee. Chart reader: Accessible visualization experiences designed with screen reader users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, page 1–18. ACM, April 2023. doi: 10.1145/3544548.3581186. URL <https://doi.org/10.1145/3544548.3581186>. 2.2.1, 2.4.1, 6.3.2, 6.3.3, 6.4, 7.3.2
- [187] Alexandra To, Angela D. R. Smith, Dilruba Showkat, Adinawa Adjabodjou, and Christina Harrington. Flourishing in the everyday: Moving beyond damage-centered design in hci for bipoc communities. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*, DIS '23, pages 917–933. ACM, July 2023. doi: 10.1145/3563657.3596057. URL <https://doi.org/10.1145/3563657.3596057>. 2.1.1, 10.2
- [188] Visa. Visa Chart Components. <https://github.com/visa/visa-chart-components>, 2022. Accessed: 2022-12-01. 2.2.1
- [189] W3C. Web content accessibility guidelines (wcag) 2.1. <https://www.w3.org/TR/WCAG21/>, 2018. Accessed: 2021-09-03. 2.2.2, 3
- [190] WAI. Understanding Success Criterion 1.3.1 — Understanding WCAG 2.0. <https://www.w3.org/TR/UNDERSTANDING-WCAG20/content-structure-separation-programmatic.html>, 2008. [Accessed 01-04-2025]. G3:
- [191] WAI. Understanding success criterion 4.1.2: name, role, value. WCAG standard, W3C, 2016. Accessed: 2022-03-04. 2.2.2
- [192] WAI. Understanding success criterion 2.1.1: keyboard. WCAG standard, W3C, 2017. URL <https://www.w3.org/WAI/WCAG21/Understanding/keyboard.html>. Accessed: 2026-03-20, <https://www.w3.org/WAI/WCAG21/Understanding/keyboard.html>. 6.3.2
- [193] Bruce N. Walker and Michael A. Nees. Theory of sonification. In Thomas Hermann, Andy Hunt, and John G. Neuhoff, editors, *The Sonification Handbook*, chapter 2, pages 9–39. Logos Verlag, Berlin, Germany, 2011. 2.2.1
- [194] Chris Weaver. Multidimensional visual analysis using cross-filtered views. In *2008 IEEE Symposium on Visual Analytics Science and Technology*, page 163–170. IEEE, October 2008. doi: 10.1109/vast.2008.4677370. URL <https://doi.org/10.1109/vast.2008.4677370>. 3, 7.2, 7.3.1
- [195] WebAIM. Webaim: The WebAIM Million - An annual accessibility analysis of the top 1,000,000 home pages. <https://webaim.org/projects/million/#wcag>, 2021. Accessed: 2021-09-03. 9.7.1, 10.3
- [196] Hadley Wickham. A layered grammar of graphics. *Journal of Computational and Graphical Statistics*, 19(1):3–28, January 2010. ISSN 1537-2715. doi: 10.1198/jcgs.2009.07098. URL <https://doi.org/10.1198/jcgs.2009.07098>. 6.2
- [197] David Gray Widder and Richmond Wong. Thinking upstream: Ethics and policy oppor-

- tunities in ai supply chains, 2023. URL <https://doi.org/10.48550/arxiv.2303.07529>. 1.2
- [198] David Gray Widder, Meredith Whittaker, and Sarah Myers West. Why ‘open’ ai systems are actually closed, and why this matters. *Nature*, 635(8040):827–833, November 2024. ISSN 1476-4687. doi: 10.1038/s41586-024-08141-1. URL <https://doi.org/10.1038/s41586-024-08141-1>. 1.2
- [199] Leland Wilkinson. The grammar of graphics. In *Handbook of Computational Statistics*, page 375–414. Springer Berlin Heidelberg, December 2011. ISBN 9783642215513. doi: 10.1007/978-3-642-21551-3_13. URL https://doi.org/10.1007/978-3-642-21551-3_13. 6.2, 9.5
- [200] Rua M. Williams and LouAnne E. Boyd. Prefigurative politics and passionate witnessing. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’19, page 262–266. ACM, October 2019. doi: 10.1145/3308561.3355617. URL <http://dx.doi.org/10.1145/3308561.3355617>. 10.2
- [201] Rua M. Williams, Kathryn Ringland, Amelia Gibson, Mahender Mandala, Arne Maibaum, and Tiago Guerreiro. Articulations toward a crip hci. *Interactions*, 28(3):28–37, April 2021. ISSN 1558-3449. doi: 10.1145/3458453. URL <http://dx.doi.org/10.1145/3458453>. 2.3, 10.2
- [202] Rua Mae Williams, Louanne Boyd, and Juan E. Gilbert. Counterventions: a reparative reflection on interventionist hci. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI ’23, page 1–11. ACM, April 2023. doi: 10.1145/3544548.3581480. URL <http://dx.doi.org/10.1145/3544548.3581480>. 10.2
- [203] B. L. Wimer, L. South, K. Wu, D. A. Szafir, M. A. Borkin, and R. A. Metoyer. Beyond vision impairments: Redefining the scope of accessible data representations. *IEEE Transactions on Visualization and Computer Graphics*, 30(12):7619–7636, December 2024. 2.2.3
- [204] Brianna L. Wimer, Ritesh Kanchi, Kaija Frierson, Venkatesh Potluri, Ronald Metoyer, Jennifer Mankoff, Miya Natsuhara, and Matt X. Wang. Nonvisual support for understanding and reasoning about data structures, 2026. URL <https://doi.org/10.48550/arxiv.2601.19168>. 6.3.3
- [205] Langdon Winner. Do artifacts have politics? In *Computer Ethics*, page 177–192. Routledge, May 2017. ISBN 9781315259697. doi: 10.4324/9781315259697-21. URL <https://doi.org/10.4324/9781315259697-21>. 1.2
- [206] Jacob O. Wobbrock, Shaun K. Kane, Krzysztof Z. Gajos, Susumu Harada, and Jon Froehlich. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing*, 3(3):1–27, April 2011. ISSN 1936-7236. doi: 10.1145/1952383.1952384. URL <https://doi.org/10.1145/1952383.1952384>. 1.2, 2.1.2, 2.4.3
- [207] Mintesnot Woldeamanuel and Andrew Kent. Measuring walk access to transit in terms of sidewalk availability, quality, and connectivity. *Journal of Urban Planning and Develop-*

- ment, 142(2), June 2016. ISSN 1943-5444. doi: 10.1061/(asce)up.1943-5444.0000296. URL [http://dx.doi.org/10.1061/\(ASCE\)UP.1943-5444.0000296](http://dx.doi.org/10.1061/(ASCE)UP.1943-5444.0000296). 10.3
- [208] R. Wood et al. “creatures of habit”: influential factors to the adoption of computer personalization and accessibility settings. *Universal Access in the Information Society*, 23(2): 927–953, March 2023. 2.4.3, 3
- [209] Rebecca Yeo and Karen Moore. Including disabled people in poverty reduction work: “nothing about us, without us”. *World Development*, 31(3):571–590, March 2003. ISSN 0305-750X. doi: 10.1016/s0305-750x(02)00218-8. URL [https://doi.org/10.1016/s0305-750x\(02\)00218-8](https://doi.org/10.1016/s0305-750x(02)00218-8). 1.1
- [210] Xiaoyi Zhang, Tracy Tran, Yuqian Sun, Ian Culhane, Shobhit Jain, James Fogarty, and Jennifer Mankoff. Interactiles: 3d printed tactile interfaces to enhance mobile touchscreen accessibility. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’18, page 131–142. ACM, October 2018. doi: 10.1145/3234695.3236349. URL <https://doi.org/10.1145/3234695.3236349>. 7.3.4, 7.5.2
- [211] Haixia Zhao, Catherine Plaisant, Ben Shneiderman, and Jonathan Lazar. Data sonification for users with visual impairment: A case study with georeferenced data. *ACM Transactions on Computer-Human Interaction*, 15(1):1–28, May 2008. ISSN 1557-7325. doi: 10.1145/1352782.1352786. URL <https://doi.org/10.1145/1352782.1352786>. 2.2.1
- [212] Jonathan Zong. Using real names of disabled participant-contributors to practice citational justice in accessibility. In *2025 IEEE Workshop on Accessible Data Visualization (AccessViz)*, page 30–33. IEEE, November 2025. doi: 10.1109/accessviz68666.2025.00011. URL <https://doi.org/10.1109/accessviz68666.2025.00011>. 2.3, 6.4
- [213] Jonathan Zong, Crystal Lee, Alan Lundgard, JiWoong Jang, Daniel Hajas, and Arvind Satyanarayan. Rich screen reader experiences for accessible data visualization. *Computer Graphics Forum*, 41(3):15–27, June 2022. doi: 10.1111/cgf.14519. URL <https://doi.org/10.1111/cgf.14519>. 2.2.1, 2.3, 2.4.1, 2.4.2, 3, 6.3.2, 6.3.3, 6.4, 7.2, 7.3.3
- [214] Jonathan Zong, Josh Pollock, Dylan Wootton, and Arvind Satyanarayan. Animated vegalite: Unifying animation with a grammar of interactive graphics. *IEEE Transactions on Visualization and Computer Graphics*, 29(1):149–159, January 2023. ISSN 2160-9306. doi: 10.1109/tvcg.2022.3209369. URL <https://doi.org/10.1109/tvcg.2022.3209369>. 6.2
- [215] Jonathan Zong, Isabella Pedraza Pineros, Mengzhu (Katie) Chen, Daniel Hajas, and Arvind Satyanarayan. Umwelt: Accessible structured editing of multi-modal data representations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI ’24, page 1–20. ACM, May 2024. doi: 10.1145/3613904.3641996. URL <https://doi.org/10.1145/3613904.3641996>. 2.2.1, 6.3.3, 6.8.4, 7.2, 7.3.2, 9.7.3